

K-FIELD FROM CELL3 THROUGH THE LUCAS SUPERGROUP

Canonical Geometry, Seven-Sheet Polarity, Reciprocal Closure, Mass Bridge, and the Minimal Complete Working Model

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This document is a stand-alone reference. It defines the K-Field from the single Cell3 through the Lucas Group and Lucas Supergroup, derives every stated count and metric from the canonical matrices, and separates intrinsic geometry from observer-dependent physical rendering.

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The numerical tables, equations, node catalogues, plane sections, reciprocal-direction catalogue, and scale hierarchy in this report constitute the archival technical definition of the working model. All diagrams are computed from the numerical records printed in the report; no decorative or illustrative geometry substitutes for the canonical data.

Abstract

The K-Field is a recursively quantized reciprocal phase geometry coupling mass-energy to motion, orientation, and closure through a fixed oblique Cell3 lattice, a seven-sheet polarity complex, and the Lucas-Supergroup hierarchy. Its intrinsic data are an ICRS/J2000 origin, a left-handed direct basis B , an independent positive-definite address-energy form W , the integer address lattice Z^3 , the laminar grade λ , alternating polarity χ , complete factor-three aggregation, and a locked projective Mass Bridge. An observer packet is added only when the fixed structure is rendered as force or energy in a specified frame.

One Cell3 contains the centered ternary address set $\{-1,0,1\}^3$. Its 27 positions are one volume center, six face centers, twelve edge centers, and eight vertices. The homomorphism $\lambda=i+j+k$ sorts the positions into seven parallel sheets with populations 1,3,6,7,6,3,1, while $\chi=(-1)^\lambda$ alternates their polarity. The kernel of λ is an embedded A_2 root lattice, so same-sheet motion is generated by the three half-edge differences $(a-b)/2$, $(a-c)/2$, and $(b-c)/2$. The 26 noncentral addresses form thirteen antipodal projective channels.

Complete cells aggregate by three. A Lucas Group contains $3 \times 3 \times 3 = 27$ Cell3 volumes. A Lucas Supergroup contains $3 \times 3 \times 3$ Lucas Groups, equivalently $9 \times 9 \times 9 = 729$ Cell3 volumes. Its complete cubical complex has 1,000 vertices, 2,700 edges, 2,430 faces, and 729 three-cells. Its half-step complex has 6,859 states on 55 sheets. Its center graph has 1,944 interfaces and cycle rank 1,216, but the full oriented cubical complex is contractible: the boundary ranks are 999, 1,701, and 729, and the Betti vector is $(1,0,0,0)$. Local graph cycles therefore become coexact when plaquettes are included.

The reciprocal spectrum is generated exactly by $R=(B^T B)^{-1}$. On the primitive radius-nine projective shell there are 864 directions, each with a distinct quadratic-form value. The locked Mass Bridge is carried by the three primitive face areas. The independent form W produces the positive phase metric $G_{\text{phase}}=4 B^T W B^{-1}$. The geometry and topology are intrinsic; force, energy, occupation, pairing, spin, isomer, and decay assignments require the declared constitutive and observer layers. The report concludes by reducing the full model to eight nonredundant axioms and one canonical formula set.

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1. Archival Definition, Completeness, and Scope

1.1 Exhaustive archival graph integrity

The canonical record is treated as a typed archival multigraph rather than as a sequence of isolated reports. Artifact, equation, numerical value, geometric object, interpretation, correction, dependency, supersession, and witness records are distinct node classes. Randomized entry through materials, propagation, energy, galaxy geometry, holonomy, rotating devices, topology, Mass Bridge, lexicon, standing waves, force reduction, and solar-system quantization converges on the same intrinsic construction.

The exhaustive July 18 graph contains 20,960 nodes, 102,390 typed edges, 2,956 hyperedges, 152 canonical terms, 1,071 reconstruction paths, and 30 conflict-repair records. The July 23-26 subcorpus contains 32 exact JSON records, one duplicate, one superseded force-reduction edition, 30 active records, 128 concepts, 140 relations, and convergence across 128 randomized re-entry passes. These counts are archival integrity metadata; the mathematical model that follows is independent of entry order.

Table 1.1. Archival graph integrity counts.

Record family	Count or status
Exhaustive graph nodes	20,960
Typed edges	102,390
Hyperedges	2,956
Canonical terms	152
Reconstruction paths	1,071
Conflict-repair records	30
Exact July 23-26 JSON records	32
Duplicate records	1
Superseded force-reduction editions	1
Active records	30
Concepts	128
Relations	140
Randomized re-entry passes	128

1.2 Authoritative dependency set

The following files form the controlling dependency set for the canonical Cell3-to-Supergroup model. The sequence is a dependency ordering, not a developmental narrative: each later record consumes the fixed objects named earlier and adds a typed layer without redefining the underlying basis.

Table 1.2. Authoritative archival dependency chain.

Order	Authoritative machine-readable record
1	K_Field_native_unit_sphere_derivation_stack_1784476462.json
2	K-Field_entire_compact_form_1784476462_machine_readable.json
3	K-Field_Mass-Bridge_is_now_a_locked_projective_ratio_system_1785176592(1).json
4	K-Field_mass_bridge_drives_volume_filling_hierarchy_1785204375.json
5	K-Field_internal_energy_hierarchy_1785190900.json
6	K-Field_Lucas_Supergroup_Analysis_1785208014.json
7	K-Field_Lucas_Supergroup_Complete_Nuclear_Closure_Domain_1785235808.json
8	K-Field_Lucas_Supergroup_metric_Hodge_program_1785236325.json
9	K-Field_Lucas_Supergroup_exact_deterministic_reciprocal_field_spectrum_1785237526.json
10	K-Field_Lucas_Supergroup_extracted_definition_1785238606.json

1.3 Canonical object

The minimal complete intrinsic object is

$$K = (O_ICRS, B, W, Z^3, \lambda, \chi, A_3, P_MB).$$

When a physical force or an observer-indexed energy is required, the optional observer packet is

```
O_packet = ( t, v_ref, kappa_F, H ).
```

Here $B=[a \ b \ c]$ is the ordered left-handed Cell3 basis in fixed ICRS/J2000 Cartesian space; W is the independent positive-definite address-energy form; Z^3 is the lossless address lattice; $\lambda(i,j,k)=i+j+k$ is the laminar grade; $\chi(i,j,k)=(-1)^{(i+j+k)}$ is the alternating polarity; A_3 is complete-cell aggregation by a factor of three; and P_{MB} is the locked Mass-Bridge projective ratio system. The observer packet contains time, a reference velocity, the matter-coupling coefficient, and the shape or occupancy tensor.

The direct geometry, reciprocal geometry, Lucas nodes, seven sheets, Emily bands, Lucas Groups, Lucas Supergroups, cubical boundaries, Hodge spectra, shell directions, and continuum reductions are derived from these data. Observer motion renders the fixed field; it does not define or rotate the field.

A fixed oblique rank-three lattice has a direct geometry, a reciprocal geometry, and an independent positive-definite address-energy form. Centered half-step sampling of one cell creates a 27-state affine ternary registry. The sum homomorphism divides that registry into seven A_2 -parallel sheets with alternating polarity. Complete cells aggregate recursively by three, producing the Lucas Group and the 9B Lucas Supergroup. The oriented cubical complex determines which local graph modes survive globally. The Mass Bridge locks the face-cofactor hierarchy to physical witness ratios, while observer motion renders the fixed structure through the phase metric rather than defining it.

1.4 Intrinsic and rendered layers

The intrinsic layer consists of B , M , R , W , the address lattice, the sheet maps, the incidence graph, the complete cubical hierarchy, and the Mass-Bridge ratios. It is invariant under a change of observer packet. The rendered layer begins when a reference velocity, angular velocity, polarization, active mass, geometry tensor, and coupling coefficient are supplied. This separation prevents a traversal direction, such as the Earth-CMB direction, from being mistaken for an intrinsic Cell3 normal.

2. Mathematical Language, Coordinates, and Dimensional Discipline

2.1 Fixed coordinate frame

All Cartesian vectors in this report are expressed in the fixed ICRS/J2000 frame unless a local plane basis is explicitly stated. A vector is a three-component column. The direct basis matrix B stores a , b , and c as columns. Because $\det(B)$ is negative, the ordered triple is left-handed. This sign is retained; the physical cell volume is its absolute value.

A direct integer address n in Z^3 maps to physical space by $x=B n$. A centered Cell3 uses local coordinates u in $[-1/2,1/2]^3$. A reciprocal integer address s maps to a physical wavevector through $B^{\wedge-T}$. Direct addresses, reciprocal addresses, and observer vectors are never interchanged.

2.2 Units and scale covariance

The primary construction is unitless and scale-covariant. The terminal dimensional witness used throughout the figures is the nanometre matrix $B_nm=\sigma B_native$ with $\sigma=16.068602900494266$ nm/native. Multiplying B by a scalar multiplies lengths by that scalar, areas by its square, and volumes by its cube. It does not alter integer addresses, incidence, sheet labels, polarity, projective face ratios, normalized reciprocal directions, or hierarchy counts.

Table 2.1. Typed quantities and units.

Object	Definition	Units in this report
B	direct basis matrix	nm
M	$B^{\wedge T} B$	$nm^{\wedge 2}$
R	$M^{\wedge -1}$	$nm^{\wedge -2}$
W	address-energy quadratic form	$nm^{\wedge 2}$ in stored normalization
G_phase	$4 B^{\wedge -T} W B^{\wedge -1}$	dimensionless
lambda	$i+j+k$	integer
chi	$(-1)^{\lambda}$	sign
k_s	$(2 \pi / 9) B^{\wedge -T} s$	$nm^{\wedge -1}$
q_R(s)	$s^{\wedge T} R s$	$nm^{\wedge -2}$
V_1	$ \det B $	$nm^{\wedge 3}$

2.3 Direct, reciprocal, and energetic tensors

Three matrices must remain distinct. M is the direct Gram matrix and measures squared direct lengths. R is the inverse Gram matrix and measures squared reciprocal lengths. W is an independent positive-definite quadratic form on address rates. G_phase is the physical-space form obtained by pushing W through the inverse direct basis.

$$M = B^{\wedge T} B; R = M^{\wedge -1} = B^{\wedge -1} B^{\wedge -T}; G_phase = 4 B^{\wedge -T} W B^{\wedge -1}.$$

3. Intrinsic Cell3 Geometry

3.1 Direct basis

The terminal nanometre rendering of the ordered Cell3 basis is

Table 3.1. Direct basis B in nanometres.

row	a column	b column	c column
1	0.32750124205589060	0.03533077584382840	-0.01012931762672882
2	-0.05489700788400644	0.20731985044019177	-0.38978770236625987
3	-0.03748750289472991	-0.54740108538850807	-0.68973440836726763

$B_{nm} = \sigma B_{native}$, $\sigma = 16.068602900494266$ nm/native.

Cell3 base geometry: $|a|=0.334180$ nm, $|b|=0.586411$ nm, $|c|=0.792320$ nm

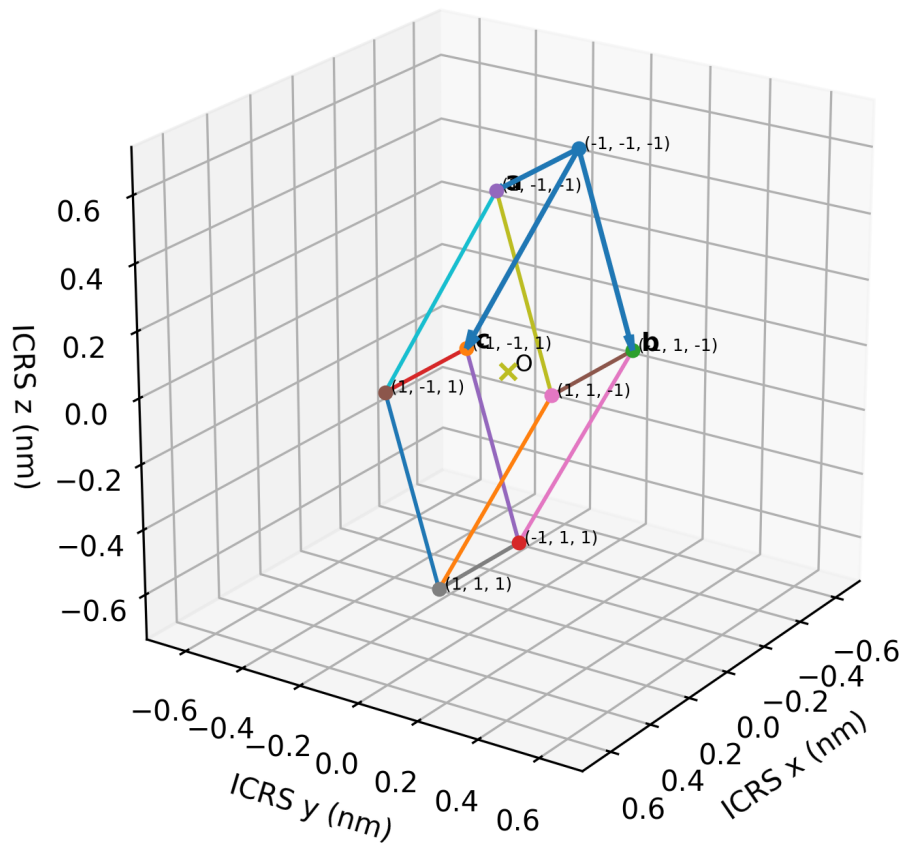


Figure 3.1. Dimensionally correct Cell3 parallelepiped in fixed ICRS/J2000 Cartesian coordinates. The twelve boundary edges and all eight vertices are computed from $B[-1/2, 1/2]^3$. The ordered basis arrows originate at the $(-1, -1, 1)$ corner. Axes are in nanometres and use equal three-dimensional scaling.

3.2 Direct and reciprocal metrics

Table 3.2. Direct Gram matrix $M=B^T B$, in nm^2 .

M	column 1	column 2	column 3
row 1	0.11167605789605009	0.02071033328060580	0.04393723509621977
row 2	0.02071033328060580	0.34387773239278702	0.29639275896152290
row 3	0.04393723509621977	0.29639275896152290	0.62777061007729584

Table 3.3. Reciprocal metric $R=M^{-1}$, in nm^{-2} .

R	column 1	column 2	column 3
row 1	9.20802552945754904	0.00153480242819270	-0.64518803569993899
row 2	0.00153480242819275	4.90339831236798673	-2.31517558441638904
row 3	-0.64518803569993899	-2.31517558441638904	2.73117127471169452

For any direct integer address n , the squared physical length is $q_M(n)=n^T M n$. For any reciprocal integer address s , the squared reciprocal magnitude before the 2π periodicity factor is $q_R(s)=s^T R s$.

3.3 Edge lengths, angles, volume, faces, and spacings

Table 3.4. Complete Cell3 metric inventory.

Quantity	Exact numerical value	Meaning
a	0.3341796790591105 nm	first primitive edge
b	0.5864108904111408 nm	second primitive edge
c	0.7923197650426852 nm	third primitive edge
theta_ab	83.93348301231273 deg	included angle between a and b
theta_ac	80.44813001833791 deg	included angle between a and c
theta_bc	50.36323181375618 deg	included angle between b and c
det B	-0.11791489191174465 nm^3	signed volume; negative means left-handed
V_1	0.11791489191174465 nm^3	physical Cell3 volume
A_ab	0.19486916546048413 nm^2	area of primitive ab face
A_ac	0.26110623579020531 nm^2	area of primitive ac face
A_bc	0.35780953912856150 nm^2	area of primitive bc face
d_a	0.3295465296954468 nm	spacing between opposite bc faces
d_b	0.4515973797212847 nm	spacing between opposite ac faces
d_c	0.6050977415185554 nm	spacing between opposite ab faces
L_G	1.3436536216759252 nm	longest body diagonal a+b+c

The opposite-face spacings follow from volume divided by face area. For example, the altitude normal to the bc face is $d_a=V_1/A_{bc}$. These are exact parallelepiped identities, not fitted lengths.

$$d_a = V_1/A_{bc} = 0.3295465296954468 \text{ nm}; \quad d_b = V_1/A_{ac} = 0.4515973797212847 \text{ nm}; \quad d_c = V_1/A_{ab} = 0.6050977415185554 \text{ nm}.$$

Primitive ab face in its true metric plane

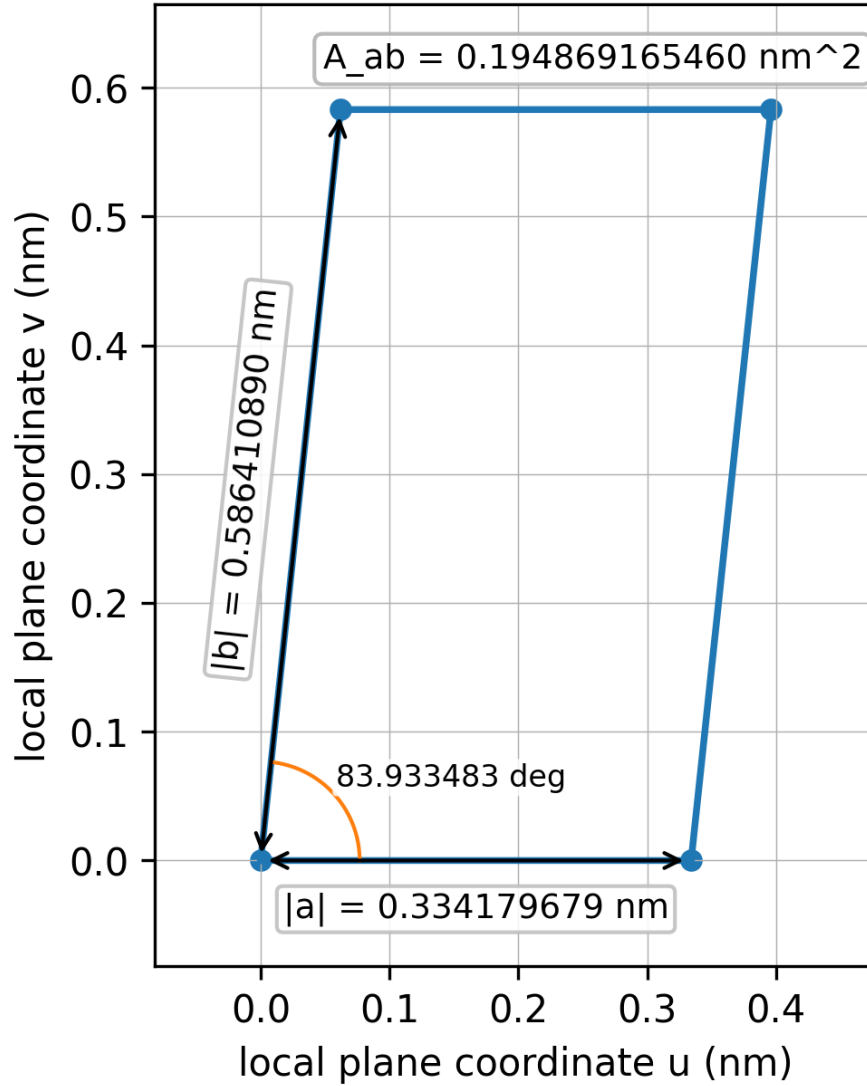


Figure 3.2. The primitive ab face unfolded into an orthonormal local plane. The plotted edge lengths, included angle, and enclosed area are computed directly from the corresponding columns of B. No visual rescaling changes the metric relationships.

Primitive ac face in its true metric plane

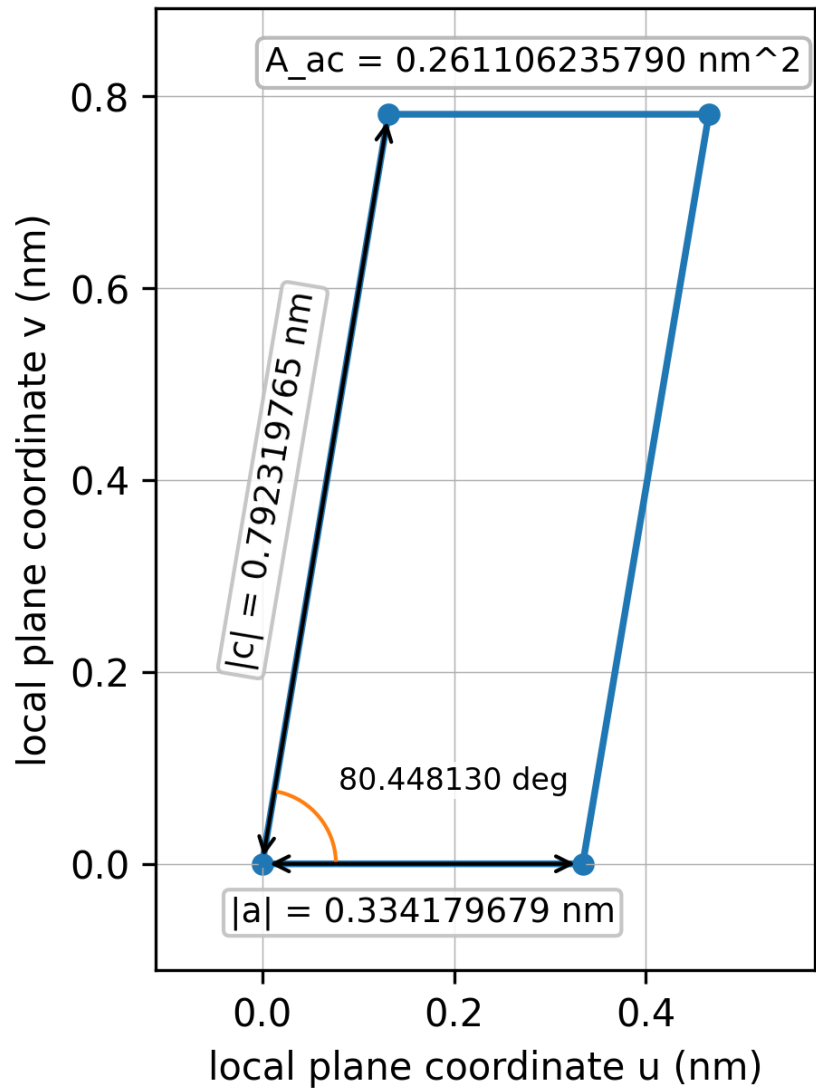


Figure 3.3. The primitive ac face unfolded into an orthonormal local plane. The plotted edge lengths, included angle, and enclosed area are computed directly from the corresponding columns of B. No visual rescaling changes the metric relationships.

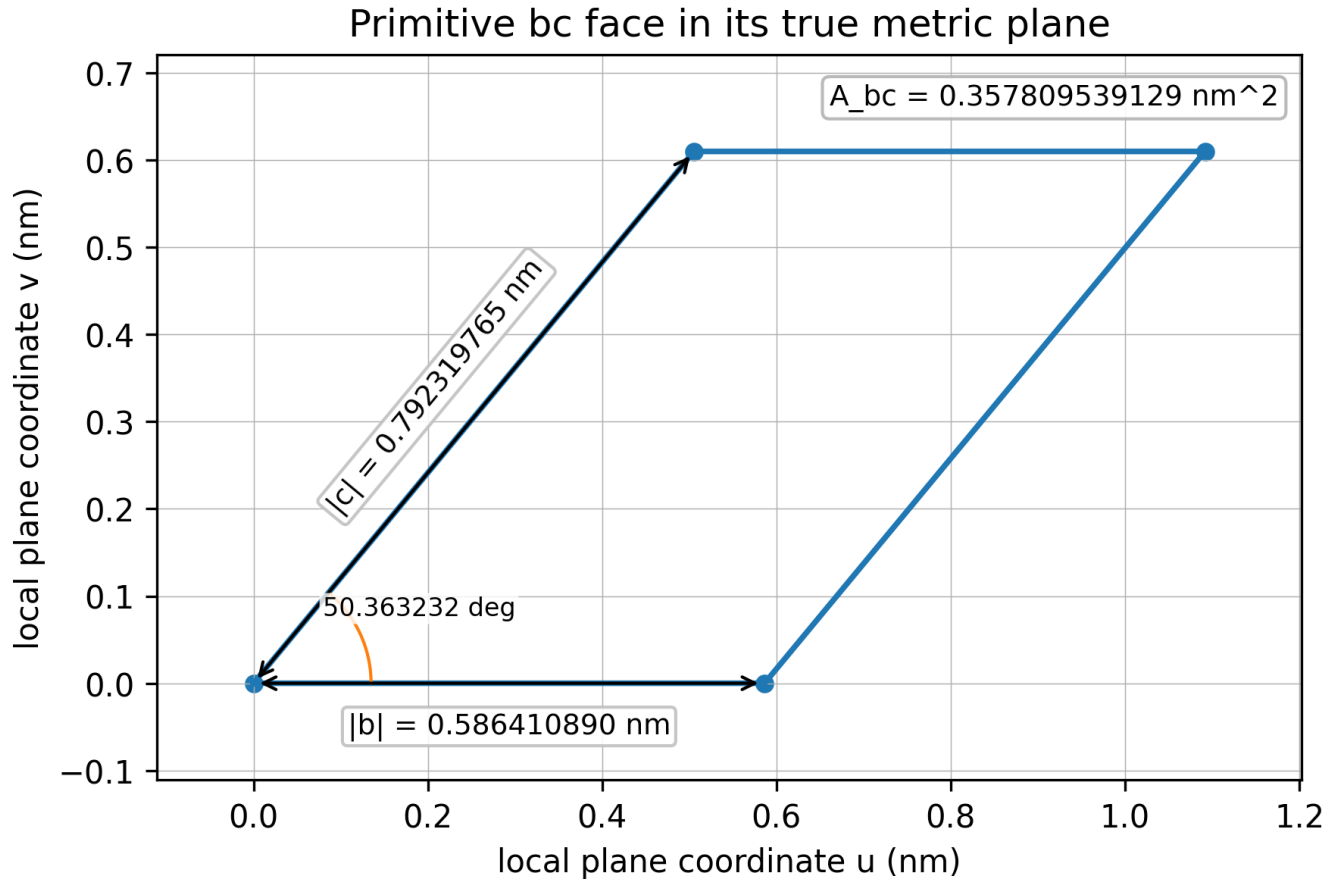


Figure 3.4. The primitive bc face unfolded into an orthonormal local plane. The plotted edge lengths, included angle, and enclosed area are computed directly from the corresponding columns of B. No visual rescaling changes the metric relationships.

3.4 Infinite direct and reciprocal lattices

$$\Lambda = B Z^3.$$

The centered Cell3 at integer address z is

$$C_z = O_{ICRS} + B (z + [-1/2, 1/2]^3), \quad z \text{ in } Z^3.$$

Using the 2π reciprocal convention, the reciprocal wave lattice is

$$\Lambda_{\text{star}} = 2\pi B^{-T} Z^3.$$

4. Lucas Nodes: The Complete Single-Cell3 Registry

4.1 Centered ternary sampling

One Cell3 contains the centered half-step address registry $L_0=\{-1,0,1\}^3$. Every address $L=(i,j,k)$ maps to the physical position $p(L)=B L/2$. The three address values on each axis are positions at the negative boundary, center, and positive boundary of one cell; they are not three complete cell intervals.

$$L_0 = \{-1,0,1\}^3; p(L) = (1/2) B L; |L_0| = 3^3 = 27.$$

A node with all three address components nonzero is a vertex. A node with two nonzero components is an edge center. A node with one nonzero component is a face center. The zero address is the volume center. The class polynomial is

$$F_1(t) = (2+t)^3 = 8 + 12 t + 6 t^2 + t^3.$$

Table 4.1. Geometric classification of the 27 Lucas nodes.

Geometric class	Nonzero address components	Count
Vertices	3	8
Edge centers	2	12
Face centers	1	6
Volume center	0	1
Total		27

The complete 27-node centered half-step Lucas registry in one Cell3

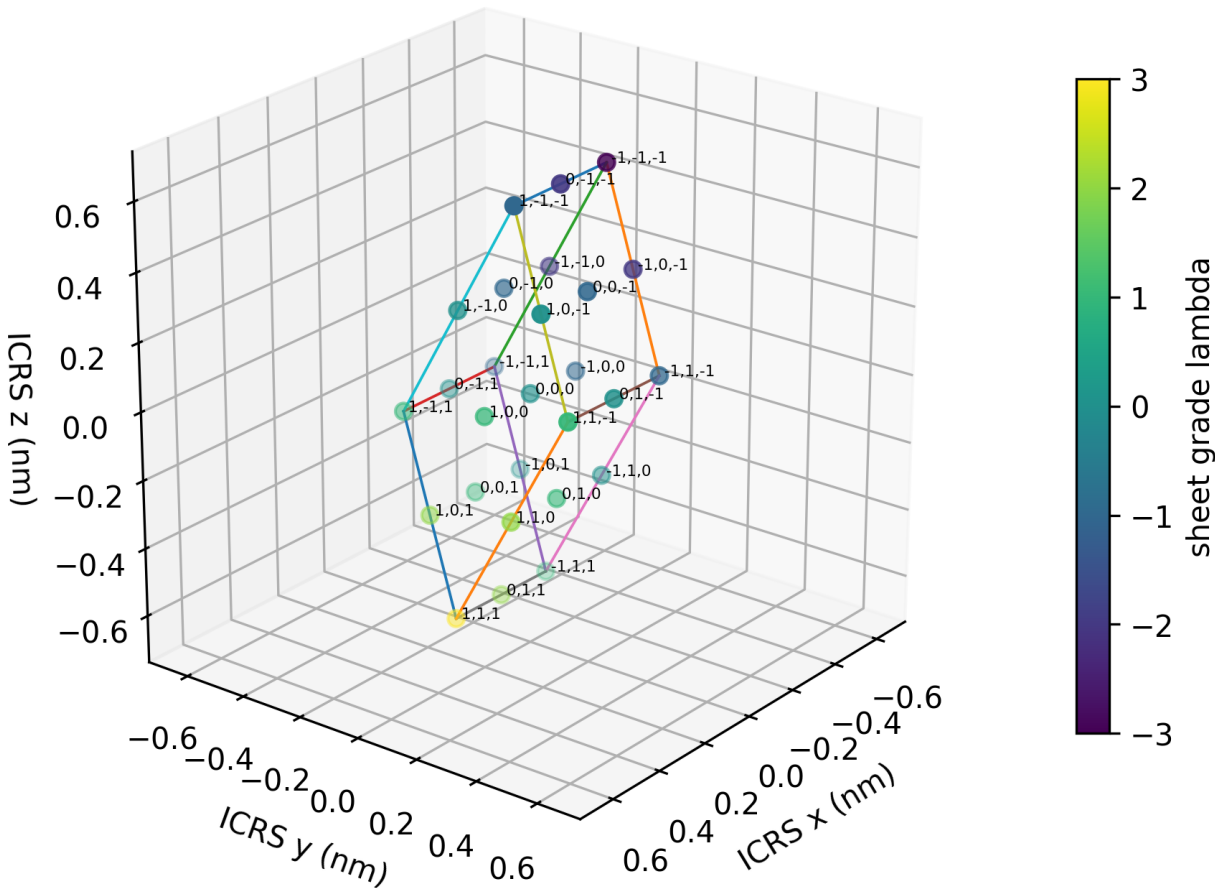


Figure 4.1. All 27 Lucas nodes plotted in the dimensionally correct Cell3. Each label is the integer address (i,j,k); color encodes the sheet grade $\lambda = i+j+k$. The plot includes all vertices, edge centers, face centers, and the central node.

4.2 Eight half-step child volumes

The three coordinate midplanes divide the single Cell3 into $2 \times 2 \times 2 = 8$ congruent half-step children. The 27 registry nodes are the complete vertex set of this refined eight-child decomposition. Thus the registry simultaneously records one parent volume, its boundary strata, and its centered subdivision.

4.3 Complete node catalogue

Table 4.2. Complete 27-node Cell3 registry. Every text cell is wrapped and every coordinate is computed from $p=B L/2$.

ID	Address	x (nm)	y (nm)	z (nm)	lambda	chi	class
L(-1,-1,-1)	(-1, -1, -1)	-0.176351350136	0.118682429905	0.637311498325	-3	-1	vertex
L(-1,-1,0)	(-1, -1, 0)	-0.181416008950	-0.076211421278	0.292444294142	-2	1	edge center
L(-1,0,-1)	(-1, 0, -1)	-0.158685962215	0.222342355125	0.363610955631	-2	1	edge center
L(0,-1,-1)	(0, -1, -1)	-0.012600729109	0.091233925963	0.618567746878	-2	1	edge center
L(-1,-1,1)	(-1, -1, 1)	-0.186480667763	-0.271105272461	-0.052422910042	-1	-1	vertex
L(-1,0,0)	(-1, 0, 0)	-0.163750621028	0.027448503942	0.018743751447	-1	-1	face center
L(-1,1,-1)	(-1, 1, -1)	-0.141020574293	0.326002280345	0.089910412937	-1	-1	vertex
L(0,-1,0)	(0, -1, 0)	-0.017665387922	-0.103659925220	0.273700542694	-1	-1	face center
L(0,0,-1)	(0, 0, -1)	0.005064658813	0.194893851183	0.344867204184	-1	-1	face center
L(1,-1,-1)	(1, -1, -1)	0.151149891919	0.063785422021	0.599823995431	-1	-1	vertex
L(-1,0,1)	(-1, 0, 1)	-0.168815279841	-0.167445347241	-0.326123452736	0	1	edge center
L(-1,1,0)	(-1, 1, 0)	-0.146085233106	0.131108429162	-0.254956791247	0	1	edge center
L(0,-1,1)	(0, -1, 1)	-0.022730046735	-0.298553776403	-0.071166661489	0	1	edge center
L(0,0,0)	(0, 0, 0)	0.000000000000	0.000000000000	0.000000000000	0	1	volume center
L(0,1,-1)	(0, 1, -1)	0.022730046735	0.298553776403	0.071166661489	0	1	edge center
L(1,-1,0)	(1, -1, 0)	0.146085233106	-0.131108429162	0.254956791247	0	1	edge center
L(1,0,-1)	(1, 0, -1)	0.168815279841	0.167445347241	0.326123452736	0	1	edge center
L(-1,1,1)	(-1, 1, 1)	-0.151149891919	-0.063785422021	-0.599823995431	1	-1	vertex
L(0,0,1)	(0, 0, 1)	-0.005064658813	-0.194893851183	-0.344867204184	1	-1	face center
L(0,1,0)	(0, 1, 0)	0.017665387922	0.103659925220	-0.273700542694	1	-1	face center
L(1,-1,1)	(1, -1, 1)	0.141020574293	-0.326002280345	-0.089910412937	1	-1	vertex
L(1,0,0)	(1, 0, 0)	0.163750621028	-0.027448503942	-0.018743751447	1	-1	face center
L(1,1,-1)	(1, 1, -1)	0.186480667763	0.271105272461	0.052422910042	1	-1	vertex
L(0,1,1)	(0, 1, 1)	0.012600729109	-0.091233925963	-0.618567746878	2	1	edge center
L(1,0,1)	(1, 0, 1)	0.158685962215	-0.222342355125	-0.363610955631	2	1	edge center
L(1,1,0)	(1, 1, 0)	0.181416008950	0.076211421278	-0.292444294142	2	1	edge center
L(1,1,1)	(1, 1, 1)	0.176351350136	-0.118682429905	-0.637311498325	3	-1	vertex

5. Seven-Sheet Polarity Complex

5.1 Grade, polarity, and exact populations

For $L=(i,j,k)$, define the laminar grade $\lambda(L)=i+j+k$ and the polarity $\chi(L)=(-1)^{i+j+k}$. Since each coordinate is -1, 0, or 1, λ takes the seven values -3 through 3. The coefficient of z^λ in the generating function gives the sheet population.

$$(z^{-1} + 1 + z)^3 = z^{-3} + 3 z^{-2} + 6 z^{-1} + 7 + 6 z + 3 z^2 + z^3.$$

Table 5.1. The complete seven-sheet partition of one Cell3.

lambda	population	chi	plane offset $N \cdot p$ (nm)	nodes
-3	1	-1	-0.453818064112	$L(-1,-1,-1)$
-2	3	1	-0.302545376075	$L(-1,-1,0), L(-1,0,-1), L(0,-1,-1)$
-1	6	-1	-0.151272688037	$L(-1,-1,1), L(-1,0,0), L(-1,1,-1), L(0,-1,0), L(0,0,-1), L(1,-1,-1)$
0	7	1	0.000000000000	$L(-1,0,1), L(-1,1,0), L(0,-1,1), L(0,0,0), L(0,1,-1), L(1,-1,0), L(1,0,-1)$
1	6	-1	0.151272688037	$L(-1,1,1), L(0,0,1), L(0,1,0), L(1,-1,1), L(1,0,0), L(1,1,-1)$
2	3	1	0.302545376075	$L(0,1,1), L(1,0,1), L(1,1,0)$
3	1	-1	0.453818064112	$L(1,1,1)$

5.2 Physical sheet normal and spacing

The address-plane equation $1^T L = \lambda$ becomes a physical-space plane after applying $p = B L/2$. The unit normal and adjacent-sheet spacing are

$$N = B^{-T} (1, 1, 1)^T / \|B^{-T} (1, 1, 1)^T\|; \quad h = 1 / (2 \|B^{-T} (1, 1, 1)^T\|).$$

$$N = (0.876976866375, 0.047223124232, -0.478206600101); \quad h = 0.151272688037 \text{ nm}.$$

Every Lucas node satisfies the exact identity

$$N \cdot p(L) = h \lambda(L).$$

Seven-sheet polarity complex; adjacent sheet spacing $h=0.151272688037$ nm

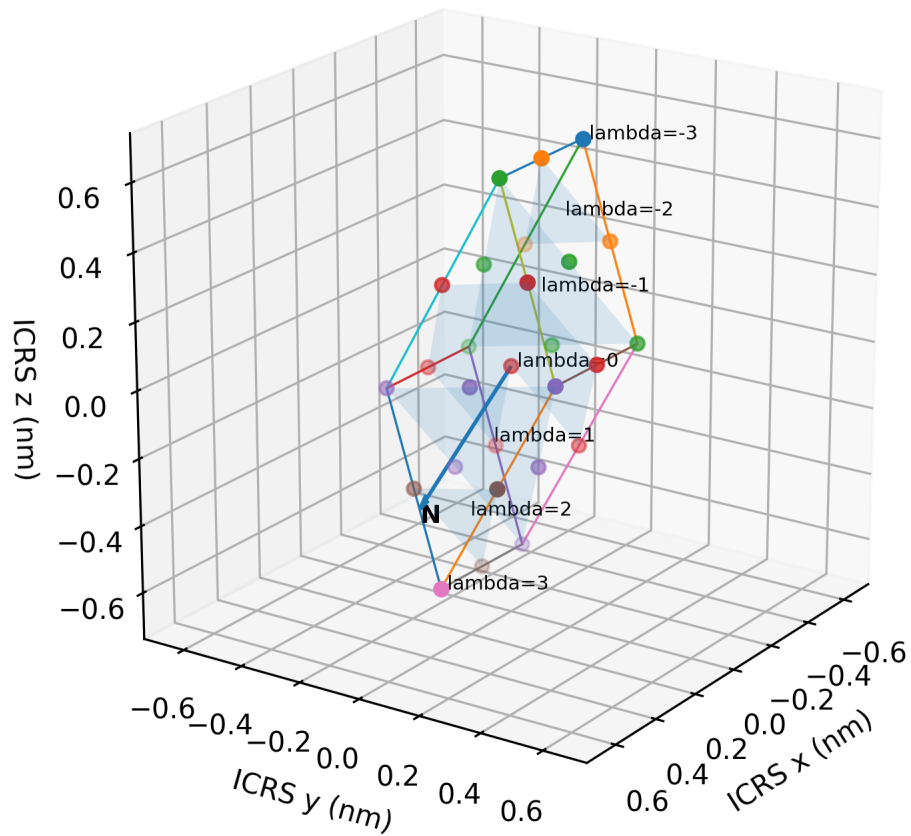


Figure 5.1. The complete seven-sheet stack inside one Cell3. Sheet polygons and nodes are computed from the exact condition $\lambda = i + j + k$. The vector N is the derived physical normal. Adjacent sheets are separated by $h=0.151272688037$ nm.

5.3 Orthographic anatomy of every sheet

The following seven figures view each plane along N. The horizontal and vertical axes are a common orthonormal basis internal to the sheet family, so distances are in nanometres and can be compared across grades. Every node is labeled. The central sheet also shows the three same-sheet root translations.

Sheet $\lambda=-3$: population=1, χ =negative (-1), offset=-0.453818064112 nm

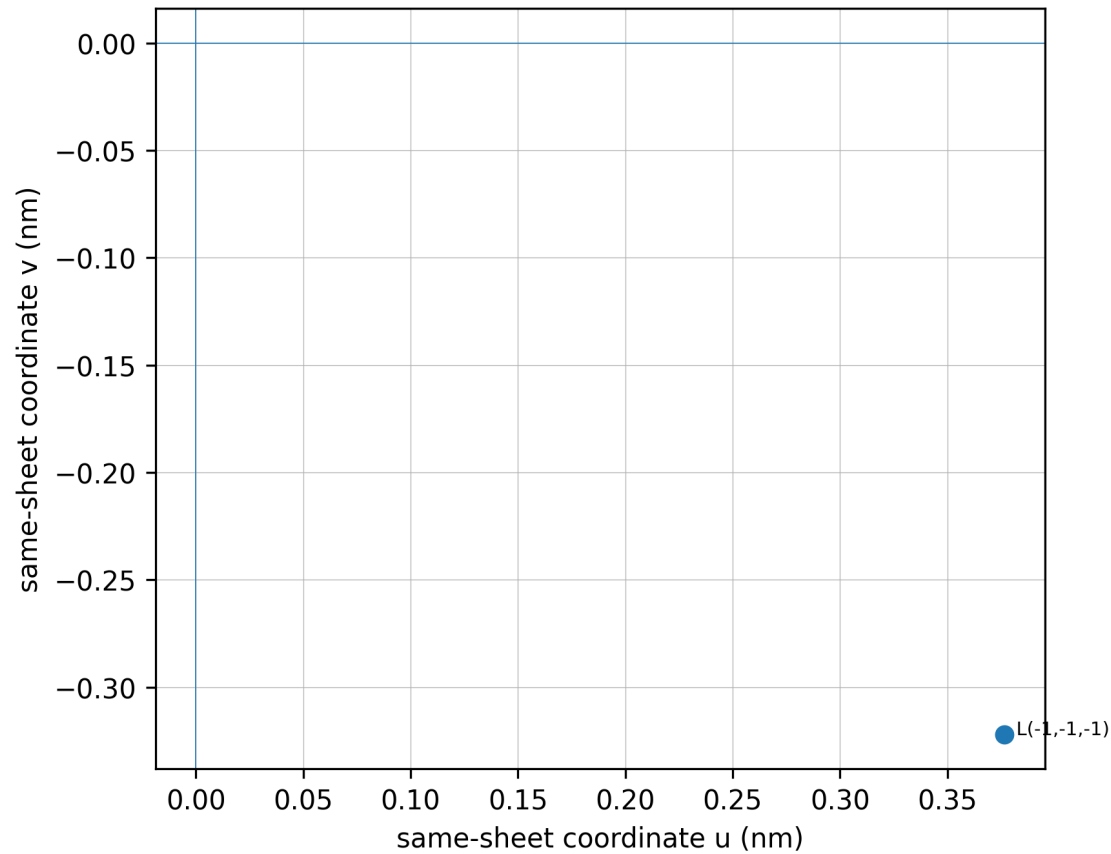


Figure 5.2. Orthographic section of sheet $\lambda=-3$. Population=1; polarity $\chi=-1$; physical offset=-0.453818064112 nm. All nodes on the sheet are shown and labeled.

Sheet $\lambda=-2$: population=3, χ =positive (+1), offset=-0.302545376075 nm

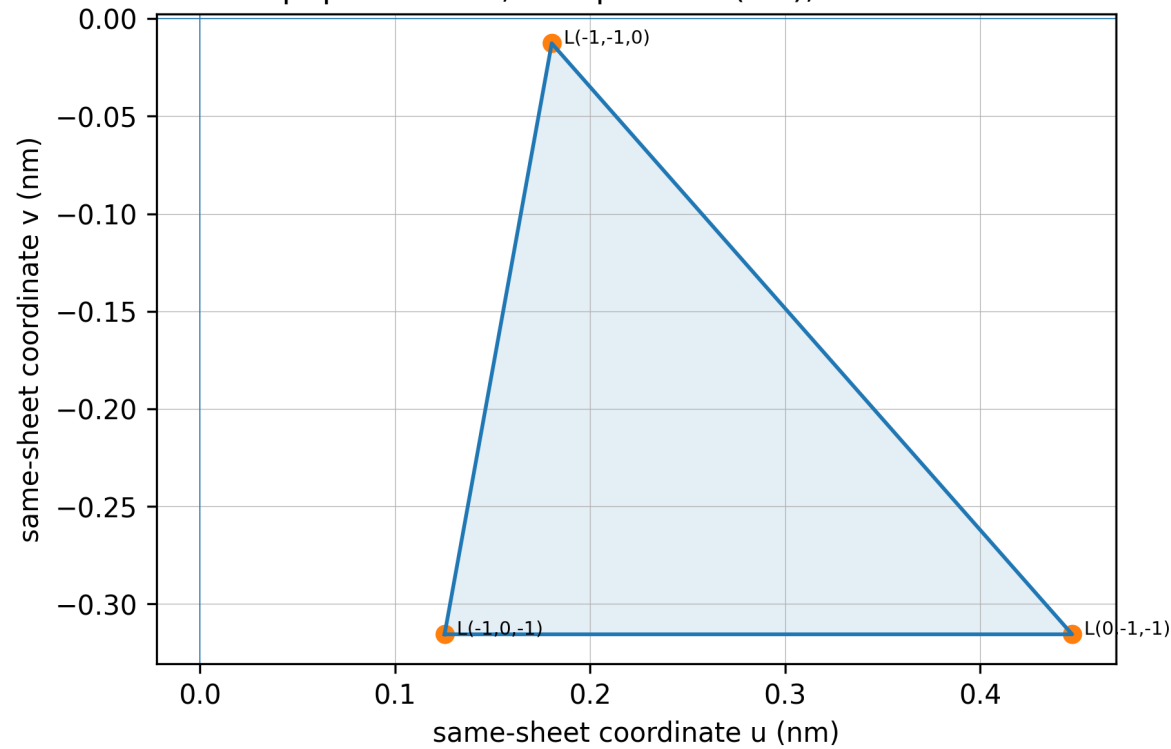


Figure 5.3. Orthographic section of sheet $\lambda=-2$. Population=3; polarity $\chi=1$; physical offset=-0.302545376075 nm. All nodes on the sheet are shown and labeled.

Sheet $\lambda=-1$: population=6, χ =negative (-1), offset=-0.151272688037 nm

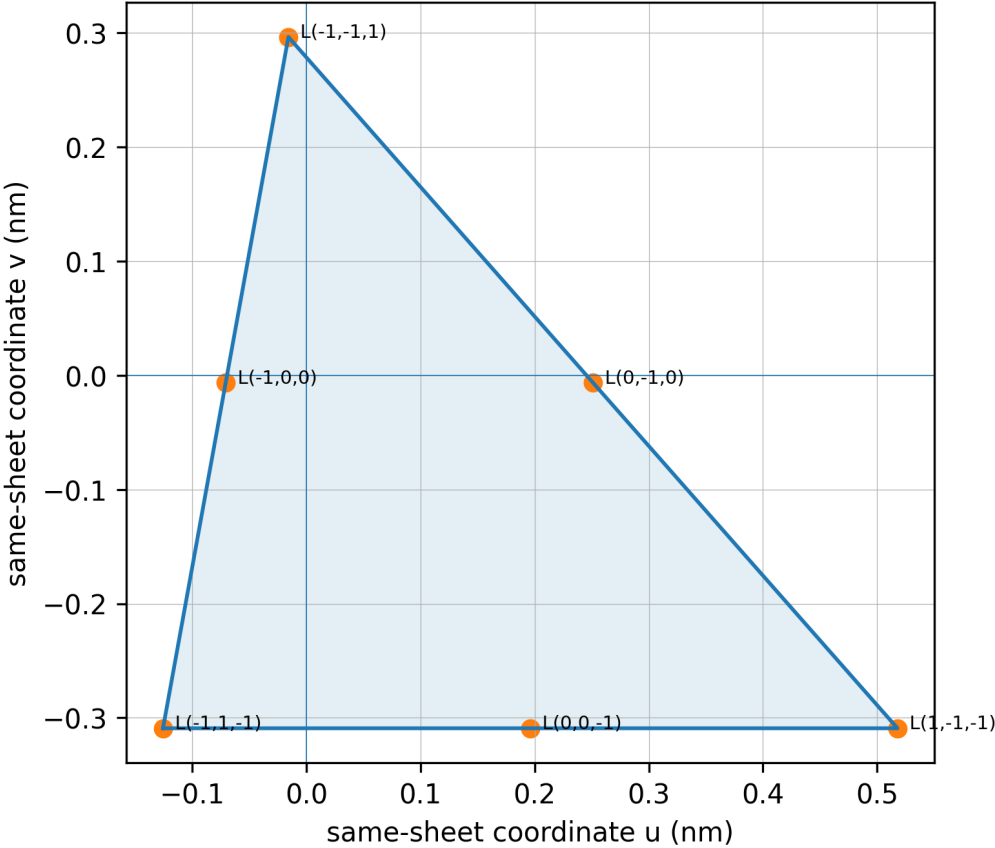


Figure 5.4. Orthographic section of sheet $\lambda=-1$. Population=6; polarity $\chi=-1$; physical offset=-0.151272688037 nm. All nodes on the sheet are shown and labeled.

Sheet $\lambda=0$: population=7, χ =positive (+1), offset=0.000000000000 nm

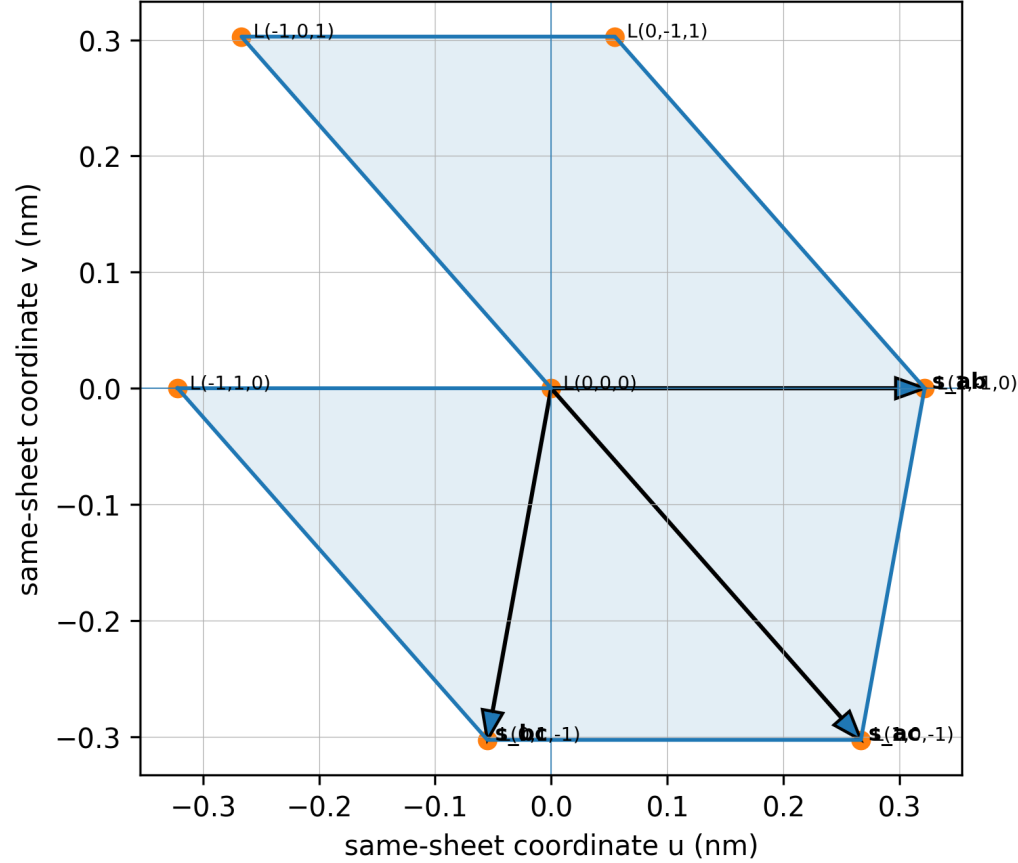


Figure 5.5. Orthographic section of sheet $\lambda=0$. Population=7; polarity $\chi=1$; physical offset=0.000000000000 nm. All nodes on the sheet are shown and labeled.

Sheet $\lambda=1$: population=6, χ =negative (-1), offset=0.151272688037 nm

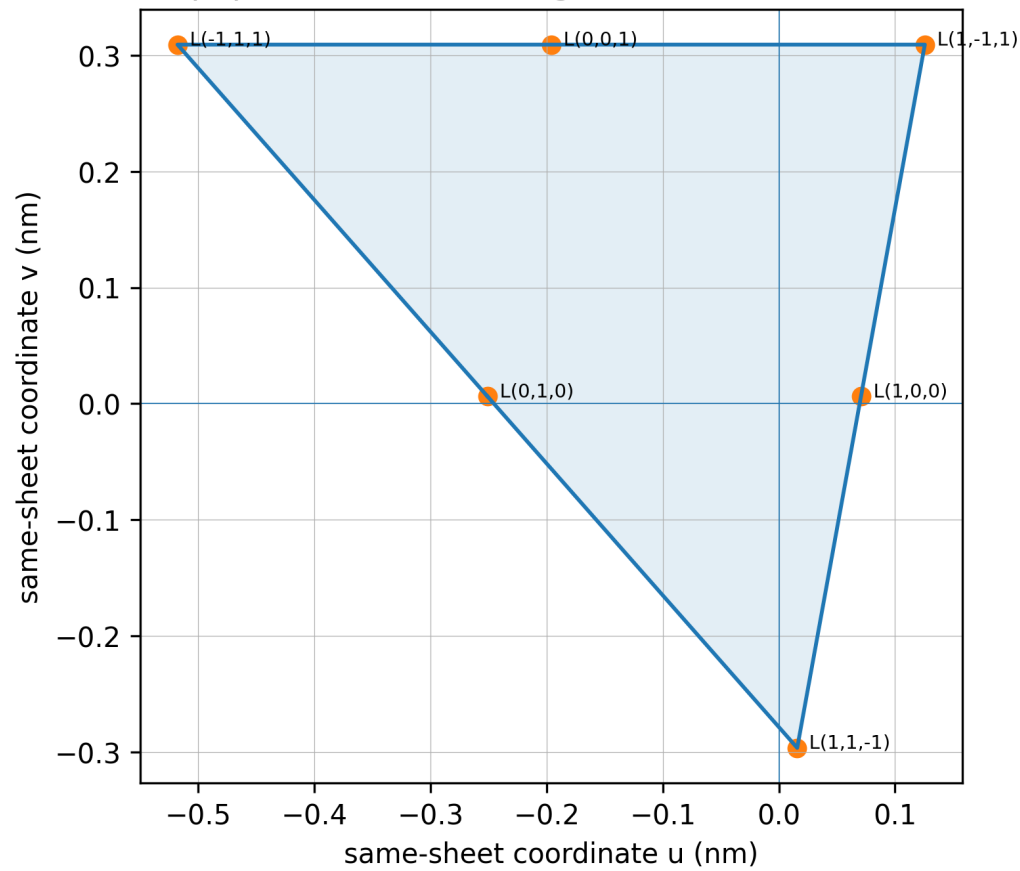


Figure 5.6. Orthographic section of sheet $\lambda=1$. Population=6; polarity $\chi=-1$; physical offset=0.151272688037 nm. All nodes on the sheet are shown and labeled.

Sheet $\lambda=2$: population=3, χ =positive (+1), offset=0.302545376075 nm

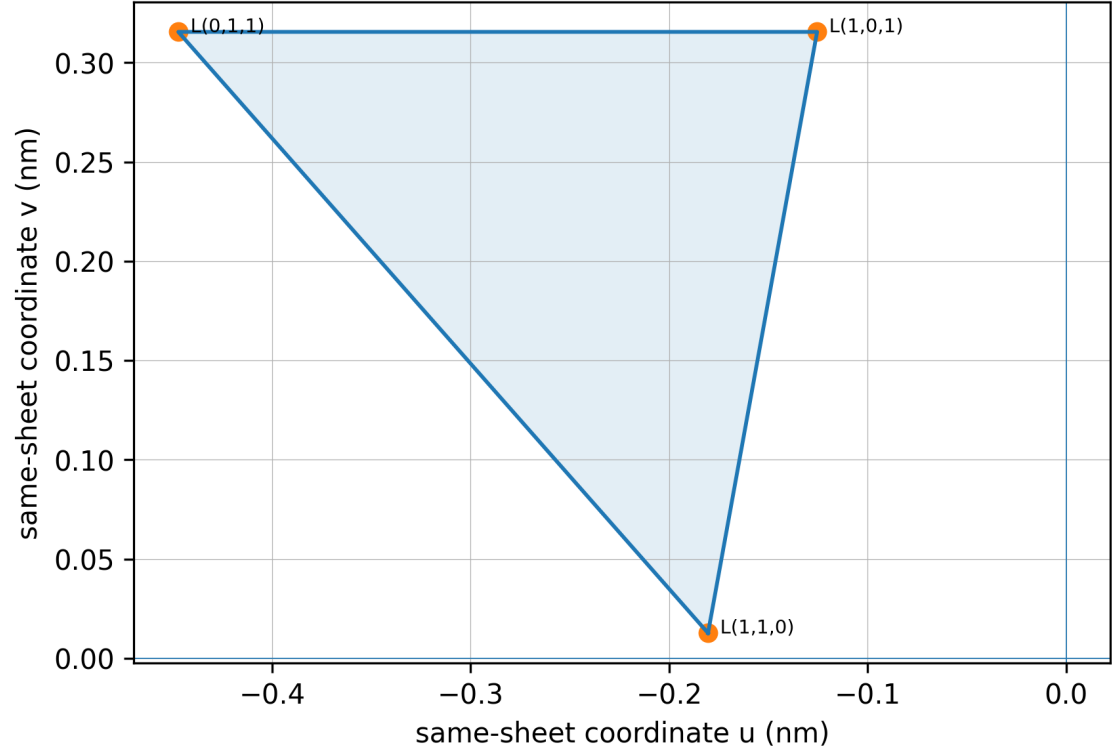


Figure 5.7. Orthographic section of sheet $\lambda=2$. Population=3; polarity $\chi=1$; physical offset=0.302545376075 nm. All nodes on the sheet are shown and labeled.

Sheet lambda=3: population=1, chi=negative (-1), offset=0.453818064112 nm

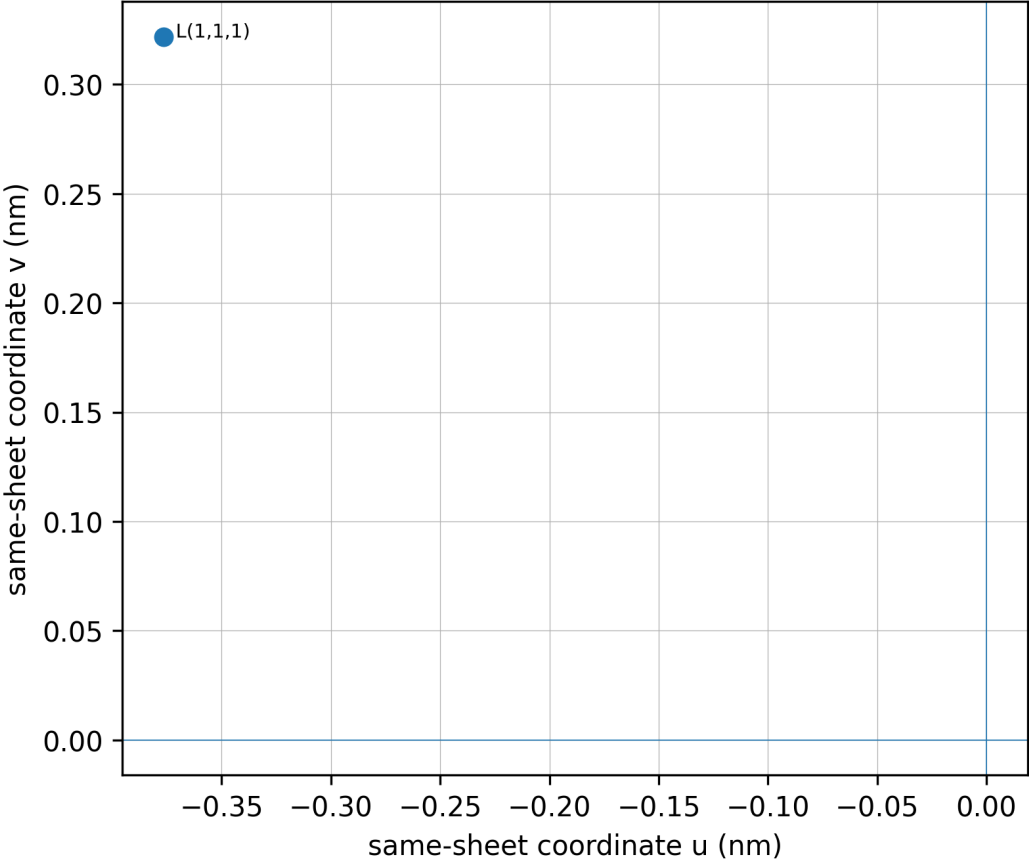


Figure 5.8. Orthographic section of sheet lambda=3. Population=1; polarity chi=-1; physical offset=0.453818064112 nm. All nodes on the sheet are shown and labeled.

5.4 A2 same-sheet lattice

The sum map lambda:Z^3 to Z has kernel A2={ (i,j,k) in Z^3 : i+j+k=0 }. This gives the exact sequence

0 -> A2 -> Z^3 --lambda--> Z -> 0.

The primitive half-step translations a/2, b/2, and c/2 change lambda by one and reverse chi. In compact form, t_a=a/2, t_b=b/2, and t_c=c/2. The difference vectors (a-b)/2, (a-c)/2, and (b-c)/2 lie in the kernel: they preserve both sheet and polarity. They are the embedded A2 root directions.

Table 5.2. Same-sheet A2 root translations.

Root	Cartesian x (nm)	Cartesian y (nm)	Cartesian z (nm)	N dot root
s_ab	0.146085233106	-0.131108429162	0.254956791247	6.705e-18
s_ac	0.168815279841	0.167445347241	0.326123452736	1.203e-17
s_bc	0.022730046735	0.298553776403	0.071166661489	5.320e-18

The finite quotient A2*/A2 is Z3. Combined with the polarity Z2, it produces a Z6 closure registry. This Z6 is not an extra spatial dimension; it is a finite algebraic registry carried by the direct lattice and its alternating sheet sign.

5.5 Thirteen antipodal projective channels

Removing the central address leaves 26 states. Identifying each state with its antipode produces 13 projective channels, the points of PG(2,3). A canonical representative is chosen by requiring the first nonzero component to be positive.

Table 5.3. Complete projective channel catalogue.

Channel	Representative address	Antipode	lambda	chi
1	(0, 0, 1)	(0, 0, -1)	1	-1
2	(0, 1, -1)	(0, -1, 1)	0	1
3	(0, 1, 0)	(0, -1, 0)	1	-1
4	(0, 1, 1)	(0, -1, -1)	2	1
5	(1, -1, -1)	(-1, 1, 1)	-1	-1
6	(1, -1, 0)	(-1, 1, 0)	0	1
7	(1, -1, 1)	(-1, 1, -1)	1	-1
8	(1, 0, -1)	(-1, 0, 1)	0	1
9	(1, 0, 0)	(-1, 0, 0)	1	-1
10	(1, 0, 1)	(-1, 0, -1)	2	1
11	(1, 1, -1)	(-1, -1, 1)	1	-1
12	(1, 1, 0)	(-1, -1, 0)	2	1
13	(1, 1, 1)	(-1, -1, -1)	3	-1

6. Local Incidence, Emily Spectrum, and Named Algebraic Layers

6.1 Nearest-neighbor graph

Nearest-neighbor movement changes one address coordinate by one step while holding the other two fixed. Along one axis the adjacency matrix is

J	col 1	col 2	col 3
row 1	0	1	0
row 2	1	0	1
row 3	0	1	0

The full Cell3 adjacency is the three-fold Kronecker sum

$$A_{Cell3} = J \text{ tensor } I \text{ tensor } I + I \text{ tensor } J \text{ tensor } I + I \text{ tensor } I \text{ tensor } J.$$

This is the Cartesian graph $P_3 \times P_3 \times P_3$. The one-axis eigenvalues are $-\sqrt{2}$, 0, and $\sqrt{2}$, so every three-axis eigenvalue is a sum of three one-axis values.

6.2 Exact Emily spectrum

Table 6.1. Exact adjacency spectrum.

Eigenvalue	Numerical value	Multiplicity
-3 $\sqrt{2}$	-4.242640687119	1
-2 $\sqrt{2}$	-2.828427124746	3
$-\sqrt{2}$	-1.414213562373	6
0	0.000000000000	7
$\sqrt{2}$	1.414213562373	6
2 $\sqrt{2}$	2.828427124746	3
3 $\sqrt{2}$	4.242640687119	1

The multiplicities 1,3,6,7,6,3,1 exactly match the seven sheet populations. Both sequences are coefficient counts of a triple three-state sum. The equality is structural and combinatorial. It does not identify an unweighted adjacency eigenvalue with a physical energy.

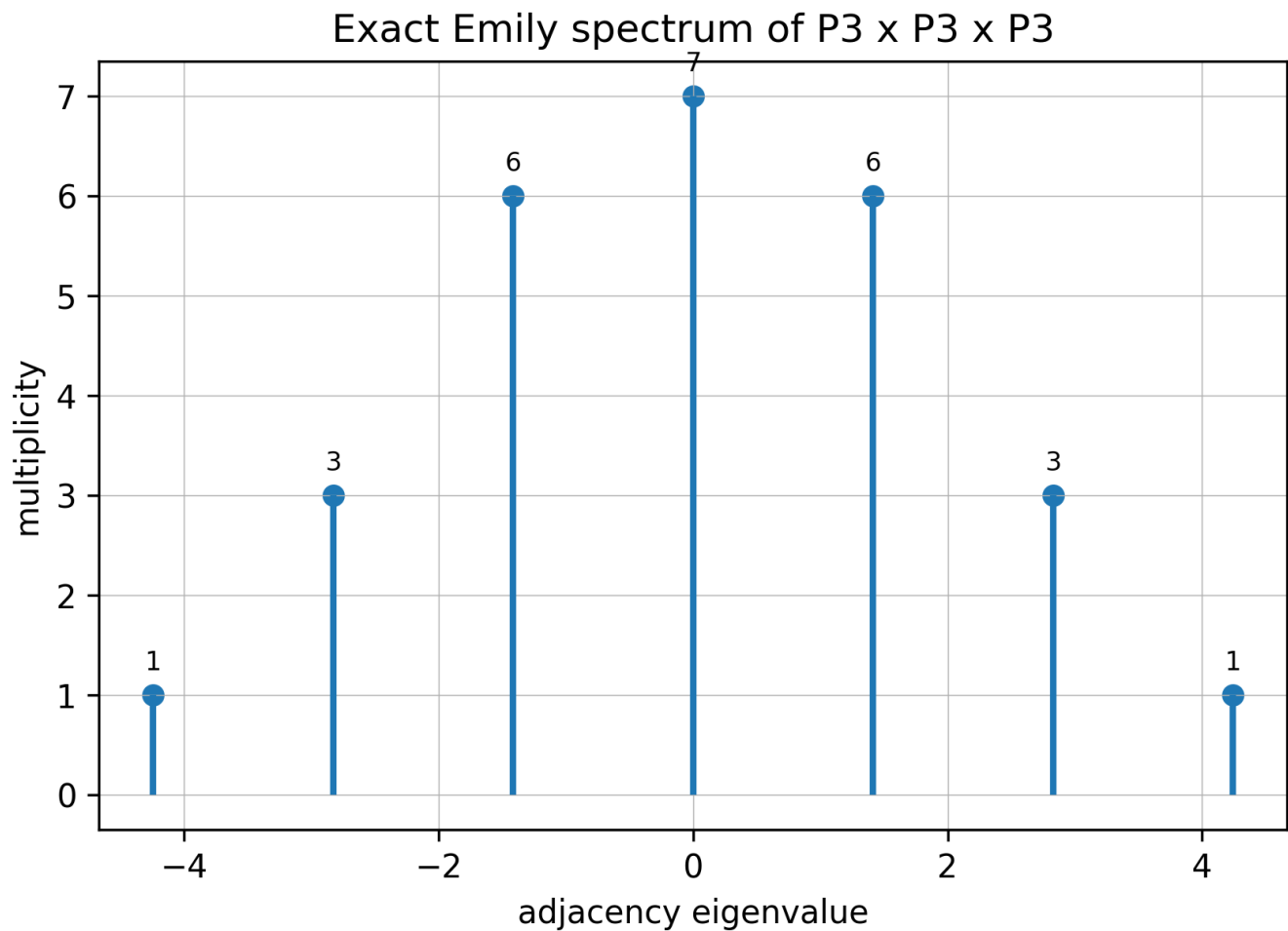


Figure 6.1. The exact Emily adjacency spectrum. The seven spectral levels have multiplicities 1,3,6,7,6,3,1, identical to the seven sheet populations. The central adjacency eigenvalue has multiplicity seven.

6.3 Adjacency, graph Laplacian, and Hodge operators are distinct

The adjacency operator has nullity seven. The unweighted graph Laplacian of the connected 27-node graph has one constant zero mode. A full cubical Hodge operator includes oriented edges, plaquettes, and volumes, and its harmonic multiplicities are determined by cohomology. These are three different operators. Their zero-mode counts must not be transferred from one operator to another.

6.4 Devon, Lucas, Emily, and Jack

Table 6.2. Named algebraic layers.

Layer	Exact role	Not an additional object
Devon	Algebraic and cofactor-dual reduction carried by B, M, R, adj(M), face areas, volume, and A2	Not a new spatial dimension
Lucas	Finite address registry and complete factor-three aggregation	Not a replacement for B
Emily	Exact Cartesian-product adjacency spectrum	Not the graph Laplacian or physical energy by itself
Jack	Perturbation quotient acting on the constructed geometry	Not required to create Cell3, the Lucas Group, or the Supergroup

Devon = (B, M, R, adj(M), A_ab, A_ac, A_bc, V, A2).

The archived Jack perturbation matrix is

J_Jack	col 1	col 2	col 3
row 1	-1	0	1

J_Jack	col 1	col 2	col 3
row 2	1	-1	0
row 3	17	-2	-3

det(J_Jack)=12; Smith normal form = diag(1,1,12); quotient = Z12.

7. Complete Factor-Three Hierarchy

7.1 Complete-cell aggregation

Let $n=3^g$ be the number of complete Cell3 volumes along each axis. The outer basis is $B_n=nB$. Only complete Cell3 volumes are aggregated; half-step registry positions are not counted as complete volumes.

$$n = 3^g; B_n = n B.$$

The exact cubical f-polynomial is

$$F_n(t) = ((n+1)+n t)^3 = (n+1)^3 + 3n(n+1)^2 t + 3n^2(n+1) t^2 + n^3 t^3.$$

Therefore $C_0=(n+1)^3$, $C_1=3n(n+1)^2$, $C_2=3n^2(n+1)$, and $C_3=n^3$.

7.2 Half-step complex and exact sheet populations

The complete half-step address cube is $U_n=\{-n,-n+1,\dots,n\}^3$. It contains $(2n+1)^3$ states and $6n+1$ sheets. The exact population $P_n(\lambda)$ is piecewise quadratic:

$$P_n(\lambda) = 3n^2+3n+1-\lambda^2 \text{ for } |\lambda| \leq n; P_n(\lambda)=C(3n+2-|\lambda|, 2) \text{ for } n < |\lambda| \leq 3n; 0 \text{ otherwise.}$$

The central population is $P_n(0)=3n^2+3n+1$. Symmetry gives $P_n(\lambda)=P_n(-\lambda)$.

7.3 Interfaces, plaquettes, cubes, and cycle rank

$$I_n = 3(n-1)n^2; E_n = 6n^2; Q_n = 3n(n-1)^2; K_n = (n-1)^3.$$

$$\text{beta1_center} = I_n - n^3 + 1 = 2n^3 - 3n^2 + 1.$$

I_n counts internal complete-cell interfaces. E_n counts exposed primitive faces. Q_n and K_n count elementary plaquettes and cubes in the graph of cell centers. The center-graph cycle rank is a graph-theoretic quantity; Chapter 9 shows why it is not a global Hodge Betti number.

7.4 Numerical hierarchy

Table 7.1. Exact hierarchy through $g=5$.

Level	g	n	Cell3 volumes	C0	C1	C2	C3	Half-step states	Sheets	Interfaces	Center cycles
Cell3	0	1	1	8	12	6	1	27	7	0	0
Lucas Group G1	1	3	27	64	144	108	27	343	19	54	28
Lucas Supergroup G2	2	9	729	1,000	2,700	2,430	729	6,859	55	1,944	1,216
G3 bulk	3	27	19,683	21,952	63,504	61,236	19,683	166,375	163	56,862	37,180
G4 bulk	4	81	531,441	551,368	1,633,932	1,614,006	531,441	4,330,747	487	1,574,640	1,043,200
G5 bulk	5	243	14,348,907	14,526,784	43,401,744	43,223,868	14,348,907	115,501,303	1,459	42,869,574	28,520,668

Lucas Group: 3 x 3 x 3 complete Cell3 volumes; outer basis 3B; volume=3.183702082 nm³

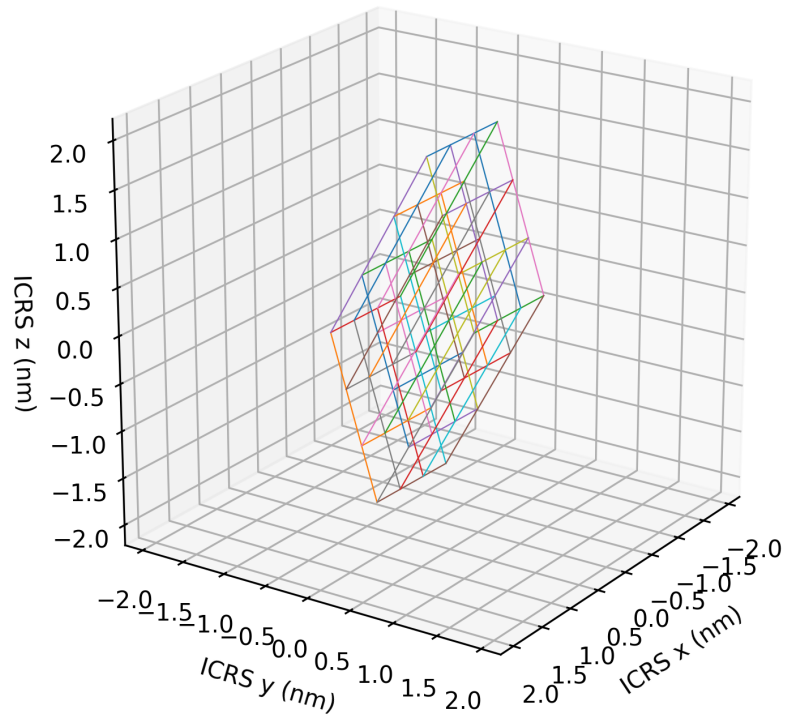


Figure 7.1. Dimensionally correct Lucas Group. The outer basis is 3B and the block contains 27 complete Cell3 volumes. Grid lines are the mapped complete-cell boundaries in the oblique basis.

Lucas Supergroup: 9 x 9 x 9 complete Cell3 volumes; 27 Lucas Groups; outer basis 9B

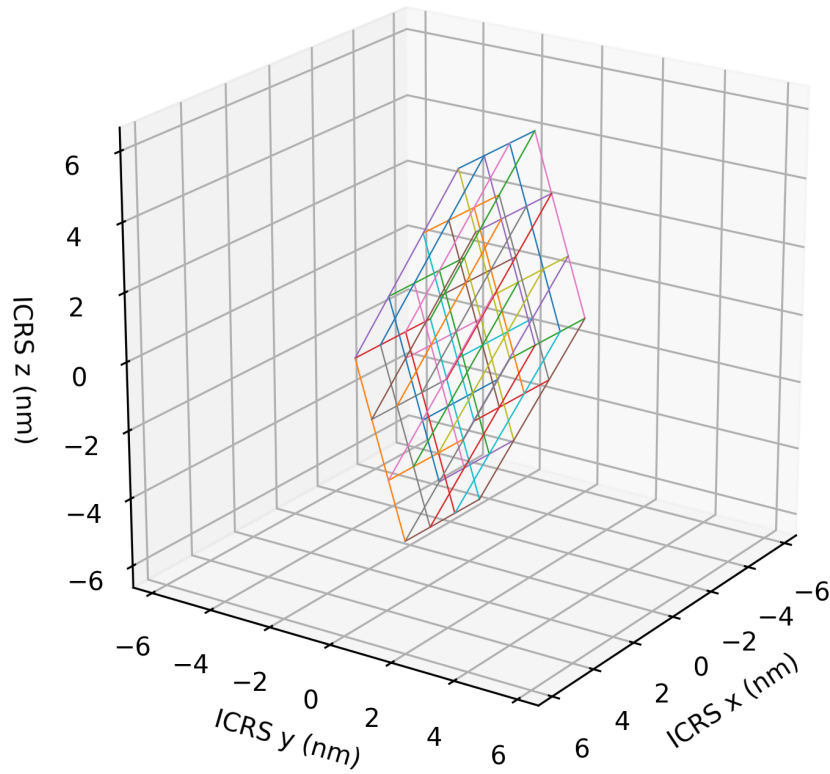


Figure 7.2. Dimensionally correct Lucas Supergroup. The outer basis is 9B. The plotted subdivision marks the 3B Lucas-Group boundaries, producing 27 complete Lucas Groups inside the 9B closure domain.

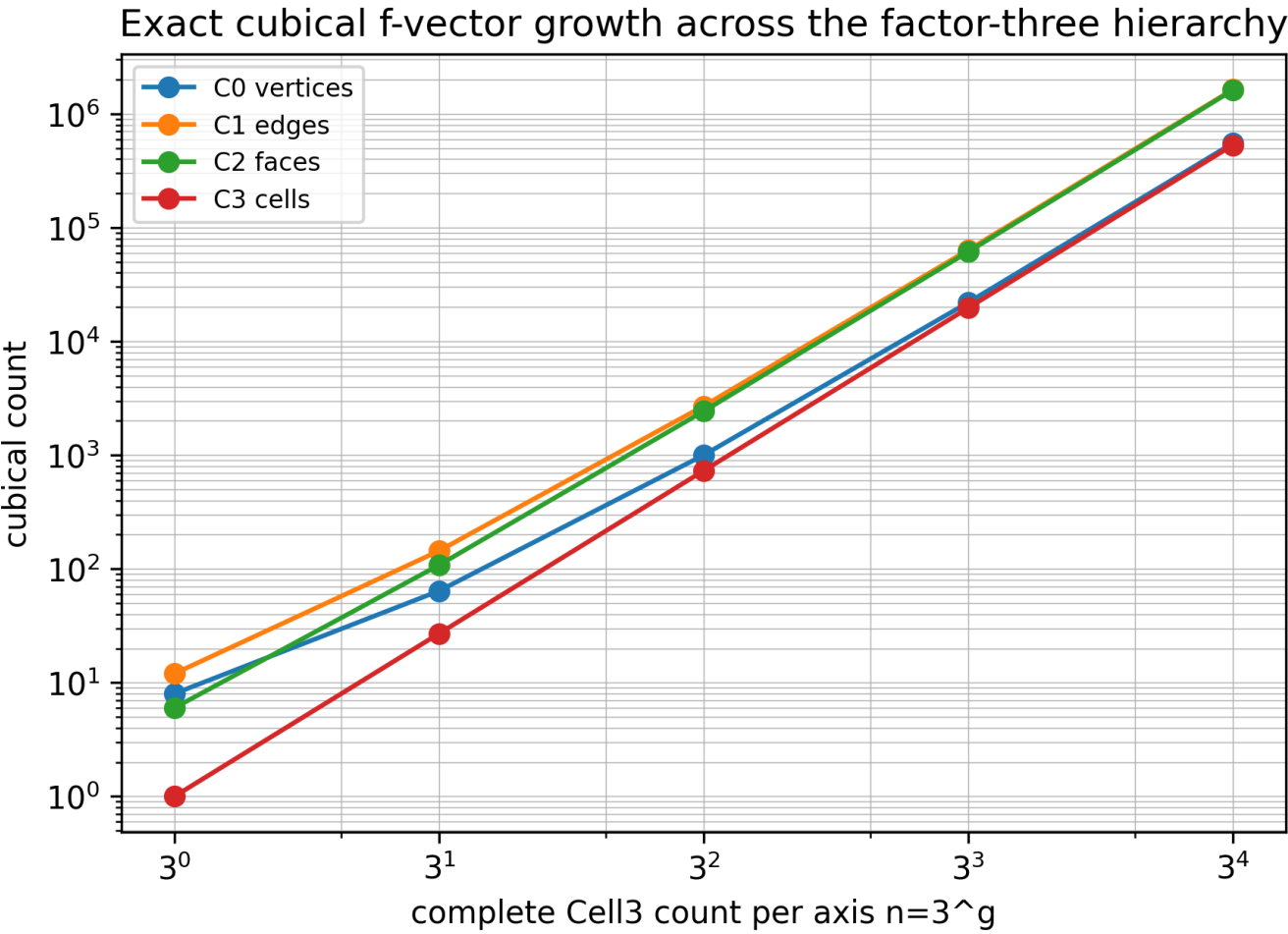


Figure 7.3. Exact growth of the cubical counts C0 through C3 across $n=1,3,9,27,81$. Both axes are logarithmic; every point is computed from the closed formulas above.

7.5 Terminology resolution

Term	Exact meaning
Cell3 Lucas window	27 registry states in one Cell3
Lucas Group	$3^3=27$ complete Cell3 volumes
Lucas Supergroup	3^3 Lucas Groups = $9^3=729$ complete Cell3 volumes

This distinction removes the principal counting ambiguity: 27 positions within one Cell3 and 27 complete Cell3 volumes are different objects, even though both arise from a three-by-three-by-three product.

8. The Lucas Supergroup as the Minimal Complete Closure Domain

8.1 Exact census at n=9

$$F_9(t) = (10+9t)^3 = 1000 + 2700 t + 2430 t^2 + 729 t^3.$$

Table 8.1. Exact Lucas-Supergroup census.

Quantity	Value	Interpretation
C0	1,000	unrefined corner nodes
C1	2,700	oriented one-cells before orientation sign
C2	2,430	oriented two-cells before orientation sign
C3	729	complete Cell3 volumes
Half-step states	6,859	19 ³ refined address positions
Sheets	55	lambda=-27 through 27
Central sheet	271	P ₉ (0)
Noncentral antipodal pairs	3,429	(6,859-1)/2
Internal interfaces	1,944	center-graph edges
Elementary plaquettes	1,728	center-graph squares
Elementary cubes	512	center-graph cubes
Center cycle rank	1,216	1,944-729+1
Exposed primitive faces	486	6 x 9 ²

8.2 Interface decomposition and the 3:1:1 law

The Supergroup consists of 27 Lucas Groups. Each Lucas Group contributes 54 intragroup interfaces, so 27 x 54=1,458 center interfaces are internal to groups. The remaining 486 are intergroup interfaces. The exterior also has 486 primitive faces. Therefore

$$\text{intragroup} : \text{intergroup} : \text{external} = 1458 : 486 : 486 = 3 : 1 : 1.$$

8.3 Outer shell identities

The Chebyshev outer shell of the 19³ half-step cube is the difference between radius nine and radius eight cubes:

$$19^3 - 17^3 = 1946 = 1944 + 2.$$

The shell contains 973 antipodal pairs. Of these, 109 are inherited from the radius-three shell and 864 are new primitive projective directions. The same count is 12 J₂(9)=12 x 72=864, where J₂ is the second Jordan totient.

8.4 Mass-Bridge grid closure

The 9B block is the first factor-three level whose unrefined vertex grid has ten nodes per edge, one hundred nodes per coordinate face, and one thousand corner nodes. This is the direct geometric realization of the Mass-Bridge registry factors 10, 100, and 1000.

$$10 \text{ nodes/edge}; 100 \text{ nodes/coordinate face}; 1000 \text{ corner nodes}.$$

It is also the first level that simultaneously contains complete local Lucas blocks, cross-block gluing interfaces, a saturated interior, a finite exterior, local cycles, gluing cycles, and a new primitive reciprocal shell. These properties make G2, rather than G1, the minimal Mass-Bridge-complete finite closure domain.

8.5 Complete 55-sheet distribution

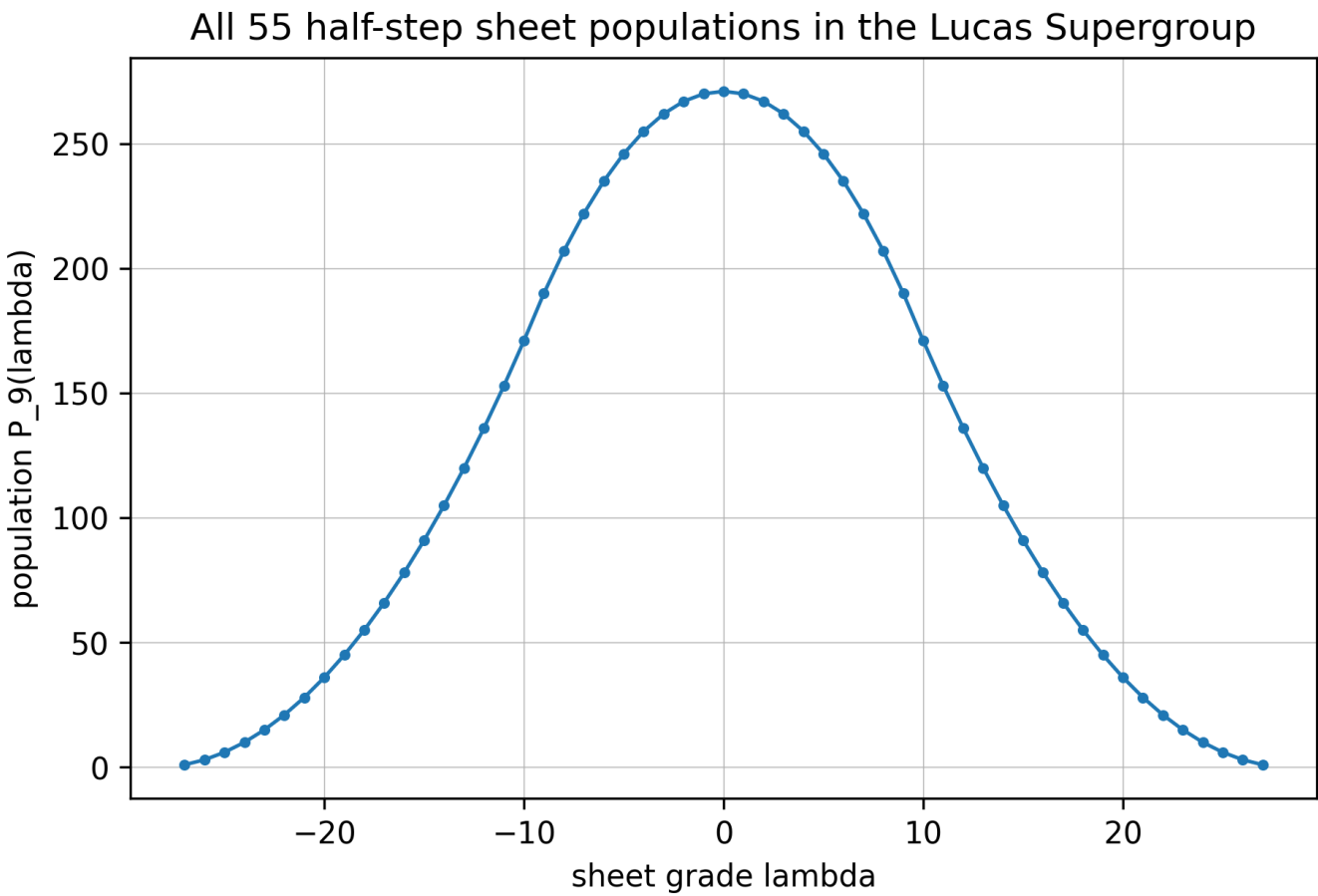


Figure 8.1. All 55 sheet populations $P_9(\lambda)$ for $\lambda=-27,\dots,27$. The central value is 271 and the total area under the discrete distribution is 6,859 states.

Table 8.2. Complete Lucas-Supergroup sheet catalogue.

lambda	population	chi	offset from central sheet (nm)
-27	1	-1	-4.084362577006
-26	3	1	-3.933089888969
-25	6	-1	-3.781817200931
-24	10	1	-3.630544512894
-23	15	-1	-3.479271824857
-22	21	1	-3.327999136820
-21	28	-1	-3.176726448782
-20	36	1	-3.025453760745
-19	45	-1	-2.874181072708
-18	55	1	-2.722908384671
-17	66	-1	-2.571635696633
-16	78	1	-2.420363008596
-15	91	-1	-2.269090320559
-14	105	1	-2.117817632522
-13	120	-1	-1.966544944484
-12	136	1	-1.815272256447
-11	153	-1	-1.663999568410
-10	171	1	-1.512726880373
-9	190	-1	-1.361454192335

lambda	population	chi	offset from central sheet (nm)
-8	207	1	-1.210181504298
-7	222	-1	-1.058908816261
-6	235	1	-0.907636128224
-5	246	-1	-0.756363440186
-4	255	1	-0.605090752149
-3	262	-1	-0.453818064112
-2	267	1	-0.302545376075
-1	270	-1	-0.151272688037
0	271	1	0.000000000000
1	270	-1	0.151272688037
2	267	1	0.302545376075
3	262	-1	0.453818064112
4	255	1	0.605090752149
5	246	-1	0.756363440186
6	235	1	0.907636128224
7	222	-1	1.058908816261
8	207	1	1.210181504298
9	190	-1	1.361454192335
10	171	1	1.512726880373
11	153	-1	1.663999568410
12	136	1	1.815272256447
13	120	-1	1.966544944484
14	105	1	2.117817632522
15	91	-1	2.269090320559
16	78	1	2.420363008596
17	66	-1	2.571635696633
18	55	1	2.722908384671
19	45	-1	2.874181072708
20	36	1	3.025453760745
21	28	-1	3.176726448782
22	21	1	3.327999136820
23	15	-1	3.479271824857
24	10	1	3.630544512894
25	6	-1	3.781817200931
26	3	1	3.933089888969
27	1	-1	4.084362577006

9. Full Cubical Topology and Hodge Closure

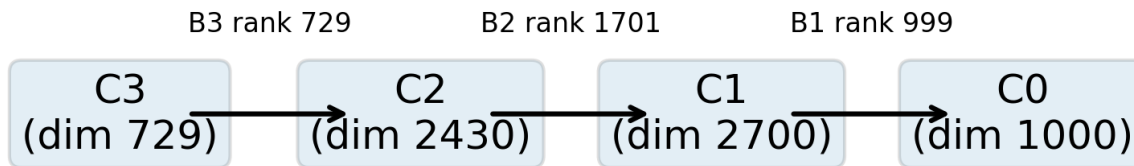
9.1 Oriented boundary maps

Let C_k be the real vector space generated by oriented k -cells of the $9 \times 9 \times 9$ cubical complex. The boundary maps are $B1:C1$ to $C0$, $B2:C2$ to $C1$, and $B3:C3$ to $C2$. Their matrix dimensions and exact ranks are

Table 9.1. Boundary-map dimensions and ranks.

Map	Matrix dimensions	Rank	Kernel dimension
B1	1000 x 2700	999	1701
B2	2700 x 2430	1701	729
B3	2430 x 729	729	0

Oriented Lucas-Supergroup cubical chain complex



Betti numbers: $(\beta_0, \beta_1, \beta_2, \beta_3) = (1, 0, 0, 0)$

Figure 9.1. The oriented chain complex from three-cells to vertices. The printed dimensions and ranks imply exactness at $C3$, $C2$, and $C1$, with one connected scalar component at $C0$.

9.2 Betti numbers and cohomology

$$(\beta_0, \beta_1, \beta_2, \beta_3) = (1, 0, 0, 0).$$

The finite Lucas Supergroup is contractible. Its cohomology is $H_0=\mathbb{R}$ and $H_1=H_2=H_3=0$. The only scalar harmonic is the constant mode.

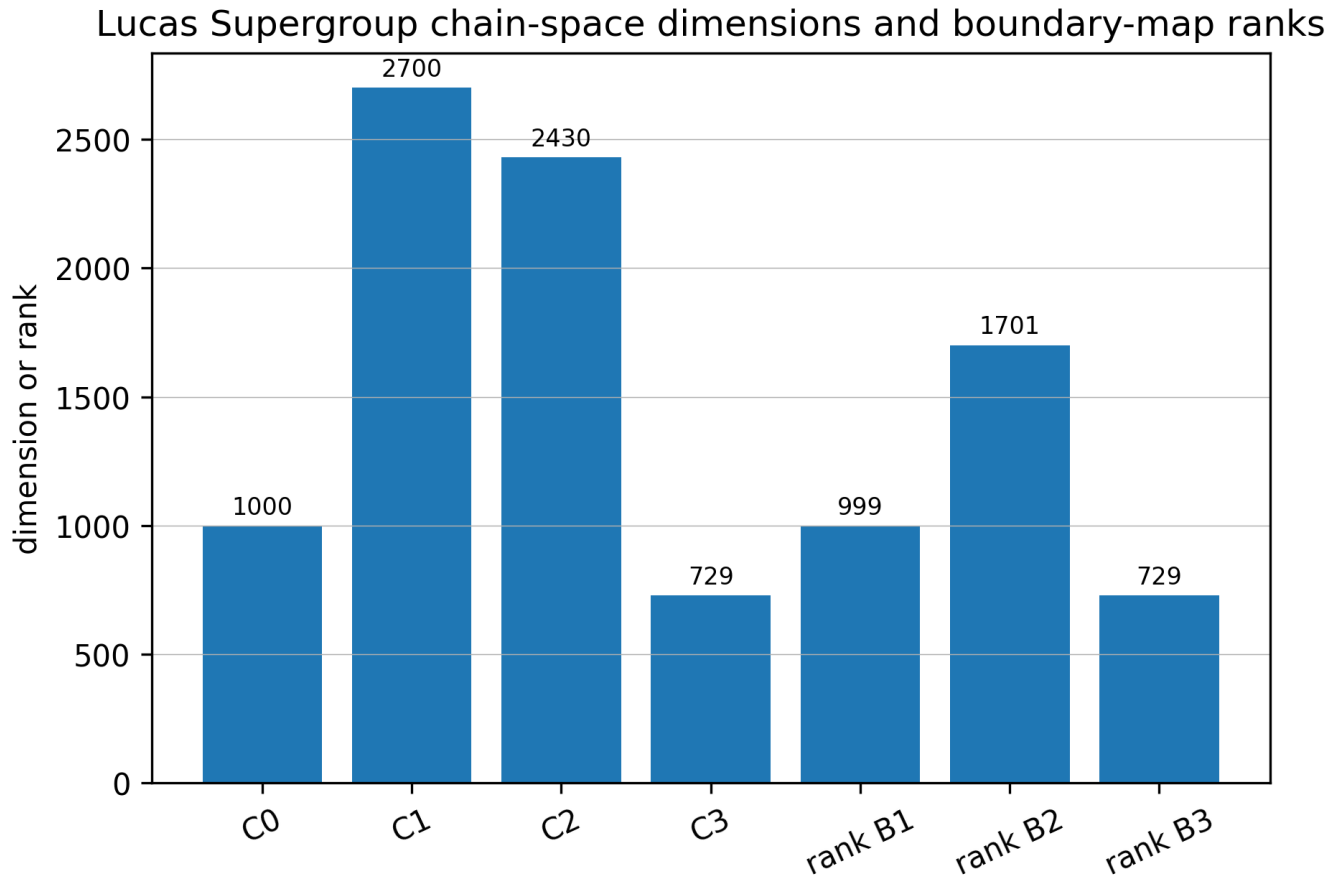


Figure 9.2. Chain-space dimensions and boundary-map ranks for the Lucas Supergroup. These data determine the Betti vector (1,0,0,0).

9.3 Why 1,216 center cycles are not protected harmonic cycles

The center graph has 729 vertices and 1,944 edges, so its graph cycle rank is $1,944 - 729 + 1 = 1,216$. These closed edge paths are genuine graph circulations. However, the complete cubical complex contains 1,728 elementary plaquettes. The oriented plaquette boundaries span the graph cycle space, giving $\ker(B1^*) = \text{im}(B2^*)$. The graph circulations are therefore coexact curl modes rather than nontrivial $H1$ classes.

9.4 Why 55 sheet modes reduce to one Hodge scalar mode

An unsigned operator that treats each sheet as disconnected can exhibit one constant mode per sheet. The oriented cubical complex contains cross-sheet incidence through primitive half-step translations. That coupling lifts 54 of the 55 sheet constants. The complete connected domain retains one global constant scalar harmonic.

10. Deterministic Reciprocal Spectrum

10.1 Reciprocal wavevectors of the 9B domain

Periodic modes of the Lucas Supergroup use the reciprocal basis of 9B. For integer reciprocal address *s*, the wavevector and squared magnitude are

$$k_s = (2 \pi / 9) B^{-T} s; \kappa_s^2 = (2 \pi / 9)^2 s^T R s.$$

The primitive radius-nine projective shell is the set of integer *s* with max(|*s*₁|,|*s*₂|,|*s*₃|)=9 and gcd(*s*₁,*s*₂,*s*₃)=1, with antipodal vectors identified. It contains exactly 864 directions.

10.2 Complete shell statistics

Table 10.1. Primitive radius-nine reciprocal statistics.

Statistic	Value
Projective directions	864
Distinct q values at 12 decimal places	864
Minimum q	130.593523477839 nm ⁻² at s=(1, 4, 9)
Maximum q	1744.361292059535 nm ⁻² at s=(9, 9, -8)
Mean q	757.916780244175 nm ⁻²
Population standard deviation	325.034198848754 nm ⁻²

Complete anisotropic reciprocal spectrum on the 864-direction primitive radius-nine shell

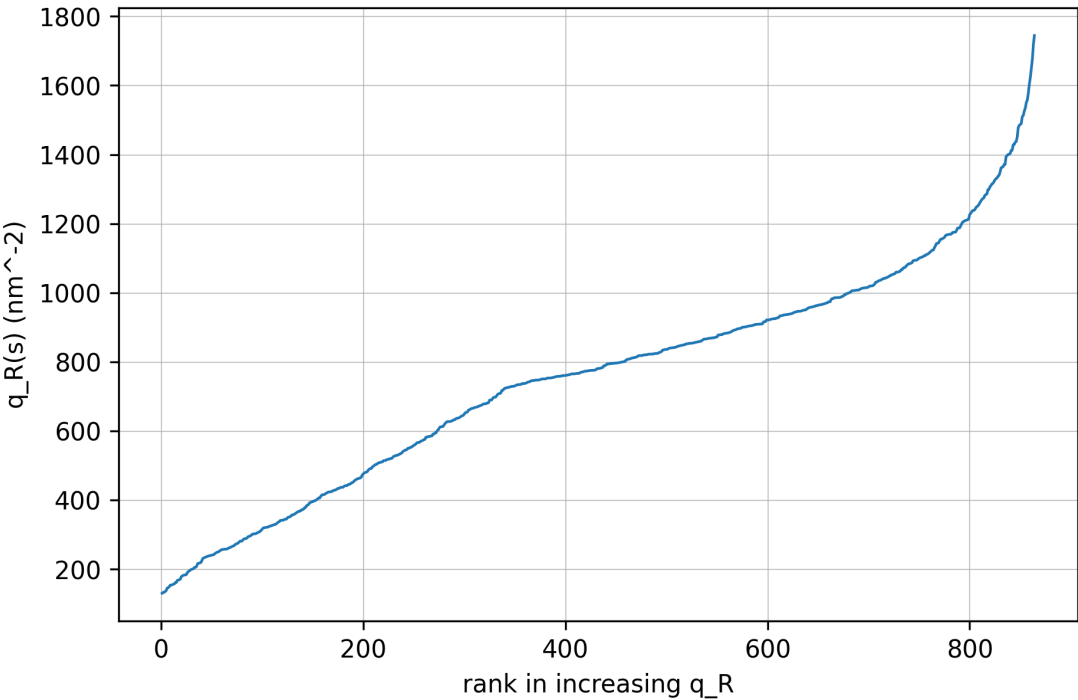


Figure 10.1. All 864 *q_R* values sorted in increasing order. The absence of repeated values at twelve decimal places records the anisotropic splitting produced by *R*.

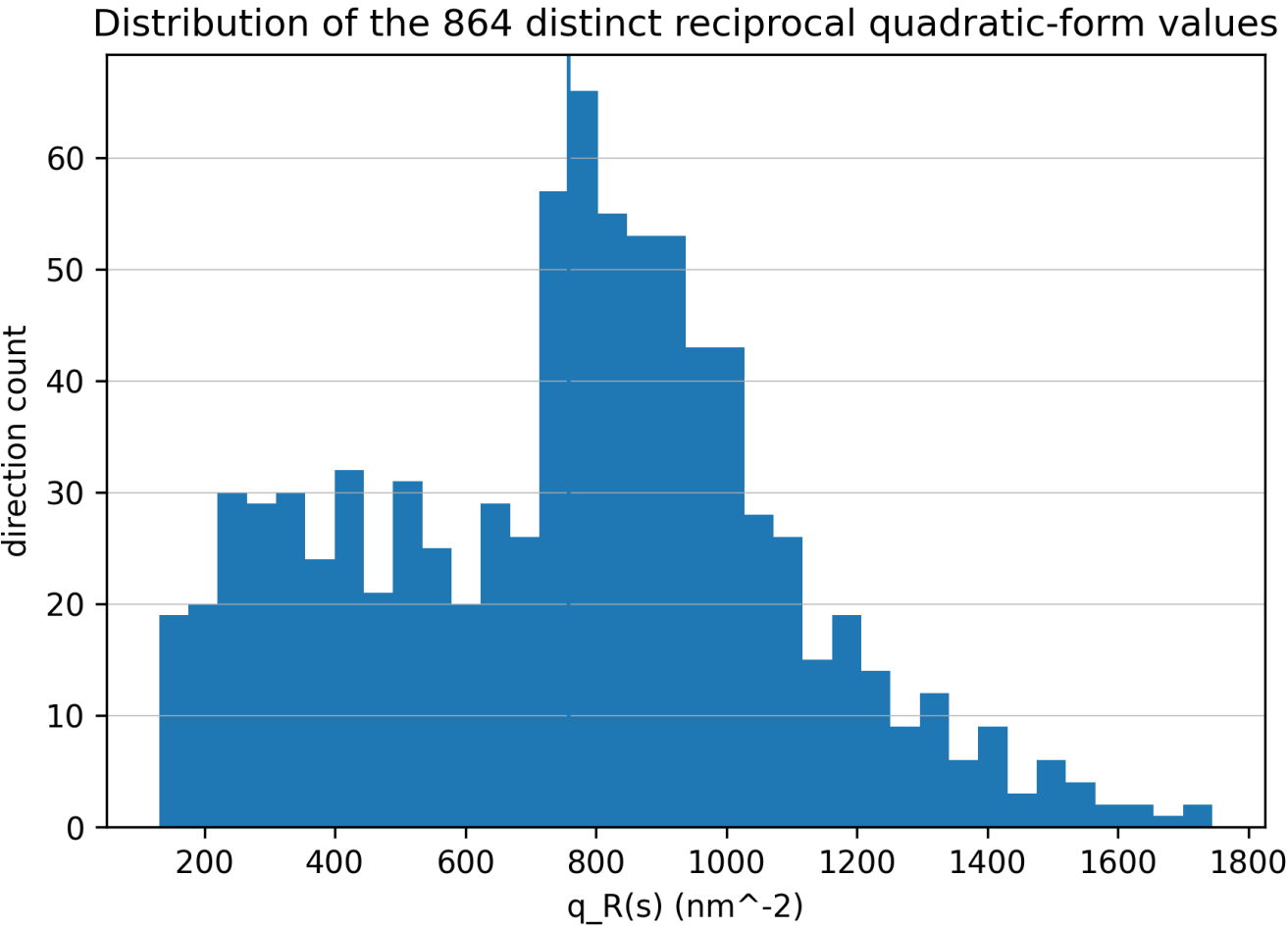


Figure 10.2. Histogram of all 864 reciprocal quadratic-form values. The vertical line marks the mean $757.916780244 \text{ nm}^{-2}$.

Physical directions of the 864 primitive projective reciprocal modes

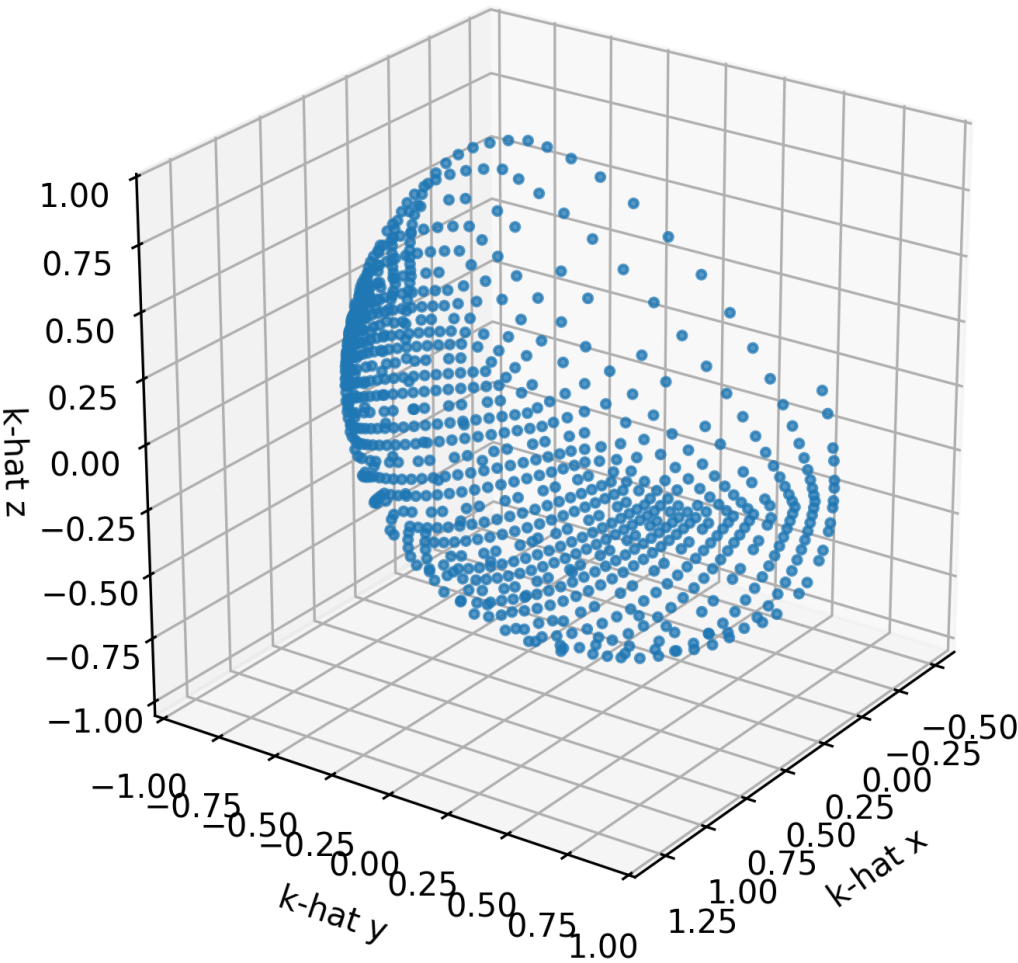


Figure 10.3. Normalized physical wave directions $k_s/|k_s|$ for the complete primitive projective shell. Each point is generated by $B^{-T}s$ and represents one antipodal direction class.

10.3 Form-degree multiplicities

For a separable metric-weighted k -form mode labeled by frequency index j , the eigenvalue is $\Lambda_j = x_j^T R x_j$. If $r(j)$ components are active, the form-degree multiplicity is the binomial coefficient $C(r(j),k)$. Geometry fixes these reciprocal levels and their tensor multiplicities. Constitutive occupation laws determine which levels are populated and how they map to measured mass, spin, lifetime, or decay channels.

10.4 Complete 864-direction archival catalogue

The following table is the full primitive projective shell, sorted by q_R . It is reproduced in full to make the reciprocal spectrum independently reconstructible from the PDF. The machine-readable companion also stores each physical k vector and normalized direction.

Table 10.2. Complete primitive radius-nine reciprocal shell.

Rank	$s=(s1,s2,s3)$	$q_R \text{ (nm}^{-2}\text{)}$	$\kappa^2 \text{ (nm}^{-2}\text{)}$	$ \kappa \text{ (nm}^{-1}\text{)}$
1	(1, 4, 9)	130.593523477839	63.649699460279	7.978076175387
2	(0, 4, 9)	132.986604171555	64.816057966297	8.050842065666
3	(1, 5, 9)	133.054017374512	64.848914343808	8.052882362472

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
4	(0, 5, 9)	135.444028463372	66.013776761648	8.124886261459
5	(1, 3, 9)	137.939826205902	67.230198311470	8.199402314283
6	(1, 6, 9)	145.321307895922	70.827842962057	8.415927932323
7	(2, 4, 9)	146.616493843039	71.459100884224	8.453348501288
8	(2, 5, 9)	149.080057344568	72.659811855931	8.524072492414
9	(1, -4, -9)	153.795735924186	74.958176402278	8.657839014574
10	(2, 3, 9)	153.959726966245	75.038103647238	8.662453673598
11	(1, 2, 9)	155.092925558701	75.590410897381	8.694274604438
12	(1, -5, -9)	156.250090611146	76.154399109451	8.726648790312
13	(0, 2, 9)	157.492145462129	76.759761579755	8.761264839037
14	(1, -3, -9)	161.148177861961	78.541667429824	8.862373690486
15	(2, 6, 9)	161.350417470834	78.640236562358	8.867933049046
16	(1, 7, 9)	167.395395042067	81.586485315026	9.032523751147
17	(1, -6, -9)	168.511241922843	82.130335551344	9.062578857662
18	(0, 7, 9)	169.779266921214	82.748355556510	9.096612312092
19	(2, 2, 9)	171.109756714187	83.396820144971	9.132185945598
20	(1, -2, -9)	178.307416424473	86.904872192091	9.322278272616
21	(3, 4, 9)	181.055515267153	88.244262238133	9.393841718814
22	(1, 1, 9)	182.052821536235	88.730337218013	9.419678190788
23	(2, 7, 9)	183.427574221835	89.400375003505	9.455177153470
24	(3, 5, 9)	183.522148373539	89.446469298017	9.457614355535
25	(0, 1, 9)	184.455111044520	89.901183988564	9.481623489074
26	(1, -7, -9)	190.579189859275	92.885985727958	9.637737583477
27	(2, -4, -9)	193.020918735732	94.076054768222	9.699281146983
28	(2, -5, -9)	195.472203817836	95.270781387217	9.760675252626
29	(2, 1, 9)	198.066583086865	96.535250377425	9.825235385344
30	(1, 8, 9)	199.276278812948	97.124841402715	9.855193625836
31	(2, -3, -9)	200.376430278364	97.661041883946	9.882360137333
32	(0, 8, 9)	201.657081087238	98.285215556022	9.913890031467
33	(1, -1, -9)	205.273451611720	100.047790689078	10.002389249028
34	(3, 2, 9)	205.542638928589	100.178989322524	10.008945465059
35	(2, -6, -9)	207.730285524676	101.245221740933	10.062068462346
36	(2, 8, 9)	215.311527597573	104.940227179372	10.244033735759
37	(2, -2, -9)	217.538738445732	106.025742734391	10.296880242792
38	(3, 7, 9)	217.875804460519	106.190024621947	10.304854420221
39	(1, 0, 9)	218.819514138506	106.649977273365	10.327147586501
40	(1, -8, -9)	222.453934420444	108.421349639292	10.412557305451
41	(2, -7, -9)	229.795163856252	111.999375829369	10.582975754927
42	(3, 1, 9)	232.496395696411	113.315923466800	10.644995230943
43	(4, 4, 9)	233.910587750182	114.005183522005	10.677320989930
44	(2, 0, 9)	234.830206084280	114.453394344599	10.698289318606
45	(4, 5, 9)	236.380290461425	115.208886670067	10.733540267315
46	(1, 9, 8)	237.497584129361	115.753442052900	10.758877360250
47	(0, 9, 8)	238.584940727395	116.283406470950	10.783478403138
48	(0, 9, 7)	239.290532126215	116.627303161082	10.799412167386
49	(1, 9, 7)	239.493551599581	116.726252390111	10.803992428270
50	(1, 9, 9)	240.963959208565	117.442911225125	10.837108065583
51	(4, 3, 9)	241.247681663676	117.581194108663	10.843486252523
52	(1, 0, -9)	242.046283423704	117.970422920786	10.861419010460
53	(2, -1, -9)	244.507843237835	119.170157319556	10.916508476594
54	(1, 9, 6)	246.951861619224	120.361342236756	10.970931694107
55	(4, 6, 9)	248.656789797403	121.192303552849	11.008737600327

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
56	(3, 8, 9)	249.762827441113	121.731372885992	11.033194137963
57	(3, -4, -9)	250.662152606193	122.169693064129	11.053039991972
58	(3, -5, -9)	253.110368083441	123.362923594947	11.106886314127
59	(2, 9, 8)	254.826278590242	124.199237564814	11.144471165776
60	(2, 9, 9)	257.002277598046	125.259793089959	11.191952157240
61	(0, 9, 5)	257.088742572124	125.301935069648	11.193834690116
62	(1, -9, -7)	257.503563711764	125.504113862016	11.202861860347
63	(1, -9, -8)	258.088348384344	125.789130818963	11.215575367272
64	(2, 9, 7)	258.112622131862	125.800961549103	11.216102779001
65	(4, 2, 9)	258.391572201905	125.936918430041	11.222161932090
66	(1, 9, 5)	259.872514188290	126.658711592835	11.254275258444
67	(2, -8, -9)	261.666838812564	127.533243652525	11.293061748371
68	(1, -9, -6)	262.381121588607	127.881376414503	11.308464812454
69	(1, -9, -9)	264.135475606348	128.736427285345	11.346207616880
70	(1, -1, 9)	265.393003365513	129.349331063436	11.373184737066
71	(2, 9, 6)	266.861308222904	130.064965042827	11.404602800748
72	(0, 1, -9)	267.801432083510	130.523170010343	11.424673737588
73	(4, 7, 9)	270.740085758118	131.955434170352	11.487185650557
74	(1, -9, -5)	272.721022014874	132.920918476424	11.529133465982
75	(0, 9, 4)	274.181361619214	133.632670288082	11.559959787477
76	(3, -2, -9)	275.186111525906	134.122373206654	11.581121414036
77	(1, 9, 4)	278.255509306780	135.618360458348	11.645529634085
78	(2, 9, 5)	281.072336863371	136.991248045985	11.704326039802
79	(2, 0, -9)	281.283744654675	137.094285639441	11.708726900882
80	(2, -1, 9)	281.400625706430	137.151252046493	11.711159295582
81	(4, 1, 9)	285.342259364871	139.072356486139	11.792894321842
82	(3, -7, -9)	287.427188912144	140.088525860743	11.835899875411
83	(1, -9, -4)	288.523264990564	140.622740047780	11.858445937296
84	(1, 1, -9)	288.625911860423	140.672768887213	11.860555167749
85	(3, 9, 8)	290.571024110039	141.620793006690	11.900453479036
86	(2, -9, -7)	294.132646356229	143.356684492913	11.973165182729
87	(3, 9, 7)	295.147743723058	143.851430638059	11.993808012389
88	(2, -9, -8)	296.007807100209	144.270615096940	12.011270336519
89	(2, -9, -6)	297.719828161672	145.105033398321	12.045955063768
90	(2, 9, 4)	300.745708053260	146.579810558577	12.107014931790
91	(1, 9, 3)	302.100846974693	147.240288833296	12.134260951261
92	(3, -1, -9)	302.158285922866	147.268283879997	12.135414450277
93	(4, 8, 9)	302.630178343568	147.498278522575	12.144886929180
94	(2, -9, -9)	303.345310393612	147.846825210401	12.159227985789
95	(5, 4, 9)	305.181711292127	148.741864735840	12.195977399776
96	(2, -9, -5)	306.769352516538	149.515661813163	12.227659703032
97	(5, 5, 9)	307.654483608226	149.947063972080	12.245287418925
98	(1, -9, -3)	309.787850515677	150.986841128570	12.287670288894
99	(5, 3, 9)	312.515735600764	152.316379234320	12.341652208449
100	(3, -1, 9)	315.824299106262	153.928932959512	12.406809942911
101	(3, -8, -9)	319.295794263600	155.620897595721	12.474810523440
102	(5, 6, 9)	319.934052549061	155.931976943040	12.487272598251
103	(3, 9, 5)	320.688210597366	156.299544429098	12.501981620091
104	(2, -9, -4)	321.281219420828	156.588569737440	12.513535461149
105	(1, -2, 9)	321.773289217255	156.828398588228	12.523114572191
106	(4, 0, 9)	322.099743152572	156.987508276957	12.529465602210
107	(0, 2, -9)	324.184787540109	158.003733623312	12.569953604660

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
108	(0, 9, 2)	324.753627361664	158.280979253255	12.580976879927
109	(2, 9, 3)	325.881421792573	158.830652580604	12.602803361975
110	(4, -4, -9)	326.719437535569	159.239091290000	12.618997237895
111	(2, 1, -9)	327.866442696251	159.798127694046	12.641128418541
112	(4, -5, -9)	329.164583407961	160.430825732640	12.666129074529
113	(5, 2, 9)	329.656556534137	160.670607467520	12.675591010581
114	(1, 9, 2)	331.408527192029	161.524496717679	12.709228801059
115	(4, -3, -9)	334.081088287914	162.827070582080	12.760371098917
116	(1, -9, -2)	336.514778590213	164.013221718794	12.806764685852
117	(2, -2, 9)	337.777841953316	164.628823483107	12.830776417782
118	(2, -9, -3)	341.255428874542	166.323757171151	12.896656821485
119	(4, -6, -9)	341.416525905088	166.402273910000	12.899700535671
120	(3, 9, 4)	341.651957858656	166.517020588770	12.904147418128
121	(5, 7, 9)	342.020418114631	166.696603648720	12.911103889626
122	(4, 9, 9)	344.327067553754	167.820836609517	12.954568175339
123	(4, 9, 8)	344.731820688750	168.018108378530	12.962179923860
124	(1, 2, -9)	345.012336921878	168.154828588360	12.967452663818
125	(3, -9, -7)	349.177780059608	170.185015053774	13.045497884472
126	(4, 9, 7)	350.598916373169	170.877659656977	13.072018193721
127	(4, -2, -9)	351.249535664995	171.194763608880	13.084141684072
128	(3, -9, -8)	352.343316874988	171.727859304879	13.104497674649
129	(2, 9, 2)	356.479478081309	173.743774112066	13.181190162958
130	(5, 1, 9)	356.604174092246	173.804549435441	13.183495342110
131	(0, 9, 1)	358.233274057024	174.598552999993	13.213574573142
132	(3, -9, -5)	359.233734077118	175.086165079866	13.232012888441
133	(4, 9, 6)	361.928354607011	176.399490444859	13.281546989898
134	(4, -7, -9)	363.475265026951	177.153435822080	13.309899917809
135	(1, 9, 1)	366.178549958789	178.470984111495	13.359303279419
136	(2, -9, -2)	366.691980877678	178.721224114297	13.368665756698
137	(4, -1, 9)	368.664023565010	179.682373802495	13.404565408938
138	(1, -9, -1)	368.704049214174	179.701881818453	13.405293052315
139	(3, -2, 9)	372.198445748292	181.405008307949	13.468667651551
140	(3, -9, -4)	372.455224910008	181.530159357064	13.473312857537
141	(5, 8, 9)	373.913580304938	182.240944089121	13.499664591727
142	(4, -1, -9)	378.224779666811	184.342170370401	13.577266675233
143	(4, 9, 5)	378.720135390277	184.583600742175	13.586154744525
144	(2, 2, -9)	384.255937362563	187.281683483371	13.685089823723
145	(3, 1, -9)	385.523024590994	187.899246430842	13.707634603783
146	(1, -3, 9)	387.960371693734	189.087179847740	13.750897419723
147	(2, 9, 1)	392.539876919469	191.319175152961	13.831817492758
148	(5, 0, 9)	393.358588275092	191.718205138081	13.846234330607
149	(6, 4, 9)	394.868885892987	192.454305879638	13.872790125985
150	(4, -8, -9)	395.340800773550	192.684311468880	13.881077460661
151	(6, 5, 9)	397.344727813942	193.661001204056	13.916213608739
152	(2, -9, -1)	397.590875430239	193.780970566877	13.920523358225
153	(3, 9, 2)	399.966480029505	194.938811436416	13.962048969847
154	(4, 9, 4)	400.974258722967	195.429990548925	13.979627697079
155	(2, -3, 9)	403.961854824938	196.886108654441	14.031611049856
156	(1, -9, 0)	406.355662387557	198.052821427546	14.073124082006
157	(1, 9, 0)	406.410915274972	198.079751014746	14.074080823086
158	(1, 3, -9)	411.205558608070	200.416602024228	14.156857067309
159	(4, 0, -9)	415.006820293364	202.269290866641	14.222140867909

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
160	(3, -9, -2)	415.285234224059	202.404986439763	14.226910642854
161	(5, 9, 9)	415.613539119981	202.564998264241	14.232533093734
162	(5, 9, 8)	417.308668326376	203.391183680333	14.261528097660
163	(6, 2, 9)	419.337591925284	204.380056434963	14.296155302562
164	(5, -4, -9)	421.192773523861	205.284249445833	14.327744045935
165	(4, -9, -7)	422.638964821902	205.989105544597	14.352320563052
166	(5, -5, -9)	423.634849791395	206.474487800296	14.369220152823
167	(4, -9, -6)	423.645394484546	206.479627155847	14.369398983807
168	(5, 9, 7)	424.466140082195	206.879648605859	14.383311461755
169	(4, -2, 9)	425.035100602183	207.156953062754	14.392948032379
170	(4, -9, -8)	427.094877708682	208.160863442782	14.427780960452
171	(5, -3, -9)	428.557493881062	208.873724826091	14.452464316721
172	(4, 9, 3)	428.690724605079	208.938659865110	14.454710646191
173	(4, -9, -5)	430.114166696613	209.632428276532	14.478688762334
174	(6, 7, 9)	431.716801530060	210.413533057052	14.505637974838
175	(2, -9, 0)	433.952112532222	211.502996528891	14.543142594670
176	(2, 9, 0)	434.062618307052	211.556855703291	14.544994180243
177	(5, -6, -9)	435.883722683666	212.444439889478	14.575473916462
178	(4, -9, -9)	437.013133144886	212.994900850401	14.594344824294
179	(5, 9, 6)	437.085954387438	213.030393040820	14.595560730606
180	(3, 9, 1)	437.317254939064	213.143126124390	14.599422116111
181	(5, -1, 9)	439.919799082673	214.411574575441	14.642799410476
182	(0, 9, -1)	441.579595096014	215.220539021771	14.670396689312
183	(3, 2, -9)	441.915588862162	215.384298308345	14.675976911550
184	(4, -9, -4)	442.045281458103	215.447508906651	14.678130293285
185	(3, -9, -1)	444.893752705219	216.835819245264	14.725346150270
186	(5, -2, -9)	445.729010862999	217.242913941069	14.739162592938
187	(6, 1, 9)	446.282139878537	217.512502314705	14.748305065827
188	(1, -9, 1)	449.469618110364	219.066040546074	14.800879722033
189	(2, 3, -9)	450.452228653610	219.544953007416	14.817049402881
190	(1, 9, -1)	452.105623140578	220.350797427432	14.844217642821
191	(5, 9, 5)	455.168111242103	221.843416985215	14.894408916946
192	(5, -7, -9)	457.939392200673	223.194105713380	14.939682249411
193	(4, -9, -3)	459.438738769017	223.924869046204	14.964119387595
194	(4, 1, -9)	461.595657544652	224.976125097602	14.999204148807
195	(4, 9, 2)	461.869533036615	225.109608690729	15.003653178167
196	(6, 8, 9)	463.613033325223	225.959369585630	15.031944970150
197	(1, -4, 9)	463.954250794948	226.125674841972	15.037475680511
198	(0, 4, -9)	466.371888327515	227.304002053412	15.076604460336
199	(5, -1, -9)	472.707324469672	230.391816790768	15.178663208292
200	(2, -9, 1)	475.775692183629	231.887302000340	15.227846269264
201	(5, 9, 4)	478.712610646193	233.318720439044	15.274773989786
202	(2, -4, 9)	479.952664321296	233.923107560495	15.294545026267
203	(2, 9, -1)	481.047702244058	234.456815763056	15.311982750874
204	(4, -9, -2)	482.294538629354	235.064508695192	15.331813614025
205	(1, 4, -9)	487.205576918997	237.458089194815	15.409675181353
206	(5, -8, -9)	489.801858342416	238.723485272003	15.450679120091
207	(4, -3, 9)	491.212974264093	239.411246057732	15.472919765116
208	(0, 9, -2)	491.446269439644	239.524951296812	15.476593659356
209	(5, -2, 9)	496.287806514990	241.884657747522	15.552641503858
210	(1, -9, 2)	498.045916382594	242.741539174036	15.580164927690
211	(4, 9, 1)	500.510684017575	243.942837025783	15.618669502419

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
212	(7, 4, 9)	502.972111552762	245.142506953400	15.657027398373
213	(1, 9, -2)	503.262673555608	245.284123349552	15.661549200177
214	(7, 5, 9)	505.451023078573	246.350698365995	15.695563015260
215	(5, 9, 3)	507.719452599705	247.456303402308	15.730743892210
216	(6, 9, 8)	508.301567022918	247.740018912099	15.739759175797
217	(5, 0, -9)	509.492434701080	248.320433375186	15.758186233675
218	(7, 3, 9)	510.299996651686	248.714029275524	15.770669905731
219	(4, -9, -1)	510.612681039114	248.866427853614	15.775500874889
220	(5, -9, -6)	514.232254234356	250.630563929555	15.831315925392
221	(3, -4, 9)	514.367128906560	250.696300208982	15.833391936316
222	(5, -9, -7)	514.516200643112	250.768955965384	15.835686153918
223	(6, 9, 7)	516.749414850137	251.857397484704	15.870015673738
224	(7, 6, 9)	517.736731229121	252.338603513311	15.885169294449
225	(4, 2, -9)	517.991291420677	252.462673063282	15.889074015287
226	(5, -9, -5)	519.410650375023	253.154451403161	15.910828118082
227	(5, -9, -8)	520.262489601292	253.569627510648	15.923869740445
228	(3, -9, 1)	520.497817315809	253.684323384569	15.927470715232
229	(2, -9, 2)	523.061614384460	254.933886981223	15.966649209563
230	(2, 4, -9)	526.455316569394	256.587936266182	16.018362471432
231	(7, 2, 9)	527.434678375346	257.065265332369	16.033254982453
232	(3, 9, -1)	528.405832406454	257.538594028643	16.048009036284
233	(6, -1, 9)	529.591625659250	258.116535278351	16.066005579432
234	(5, -9, -4)	530.051389065113	258.340618386201	16.072977894161
235	(5, -9, -9)	531.471121108895	259.032578565345	16.094489074380
236	(2, 9, -2)	533.495128730488	260.019055332255	16.125106366541
237	(6, -4, -9)	534.082160571067	260.305167531630	16.133975564988
238	(6, -5, -9)	536.521167233745	261.493909797915	16.170773320961
239	(7, 7, 9)	539.829236004404	263.106222395347	16.220549386360
240	(5, 9, 2)	542.188637102641	264.256165875006	16.255957857813
241	(4, -9, 0)	544.393165998298	265.330626521470	16.288972543456
242	(4, 9, 0)	544.614177547958	265.438344870271	16.292278688700
243	(5, -9, -3)	546.154470304627	266.189064878675	16.315301556474
244	(1, -5, 9)	549.754926520899	267.943883570924	16.368991525776
245	(6, 9, 5)	550.032138152844	268.078993158217	16.373118003551
246	(1, -9, 3)	552.084557204248	269.079317311433	16.403637319553
247	(0, 5, -9)	552.175633658322	269.123706870542	16.404990303884
248	(7, 1, 9)	554.376156723743	270.196215123933	16.437646276883
249	(5, 1, -9)	556.084341557225	271.028763694324	16.462951244972
250	(6, -2, -9)	558.624537119918	272.266824203222	16.500509816464
251	(1, 9, -3)	559.882066520062	272.879728781106	16.519071668260
252	(5, -3, 9)	562.462610572043	274.137454654323	16.557096806334
253	(2, -5, 9)	565.750270442391	275.739820201269	16.605415387797
254	(3, -9, 2)	566.493363445240	276.101994718373	16.616317122587
255	(4, -4, 9)	567.197644550738	276.445252787431	16.626642859803
256	(5, -9, -2)	567.719894093564	276.699790880584	16.634295623217
257	(6, -7, -9)	570.819570433311	278.210535534644	16.679644346767
258	(7, 8, 9)	571.728537404424	278.653555012103	16.692919307662
259	(1, 5, -9)	573.012391854660	279.279290100123	16.711651327745
260	(6, 9, 4)	574.867013628334	280.183210259126	16.738674089041
261	(2, -9, 3)	575.809879134713	280.642751471540	16.752395395034
262	(5, 9, 1)	582.120164155000	283.718307857138	16.843939796174
263	(3, 9, -2)	582.143634964283	283.729747244921	16.844279362588

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
264	(4, -9, 1)	583.635993506905	284.457104698761	16.865856180424
265	(3, 4, -9)	584.121107278706	284.693543267511	16.872864109792
266	(4, 3, -9)	584.193721921437	284.728934763683	16.873912846868
267	(6, -1, -9)	585.605920331447	285.417223141098	16.894295579902
268	(6, -2, 9)	585.956563486711	285.588122362254	16.899352720215
269	(7, 0, 9)	591.124431696875	288.106878650218	16.973711398814
270	(2, 9, -3)	591.404897766342	288.243574410888	16.977737611675
271	(4, 9, -1)	594.180013627764	289.596132224193	17.017524268359
272	(5, -9, -1)	594.747660431924	289.872796391927	17.025651129749
273	(3, -5, 9)	600.161665422797	292.511516761578	17.102968068776
274	(6, -8, -9)	602.678966970197	293.738419005089	17.138798645328
275	(0, 9, -4)	607.566645775174	296.120614375198	17.208155461153
276	(1, -9, 4)	611.585540575325	298.079374958263	17.264975382498
277	(2, 5, -9)	612.265201109914	298.410633259667	17.274566080214
278	(5, 2, -9)	612.483045038106	298.516807748183	17.277638951783
279	(7, 9, 9)	613.434635429179	298.980601363579	17.291055530637
280	(7, 9, 8)	617.710516778374	301.064614073828	17.351213619624
281	(1, 9, -4)	621.963802033939	303.137613722095	17.410847587699
282	(6, -9, -7)	624.809487523237	304.524566316134	17.450632261214
283	(6, -9, -5)	627.123185112348	305.652234459753	17.482912642342
284	(5, -9, 0)	627.237769319708	305.708081412704	17.484509756144
285	(7, 9, 7)	627.448740676993	305.810906293512	17.487449965433
286	(5, 9, 0)	627.514033756783	305.842729348705	17.488359824429
287	(4, -9, 2)	628.341163564936	306.245862385486	17.499881782043
288	(8, 4, 9)	629.491388271451	306.806467957124	17.515891868732
289	(6, -9, -8)	631.846152552816	307.954151508477	17.548622496039
290	(8, 5, 9)	631.973369402119	308.016155457898	17.550389040073
291	(2, -9, 4)	634.020486434390	309.013895471292	17.578791069675
292	(6, -9, -4)	636.473547731038	310.209487795714	17.612764910590
293	(8, 3, 9)	636.816203765519	310.376494191071	17.617505333931
294	(7, -1, 9)	637.679503294743	310.797255911223	17.629442870131
295	(5, -4, 9)	638.444211253832	311.169965295844	17.640010354188
296	(6, 9, 2)	640.923792227582	312.378482989245	17.674232175380
297	(7, 9, 6)	642.649307125035	313.219478022631	17.698007741625
298	(8, 6, 9)	644.262147157523	314.005556693391	17.720201937150
299	(1, -6, 9)	645.362398871585	314.541806034597	17.735326499239
300	(4, 9, -2)	649.208192256993	316.416199087550	17.788091496491
301	(4, -5, 9)	652.989111462119	318.258973251850	17.839814271787
302	(8, 2, 9)	653.947815884323	318.726234159738	17.852905482294
303	(2, 9, -4)	654.777009351618	319.130372998956	17.864220469949
304	(4, 4, -9)	660.202949046934	321.774910198804	17.938085466370
305	(2, -6, 9)	661.354673188221	322.336246576764	17.953725144848
306	(7, 9, 5)	663.312216122501	323.290329261184	17.980276117490
307	(5, -9, 1)	665.190220756916	324.205645942916	18.005711481164
308	(7, -4, -9)	665.387598677189	324.301845547391	18.008382646629
309	(8, 7, 9)	666.357721537663	324.774671663605	18.021505810104
310	(7, -5, -9)	667.823535735011	325.489091725497	18.041316241491
311	(1, 6, -9)	668.626003415059	325.880204740151	18.052152357549
312	(6, 1, -9)	668.989076628714	326.057162221010	18.057052977189
313	(3, 5, -9)	669.934061424082	326.517736349175	18.069801779466
314	(6, -9, -2)	671.561300616689	327.310832995939	18.091733830563
315	(7, -3, -9)	672.758458244102	327.894313104004	18.107852249894

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
316	(0, 9, -5)	673.820347767074	328.411865178542	18.122137434049
317	(3, -9, 4)	674.871483352371	328.924175914285	18.136266868192
318	(1, -9, 5)	676.548866495825	329.741712114529	18.158791592904
319	(5, 9, -1)	678.370245907989	330.629430349707	18.183218371611
320	(4, -9, 3)	678.508676172390	330.696899581646	18.185073537977
321	(5, 3, -9)	678.688545143723	330.784565536762	18.187483760454
322	(7, -6, -9)	680.066269417569	331.456051638324	18.205934517028
323	(8, 1, 9)	680.886224627863	331.855687863125	18.216906649130
324	(6, 9, 1)	682.145695351341	332.469538618457	18.233747245656
325	(7, 9, 4)	689.437467669390	336.023460009171	18.330942692867
326	(1, 9, -5)	689.507880097239	336.057778172518	18.331878740940
327	(7, -2, -9)	689.936114435752	336.266494395338	18.337570569608
328	(7, -2, 9)	694.041371517347	338.267346906948	18.392045751002
329	(6, -9, -1)	697.298690883650	339.854924860203	18.435154592794
330	(2, -9, 5)	697.693436283491	340.047318980479	18.440371985957
331	(8, 8, 9)	698.260092542539	340.323500368539	18.447858964350
332	(7, -7, -9)	702.115799724863	342.202725285871	18.498722260899
333	(3, 9, -4)	706.006267728213	344.098892205780	18.549902754618
334	(2, 6, -9)	707.881882275169	345.013043987873	18.574526750038
335	(5, -9, 2)	708.605014743546	345.365489982563	18.584011676238
336	(4, 9, -3)	709.698713435646	345.898545460341	18.598347922876
337	(7, -1, -9)	716.920567252138	349.418389421391	18.692736274323
338	(8, 0, 9)	717.631429996139	349.764855301232	18.702001371544
339	(7, 9, 3)	721.025061765702	351.418870266592	18.746169482499
340	(2, 9, -5)	723.611463486318	352.679451096458	18.779761742271
341	(5, -5, 9)	724.232608560357	352.982189672085	18.787820248025
342	(6, 2, -9)	725.390849714451	353.546702363046	18.802837614654
343	(9, 2, 4)	725.721306497278	353.707762991104	18.807120007888
344	(9, 1, 3)	726.630426680247	354.150857142163	18.818896278532
345	(9, 2, 3)	727.477194554560	354.563561534214	18.829858245197
346	(6, -4, 9)	728.106829015840	354.870437734220	18.838005142112
347	(9, 1, 2)	729.218306118120	355.412158193498	18.852378051416
348	(9, 2, 5)	729.427760989419	355.514243957428	18.855085360651
349	(9, 1, 4)	729.504889791797	355.551835600263	18.856082191173
350	(9, 3, 5)	730.820623150802	356.193108081779	18.873078924271
351	(9, 3, 4)	731.744519827494	356.643404116664	18.885004742299
352	(9, 0, 2)	733.547983699710	357.522390534957	18.908262493814
353	(7, -8, -9)	733.972126656893	357.729112668138	18.913728153596
354	(4, -9, 4)	734.138531329267	357.810216287240	18.915872073136
355	(5, 9, -2)	734.688800608619	358.078410860142	18.922959886343
356	(9, 2, 2)	734.695425161266	358.081639586758	18.923045198560
357	(9, 0, 1)	736.967854518174	359.189194094516	18.952287305086
358	(3, -9, 5)	737.254057130072	359.328685776392	18.955967022982
359	(9, 1, 1)	737.268528105417	359.335738754266	18.956153057893
360	(9, 1, 5)	737.841695452771	359.615093567798	18.963520073230
361	(9, 2, 6)	738.596558030983	359.983004433187	18.973218083214
362	(8, 9, 9)	739.969260172151	360.652042808192	18.990841024246
363	(9, 4, 6)	741.928376640820	361.606892414189	19.015964146322
364	(9, 4, 5)	742.020281936921	361.651685940850	19.017141897269
365	(9, 0, 4)	743.095269711053	362.175621944143	19.030912273040
366	(9, 3, 7)	745.359857445688	363.279354540311	19.059888628749
367	(8, 9, 8)	745.535517592746	363.364969165521	19.062134433623

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
368	(4, 5, -9)	746.018972797166	363.600599368645	19.068314014843
369	(9, -1, 1)	746.473977555667	363.822363169486	19.074128110335
370	(1, -9, 6)	746.974534965749	364.066328780229	19.080522235521
371	(9, 4, 7)	747.298813894142	364.224378396962	19.084663434207
372	(9, 2, 1)	747.375998317396	364.261997148737	19.085648984217
373	(9, 4, 4)	747.574529782446	364.358758976945	19.088183752703
374	(9, -1, 2)	747.684457906036	364.412336611137	19.089587125214
375	(4, -6, 9)	748.587374998237	364.852407450989	19.101110110436
376	(9, 3, 2)	749.979340829148	365.530834714739	19.118860706505
377	(7, -9, -6)	750.654126910720	365.859717266861	19.127459770363
378	(9, -1, 0)	750.725839754722	365.894669237270	19.128373408036
379	(1, -7, 9)	750.776667847008	365.919442232989	19.129020942876
380	(9, 1, 0)	750.781092642137	365.921598824470	19.129077312418
381	(9, 1, 6)	751.640843663168	366.340631044766	19.140026934275
382	(0, 7, -9)	753.203514194144	367.102257708962	19.159912779263
383	(9, 2, 7)	753.227697621970	367.114044418380	19.160220364557
384	(7, -9, -5)	753.251770908588	367.125777446309	19.160526544078
385	(7, -9, -7)	753.518825462277	367.255936596848	19.163922787280
386	(7, 0, -9)	753.711816693259	367.349998182165	19.166376761980
387	(9, -1, 3)	754.357280805829	367.664589562223	19.174581861470
388	(5, 4, -9)	754.700841874076	367.832037060060	19.178947756852
389	(9, 0, 5)	756.062426540859	368.495656912887	19.196240697410
390	(8, 9, 7)	756.564117562765	368.740175032284	19.202608547598
391	(5, -9, 3)	757.482151279600	369.187613531643	19.214255476902
392	(7, 9, 2)	758.074998411438	369.476560033448	19.221773072052
393	(9, 4, 8)	758.131593696887	369.504143889170	19.222490574563
394	(9, 5, 6)	758.304480882843	369.588407006770	19.224682234221
395	(9, 4, 3)	758.591120177394	369.728111522475	19.228315358410
396	(9, 5, 7)	759.044566967332	369.949115988333	19.234061349292
397	(9, 0, -1)	760.194623803372	370.509639741937	19.248626957317
398	(7, -3, 9)	760.210036364688	370.517151637394	19.248822084413
399	(9, -1, -1)	760.440044503200	370.629254814487	19.251733813205
400	(9, 3, 8)	760.822988417266	370.815897033729	19.256580616343
401	(7, -9, -4)	761.311757455878	371.054117135191	19.262765043866
402	(7, -9, -8)	761.845866563256	371.314435436269	19.269520892754
403	(1, 9, -6)	762.514300709962	371.640222132376	19.277972459063
404	(9, 5, 5)	763.026737347777	371.889977534641	19.284449111516
405	(8, -1, 9)	764.183431989151	372.453736474059	19.299060507550
406	(6, -9, 1)	765.160499065841	372.929947117035	19.311394230273
407	(9, 5, 8)	765.246995601245	372.972104479331	19.312485714670
408	(9, -2, 0)	765.408408248118	373.050775045750	19.314522387203
409	(9, 2, 0)	765.518914022948	373.104634220150	19.315916603158
410	(9, -2, 1)	765.786897217897	373.235245979176	19.319297243409
411	(9, -1, 4)	766.492446255045	373.579122022742	19.328195001674
412	(2, -7, 9)	766.765872558787	373.712386686979	19.331642110462
413	(2, -9, 6)	766.828728682015	373.743021999099	19.332434456092
414	(9, 3, 1)	767.290265154110	373.967969277927	19.338251453478
415	(9, 1, -1)	769.755999728280	375.169738404107	19.369298861965
416	(9, -2, -1)	770.492261827764	375.528583621758	19.378559895456
417	(9, 1, 7)	770.902334422988	375.728448031170	19.383716053202
418	(9, -2, 2)	771.6277278737098	376.081996422037	19.392833635703
419	(8, 9, 6)	773.055060082207	376.777660408481	19.410761458750

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
420	(9, 5, 4)	773.211336362134	376.853827571946	19.412723342487
421	(9, 2, 8)	773.321179762381	376.907363913008	19.414102191783
422	(1, 7, -9)	774.046411600195	377.260833114899	19.423203472005
423	(9, 4, 9)	774.426716049056	377.446188890812	19.427974389802
424	(7, -9, -3)	774.834086552592	377.644736333507	19.433083551858
425	(9, 4, 2)	775.070053121766	377.759743577439	19.436042384638
426	(9, -1, -2)	775.616591801101	378.026119901140	19.442893814994
427	(7, -9, -9)	775.635250213659	378.035213785125	19.443127674968
428	(4, 9, -4)	775.651577163723	378.043171342567	19.443332310655
429	(3, 9, -5)	776.131097934313	378.276883950361	19.449341478579
430	(9, 5, 9)	776.911766784581	378.657372479763	19.459120547439
431	(9, 0, -2)	780.001522270106	380.163281829799	19.497776330387
432	(9, 6, 7)	780.597116665258	380.453567314424	19.505218976326
433	(6, 9, -1)	780.976529247129	380.638488405183	19.509958698193
434	(9, -2, -2)	781.038457956832	380.668671707200	19.510732218633
435	(9, 6, 8)	782.169194130338	381.219778804212	19.524850288906
436	(9, -2, 3)	782.930902805724	381.591026374333	19.534355028368
437	(9, -1, 5)	784.089954253684	382.155933992697	19.548809017244
438	(9, 5, 3)	788.858277925915	384.479957118686	19.608160472586
439	(9, 2, -1)	789.124172277925	384.609550800998	19.611464779587
440	(9, -3, -1)	790.351275777063	385.207626163748	19.626706961784
441	(7, -9, -2)	793.818758198730	386.897635041258	19.669713649193
442	(9, 6, 5)	793.839989383368	386.907982863152	19.669976686899
443	(9, 1, -2)	794.193249363847	387.080157493179	19.674352784607
444	(9, -3, 1)	794.906613504862	387.427842523586	19.683186797965
445	(8, 9, 5)	795.008345151072	387.477425294113	19.684446278575
446	(4, -9, 5)	795.230729035568	387.585812502268	19.687199204109
447	(9, 1, 8)	795.626167732232	387.778544527007	19.692093452119
448	(9, -1, -3)	796.255481648426	388.085264497226	19.699879809208
449	(9, -3, -2)	796.267120737299	388.090937247980	19.700023788005
450	(5, 9, -3)	796.469697858671	388.189670880013	19.702529555365
451	(9, 4, 1)	797.011328615561	388.453655141838	19.709227664773
452	(9, -2, -3)	797.046996635324	388.471039302076	19.709668675604
453	(2, 9, -6)	797.908260170442	388.890808703395	19.720314619787
454	(9, 0, 7)	798.383767848742	389.122565378679	19.726189834296
455	(9, 2, 9)	798.877004452215	389.362962917070	19.732282253127
456	(9, -2, 4)	799.696419423772	389.762335836062	19.742399444750
457	(7, 1, -9)	800.309862759117	390.061320677659	19.749970143716
458	(7, 9, 1)	800.587277606597	390.196529309739	19.753392855652
459	(3, -7, 9)	801.171128329481	390.481091070932	19.760594400749
460	(9, -3, 2)	805.377796192897	392.531369967657	19.812404446903
461	(9, -1, 6)	807.149804801746	393.395025472085	19.834188298796
462	(6, -9, 2)	807.284916981072	393.460877509602	19.835848293169
463	(9, 6, 4)	808.654939566559	394.128609901667	19.852672613572
464	(9, 7, 8)	808.898189284167	394.247166863814	19.855658308498
465	(9, 5, 2)	809.967562039120	394.768366174860	19.868778678491
466	(9, 7, 9)	811.302258129837	395.418880861826	19.885142213769
467	(5, -9, 4)	811.821630365078	395.672016590158	19.891506141822
468	(9, 7, 7)	811.956462987920	395.737732375236	19.893157928676
469	(2, 7, -9)	813.305360065161	396.395168450799	19.909675247246
470	(6, -5, 9)	813.892156717509	396.681166022283	19.916856328806
471	(8, -4, -9)	815.109087842225	397.274283493114	19.931740603698

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
472	(8, -5, -9)	817.541955295191	398.460033583043	19.961463713442
473	(9, 2, -2)	818.191773082324	398.776746891279	19.969395256023
474	(7, -9, -1)	818.265772394290	398.812813258443	19.970298276652
475	(9, 3, -1)	818.299141452305	398.829076932608	19.970705469077
476	(9, -2, -4)	818.517877863239	398.935686406388	19.973374437145
477	(5, -6, 9)	819.827802491617	399.574127783046	19.989350359205
478	(9, -4, -1)	820.017086351099	399.666382440458	19.991657821213
479	(9, 7, 6)	820.477079241096	399.890577396092	19.997264247794
480	(8, -2, 9)	820.542230606899	399.922331381606	19.998058190275
481	(9, -4, -2)	821.302580142502	400.292916523481	20.007321572951
482	(9, -2, 5)	821.924278591244	400.595924807227	20.014892575461
483	(9, -1, -4)	822.356714045173	400.806688602747	20.020157057395
484	(8, 9, 4)	822.423972769361	400.839469689179	20.020975742685
485	(8, -3, -9)	822.483017013995	400.868247137905	20.021694412260
486	(0, 9, -7)	822.714779399145	400.981205313534	20.024515108075
487	(1, -9, 7)	822.862545985096	401.053224955363	20.026313314122
488	(9, 1, -3)	824.092841548837	401.652856091686	20.041278803801
489	(9, -4, 0)	824.193935109119	401.702127866870	20.042508023370
490	(9, 4, 0)	824.414946658779	401.809846215671	20.045195090487
491	(9, -3, -4)	824.485838306040	401.844397944748	20.046056917627
492	(9, 1, 9)	825.812343590899	402.490920532279	20.062176365795
493	(9, 0, 8)	827.737952326819	403.429438875727	20.085552989045
494	(9, -4, -3)	828.050416483328	403.581730115937	20.089343695500
495	(8, -6, -9)	829.781619372893	404.425497407692	20.110333100366
496	(9, -4, 1)	833.833126416563	406.400152802717	20.159368859236
497	(9, 7, 5)	834.460038043696	406.705701926383	20.166945775858
498	(9, -1, 7)	835.671997899232	407.296396460909	20.181585578465
499	(9, 0, -4)	836.002346851844	407.457404533827	20.185574169040
500	(7, -4, 9)	836.185497836764	407.546670102559	20.187785170805
501	(9, 5, 1)	836.539188701747	407.719054740468	20.192054247661
502	(6, 9, -2)	838.585460019159	408.716382562698	20.216735210283
503	(8, -2, -9)	839.663742810501	409.241924517417	20.229728730693
504	(9, -4, -4)	840.260595373578	409.532823217829	20.236917334857
505	(5, 5, -9)	840.519935229165	409.659222318079	20.240040076988
506	(1, 9, -7)	840.983063872109	409.884945601668	20.245615466112
507	(2, -9, 7)	841.426363629962	410.101004527155	20.250950706748
508	(4, 6, -9)	841.641793172134	410.206002273207	20.253542956066
509	(9, -3, 4)	842.707189217236	410.725263384102	20.266357921050
510	(9, 8, 9)	843.207698739569	410.969205654938	20.272375431975
511	(9, 8, 8)	845.433981062732	412.054268658135	20.299119898610
512	(9, -2, -5)	845.451101640577	412.062613020133	20.299325432638
513	(9, -3, -5)	846.788710914546	412.714547557283	20.315377120725
514	(4, 9, -5)	847.066783441222	412.850076734227	20.318712477276
515	(7, -9, 0)	848.175129139275	413.390270985063	20.332001155446
516	(7, 9, 0)	848.561899351179	413.578778095464	20.336636351557
517	(9, -4, 2)	848.934660273431	413.760457247998	20.341102655658
518	(9, -2, 6)	849.614480308139	414.091793287825	20.349245521341
519	(8, -7, -9)	851.828080075330	415.170674967061	20.375737409160
520	(9, 3, -2)	851.997093425537	415.253050024100	20.377758709537
521	(9, 2, -3)	852.721716436147	415.606222490996	20.386422503495
522	(9, 8, 7)	853.122605935317	415.801611170767	20.391214068092
523	(9, 7, 4)	853.905339395719	416.183105966108	20.400566314838

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
524	(9, -1, -5)	853.920288991345	416.190392217702	20.400744893697
525	(4, -7, 9)	853.992435159090	416.225555384848	20.401606686358
526	(9, 6, 2)	854.671867581209	416.556702507001	20.409720784641
527	(8, 9, 3)	855.301942937073	416.863793593680	20.417242556077
528	(9, -5, -2)	856.144836172440	417.274609533701	20.427300593414
529	(7, 2, -9)	856.714705449711	417.552356907873	20.434097898069
530	(9, 4, -1)	857.280907251421	417.828316798938	20.440849219123
531	(9, -4, -5)	857.933116813251	418.146195829154	20.448623323568
532	(9, -5, -3)	858.262321344434	418.306646124948	20.452546201511
533	(9, 1, -4)	859.454776283251	418.887834199627	20.466749478108
534	(9, -5, -1)	859.489693549871	418.904852451889	20.467165227551
535	(4, -9, 6)	861.785269291292	420.023688226731	20.494479457325
536	(5, 9, -4)	863.712937658148	420.963210409317	20.517388001627
537	(9, -5, -4)	865.842149065851	422.000962225629	20.542662004366
538	(1, -8, 9)	865.997733447166	422.076792166101	20.544507591230
539	(9, 8, 6)	866.273573357327	422.211233192833	20.547779276429
540	(8, -1, -9)	866.651265231744	422.395315631648	20.552258163804
541	(6, 4, -9)	867.614785760133	422.864923851279	20.563679725460
542	(9, -5, 0)	868.296893476724	423.197374879511	20.571761589118
543	(0, 8, -9)	868.427649399158	423.261103730253	20.573310470856
544	(9, 5, 0)	868.573157913798	423.332022815511	20.575033968757
545	(9, -4, 3)	869.498536679721	423.783041202713	20.585991382557
546	(9, -3, 5)	869.565399553540	423.815629356477	20.586782880200
547	(9, -1, 8)	869.656533546141	423.860046959166	20.587861641248
548	(3, 7, -9)	870.980359589042	424.505263716662	20.603525516684
549	(5, -9, 5)	871.623451999979	424.818699158108	20.611130467738
550	(9, 0, -5)	872.196272966848	425.097885149992	20.617902055010
551	(2, 9, -7)	877.667399403988	427.764445819765	20.682467111536
552	(9, -2, -6)	877.846667967339	427.851819143313	20.684579259519
553	(3, -9, 7)	878.406232333743	428.124544028910	20.691170678067
554	(9, 7, 3)	878.812983297165	428.322789515268	20.695960705299
555	(9, -5, -5)	878.884319336691	428.357557835745	20.696800666667
556	(9, -4, -6)	881.067980802347	429.421847949914	20.722496180478
557	(2, -8, 9)	881.983868554089	429.868240531913	20.733264107031
558	(9, -5, 1)	882.566435953000	430.152176816567	20.740110337618
559	(9, -2, 7)	882.767024574458	430.249941277858	20.742467097186
560	(7, -9, 1)	883.546828433682	430.630008221117	20.751626640365
561	(8, -8, -9)	883.681337402504	430.695566261150	20.753206168232
562	(9, 8, 5)	884.886883328759	431.283134724334	20.767357432383
563	(9, 6, 1)	885.873845412670	431.764168073819	20.778935681931
564	(8, -3, 9)	886.707825849383	432.170640023874	20.788714246530
565	(1, 8, -9)	889.273616410066	433.421175224367	20.818769781723
566	(9, -1, -6)	890.946206486940	434.236375342092	20.838339073498
567	(9, 9, 8)	891.776569466033	434.641084187177	20.848047491004
568	(9, 2, -4)	892.714002339393	435.097977600147	20.859002315551
569	(8, 9, 2)	893.642255654209	435.550397007615	20.869844201805
570	(9, -4, 4)	895.524755635435	436.467904666862	20.891814298114
571	(9, 4, -2)	895.609210393486	436.509066891640	20.892799402944
572	(8, -9, -6)	896.489139837275	436.937933830459	20.903060393886
573	(9, -5, -6)	897.388832156955	437.376432955295	20.913546637414
574	(8, -9, -5)	897.796407763743	437.575080362827	20.918295350311
575	(9, 1, -5)	900.279053567088	438.785091817002	20.947197707975

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
576	(8, -9, -7)	900.644214460232	438.963066807524	20.951445458668
577	(9, -6, -2)	900.793888827115	439.036016278642	20.953186303726
578	(9, -6, -4)	901.230499382860	439.248814968150	20.958263643922
579	(7, 9, -1)	901.998863645185	439.623306390623	20.967195959179
580	(9, -5, 2)	902.298320978700	439.769258263058	20.970676151785
581	(8, 0, -9)	903.445584277722	440.328420480600	20.984003919190
582	(9, 9, 7)	904.095545507451	440.645203701019	20.991550769322
583	(1, -9, 8)	904.212899553866	440.702400639931	20.992913105139
584	(8, -9, -4)	904.566018239634	440.874506404630	20.997011844656
585	(0, 9, -8)	905.355509039315	441.259294645181	21.006172774810
586	(9, 5, -1)	906.069469675273	441.607270399989	21.014453844913
587	(9, -3, -7)	907.781483779828	442.441685310658	21.034297832603
588	(6, -9, 4)	907.920780459804	442.509576823040	21.035911599525
589	(9, -6, -1)	908.769097373378	442.923036198039	21.045736770140
590	(9, 8, 4)	908.962535849615	443.017315765269	21.047976524247
591	(9, -1, 9)	909.103411742474	443.085976966858	21.049607525245
592	(9, 7, 2)	909.182969748035	443.124752573862	21.050528558064
593	(9, -6, -5)	909.642318484868	443.348633577056	21.055845591594
594	(9, -4, -7)	909.665187340867	443.359779580108	21.056110267096
595	(8, -9, -8)	910.261631632611	443.650479294025	21.063012113514
596	(9, -2, -7)	915.704576843525	446.303304775927	21.125891810192
597	(3, -8, 9)	916.386054719926	446.635448827689	21.133751413975
598	(8, -9, -3)	916.797971264948	446.836211955868	21.138500702648
599	(9, -5, -7)	921.355687526642	449.057587584279	21.190978919915
600	(9, -2, 8)	921.381911390200	449.070368777326	21.191280489327
601	(2, -9, 8)	921.486341127333	449.121266564644	21.192481368746
602	(7, -5, 9)	921.967755933576	449.355902302444	21.198016470945
603	(7, 3, -9)	922.926344765040	449.823106872808	21.209033614779
604	(4, 9, -6)	923.944332268146	450.319261635321	21.220727170277
605	(7, -9, 2)	924.380870277513	450.532024966605	21.225739680082
606	(1, 9, -8)	924.914169583679	450.791948580394	21.231861637181
607	(5, -7, 9)	925.229793047614	450.945779628727	21.235483974441
608	(8, -9, -9)	925.341391354414	451.000171289959	21.236764614459
609	(9, -4, 5)	927.013317140573	451.815047640446	21.255941466810
610	(9, -5, 3)	927.492548553824	452.048619218983	21.261435022570
611	(2, 8, -9)	928.535634479889	452.557006648445	21.273387286665
612	(3, 9, -7)	932.767785994783	454.619705967826	21.321812914662
613	(9, -1, -7)	933.434466531958	454.944637975916	21.329431262364
614	(4, -9, 7)	933.802152096439	455.123843460629	21.333631745688
615	(8, -9, -2)	934.492266839685	455.460197016540	21.341513465932
616	(9, 3, -4)	935.780025020272	456.087834735387	21.356213024209
617	(5, 6, -9)	936.145825208990	456.266121310818	21.360386731303
618	(5, 9, -5)	936.418520007047	456.399029448056	21.363497593982
619	(5, -9, 6)	936.887616184303	456.627661235492	21.368847915494
620	(8, 9, 1)	937.444910920767	456.899279930984	21.375202453567
621	(9, 2, -5)	938.168630792063	457.252012218732	21.383451831235
622	(9, 8, 3)	938.500530919894	457.413776315639	21.387233956630
623	(9, 4, -3)	939.399856084974	457.852096493776	21.397478741519
624	(9, -3, 7)	939.668847874420	457.983199829528	21.400542045227
625	(9, -6, 1)	941.106542114174	458.683914565137	21.416907212881
626	(9, -6, -7)	942.852984337153	459.535109323170	21.436770030095
627	(9, -4, -8)	943.724736428810	459.959990719737	21.446677848090

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
628	(9, 7, 1)	945.015298748328	460.588995141890	21.461337217002
629	(9, 9, 5)	945.120525238558	460.640281257006	21.462532032754
630	(9, -7, -4)	946.425646324606	461.276381445391	21.477345772823
631	(9, -3, -8)	946.471384036604	461.298673451496	21.477864732126
632	(9, 1, -6)	946.565673400348	461.344628943812	21.478934539306
633	(4, 7, -9)	947.071410171839	461.591118912488	21.484671719914
634	(9, -7, -3)	948.106520940854	462.095619347130	21.496409452444
635	(9, 5, -2)	949.028123986171	462.544797493901	21.506854662965
636	(8, 1, -9)	950.046699948436	463.041239064272	21.518393040938
637	(9, -7, -5)	950.207114257780	463.119423053087	21.520209642406
638	(9, -5, -8)	950.784885445752	463.401021722698	21.526751304428
639	(6, 5, -9)	953.436948720079	464.693605197476	21.556753122803
640	(9, -7, -2)	955.249738106526	465.577136758303	21.577236541279
641	(3, -9, 8)	957.175833759714	466.515892419321	21.598978967056
642	(8, -9, -1)	957.648904963846	466.746461586646	21.604315809269
643	(9, -5, 4)	958.149118678371	466.990259684342	21.609957419772
644	(9, -2, -8)	959.024828269133	467.417069917976	21.619830478475
645	(9, -7, -6)	959.450924740378	467.624744170217	21.624632810067
646	(7, 9, -2)	960.898170488614	468.330114195217	21.640936074838
647	(9, 0, -7)	960.971152845127	468.365684910626	21.641757897884
648	(8, -4, 9)	962.680217716603	469.198662400861	21.660994030766
649	(2, 9, -8)	962.888881186959	469.300362445571	21.663341442298
650	(9, -4, 6)	963.964221195134	469.824470123465	21.675434715905
651	(9, 6, -1)	964.664828723861	470.165937735760	21.683310119439
652	(9, -2, 9)	965.459140755366	470.553075786227	21.692235380113
653	(9, -6, 2)	965.468778308706	470.557773012838	21.692343649611
654	(6, -9, 5)	966.432226023305	471.027345743911	21.703164417750
655	(9, -6, -8)	967.651831087430	471.621766460379	21.716854432914
656	(9, -7, -1)	967.855297821622	471.720933678910	21.719137498504
657	(4, -8, 9)	969.204291944679	472.378417053427	21.734268265884
658	(6, 9, -4)	970.190349211488	472.859009406030	21.745321552142
659	(7, -9, 3)	970.677254670767	473.096321221528	21.750777485449
660	(9, 8, 2)	973.500868539597	474.472516375443	21.782390051954
661	(9, 9, 4)	973.826528928247	474.631239299151	21.786033124439
662	(9, -7, -7)	974.157077772400	474.792344796781	21.789730259844
663	(9, -1, -8)	981.385069126399	478.315180119175	21.870417922828
664	(9, -3, 8)	982.914085858995	479.060404330205	21.887448556883
665	(9, -4, -9)	983.246628066177	479.222481368801	21.891150754787
666	(9, -5, -9)	985.676425914286	480.406735370551	21.918182757030
667	(9, 3, -5)	985.865004641774	480.498646355182	21.920279340263
668	(9, -7, 0)	985.923200086140	480.527010108952	21.920926305906
669	(3, 8, -9)	986.213703608626	480.668598002485	21.924155582427
670	(8, -9, 0)	986.267885637430	480.695005666186	21.924757824573
671	(9, 7, 0)	986.309970298045	480.715517219353	21.925225591071
672	(8, 9, 0)	986.709908736750	480.910442363788	21.929670366054
673	(9, 4, -4)	988.652844325886	481.857405605347	21.951250661531
674	(9, 2, -6)	989.085601794156	482.068326346751	21.956054434865
675	(1, -9, 9)	991.025595672060	483.013855833934	21.977576204712
676	(9, -5, 5)	994.268031352341	484.594179659137	22.013499941153
677	(9, -7, -8)	994.325573353845	484.62224932780	22.014136933634
678	(9, 5, -3)	997.449120846492	486.144604097247	22.048687128653
679	(9, 1, -7)	998.314635783032	486.566445580056	22.058251190429

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
680	(7, 4, -9)	998.944780705106	486.873570572462	22.065211772663
681	(9, -8, -5)	1000.578706655429	487.669926263838	22.083249902671
682	(9, -8, -4)	1001.427589891087	488.083661657352	22.092615545864
683	(9, -8, -6)	1005.192165969194	489.918470379758	22.134101978164
684	(4, 9, -7)	1006.284223644492	490.450726045850	22.146122144652
685	(9, -4, 7)	1006.377467799118	490.496172115918	22.147148171174
686	(8, 2, -9)	1006.454612243886	490.533771382664	22.147997006110
687	(2, -9, 9)	1007.008661174127	490.803808111568	22.154092355851
688	(5, -9, 7)	1007.614122918051	491.098902822310	22.160751404732
689	(9, -8, -3)	1007.738815676168	491.159676560301	22.162122564418
690	(9, -2, -9)	1007.807422244166	491.193114569459	22.162876947036
691	(9, -7, 1)	1009.453444900083	491.995366048428	22.180968555237
692	(4, -9, 8)	1011.281377451010	492.886278203960	22.201042277424
693	(9, 6, -2)	1012.253834203592	493.360241830881	22.211714067826
694	(9, 0, -8)	1013.552106608401	493.993004055095	22.225953389115
695	(9, 8, 1)	1013.963548708723	494.193535944681	22.230464141459
696	(1, 9, -9)	1014.307617844673	494.361231068555	22.234235562946
697	(5, 9, -6)	1014.586444905370	494.497127996229	22.237291381736
698	(6, -7, 9)	1014.883201995054	494.641763802570	22.240543244322
699	(9, -8, -7)	1015.267967832383	494.829294005113	22.244758798538
700	(7, -6, 9)	1017.556810655124	495.944848237050	22.269819223268
701	(3, 9, -8)	1019.279643849153	496.784536240710	22.288663850503
702	(9, -8, -2)	1019.512384010673	496.897970972683	22.291208378477
703	(9, -7, -9)	1019.956411484713	497.114384578214	22.296062086795
704	(8, -9, 1)	1020.349208860438	497.305829255162	22.300354913211
705	(7, -9, 4)	1022.435981613445	498.322896985885	22.323147112042
706	(7, 9, -3)	1025.259819881467	499.699201509245	22.353952704371
707	(9, -6, 4)	1030.580278346042	502.292328436543	22.411879181286
708	(9, -8, -8)	1030.806112244995	502.402397139902	22.414334635226
709	(9, 7, -1)	1033.066984397185	503.504318806250	22.438901907318
710	(9, -1, -9)	1034.798014270264	504.348001771868	22.457693598673
711	(9, -5, 6)	1035.849286575735	504.860379143365	22.469098316207
712	(9, -8, -1)	1036.748294894601	505.298544894501	22.478846609524
713	(9, -7, 2)	1038.446032263448	506.126001497339	22.497244308967
714	(5, -8, 9)	1040.438580228347	507.097145209129	22.518817580173
715	(8, 9, -1)	1041.437249102155	507.583884306026	22.529622373800
716	(5, 7, -9)	1041.578511813550	507.652734038277	22.531150304374
717	(9, 4, -5)	1043.368175116221	508.524994226352	22.550498757818
718	(6, 9, -5)	1044.186307631787	508.923742091848	22.559338245876
719	(9, 2, -7)	1045.464915345673	509.546919984205	22.573145992179
720	(9, 9, 2)	1047.625563955894	510.599993911744	22.596459765011
721	(8, -5, 9)	1048.459406208558	511.006398512569	22.605450637237
722	(9, 5, -4)	1051.332460256237	512.406690210027	22.636401883030
723	(9, -8, -9)	1051.806599207030	512.637779784125	22.641505687214
724	(2, 9, -9)	1053.572705519352	513.498558580810	22.660506582617
725	(9, -4, 8)	1054.253056952526	513.830153617805	22.667821986636
726	(9, 1, -8)	1055.525940715139	514.450541725734	22.681502192883
727	(9, -8, 0)	1059.446548327953	516.361398325753	22.723586827914
728	(9, 8, 0)	1059.888571427272	516.576835023354	22.728326709711
729	(8, -9, 2)	1059.892874632869	516.578932353571	22.728372848789
730	(9, -9, -5)	1060.757095677813	517.000143209309	22.737637150973
731	(4, 8, -9)	1062.307823796279	517.755949286489	22.754251235461

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
732	(9, -9, -7)	1066.185654517102	519.645956948164	22.795744272740
733	(9, -9, -4)	1066.236330082304	519.670655604033	22.796286004611
734	(9, -6, 5)	1071.329542188845	522.153025412547	22.850667942372
735	(8, 3, -9)	1072.669321164072	522.806017435776	22.864951726076
736	(9, -7, 3)	1072.900962176237	522.918916455684	22.867420415423
737	(9, -9, -8)	1077.093447760881	524.962283081743	22.912055409364
738	(7, -9, 5)	1079.657051105546	526.211752259677	22.939305836482
739	(9, -5, 7)	1082.892884348552	527.788858137028	22.973655741676
740	(5, -9, 8)	1083.802972201222	528.232423918563	22.983307506070
741	(7, 5, -9)	1084.770013269908	528.703748006836	22.993558837353
742	(9, 7, -2)	1085.286341045748	528.95539902582	22.999030412228
743	(9, -8, 1)	1087.607144310728	530.086531266439	23.023608128754
744	(9, 9, 1)	1092.718595293853	532.577790482192	23.077646987555
745	(9, -9, -2)	1093.581826539556	532.998518921785	23.086760684899
746	(4, 9, -8)	1094.086457570262	533.244469965813	23.092086739093
747	(4, -9, 9)	1094.222945355004	533.310992456726	23.093527068352
748	(7, 9, -4)	1095.083811823743	533.730568332707	23.102609556773
749	(5, 9, -7)	1098.216712353117	535.257506053837	23.135632821556
750	(6, -9, 7)	1099.842144798577	536.049722113955	23.152747614785
751	(8, 9, -2)	1101.626932016984	536.919605757699	23.171525753772
752	(9, 3, -7)	1102.421991533049	537.307108123075	23.179885852244
753	(9, 4, -6)	1103.545848455980	537.854862356792	23.191698134393
754	(8, -9, 3)	1104.898882954723	538.514314961415	23.205911207307
755	(9, 2, -8)	1107.306571446612	539.687793131094	23.231181483754
756	(9, -4, 9)	1107.590988655357	539.826414629127	23.234164814538
757	(9, 5, -5)	1110.678142215405	541.331055832242	23.266522211801
758	(9, 8, -1)	1111.275936695245	541.622413611461	23.272782678731
759	(9, -7, 4)	1112.818234638450	542.374110923463	23.288926787713
760	(9, -9, -1)	1115.448088592317	543.655869844812	23.316429182978
761	(9, 1, -9)	1118.199588196669	544.996917380847	23.345169037316
762	(9, -8, 2)	1121.230082842926	546.473943716559	23.376782150599
763	(7, -7, 9)	1122.952662001408	547.313507906376	23.394732482043
764	(9, 6, -4)	1123.818872811323	547.735688549428	23.403753727755
765	(6, -8, 9)	1130.088919570930	550.791633294794	23.468950408887
766	(9, -5, 8)	1135.398824670792	553.379616640125	23.524022118680
767	(7, -9, 6)	1142.340463147071	556.762887042903	23.595823508471
768	(9, 7, -3)	1142.968040243735	557.068760508348	23.602304135579
769	(8, -6, 9)	1144.045391325250	557.593848358997	23.613425172113
770	(8, 4, -9)	1148.690826708994	559.857977223608	23.661318163272
771	(6, 7, -9)	1154.501664514177	562.690109094030	23.721089964292
772	(8, -9, 4)	1155.367233826001	563.111977078693	23.729980553694
773	(5, 8, -9)	1156.817995042847	563.819060500457	23.744874404815
774	(9, -7, 5)	1158.197849650086	564.491584900677	23.759031649052
775	(9, -8, 3)	1160.315363924548	565.523635676114	23.780740856334
776	(5, -9, 9)	1165.454164033816	568.028224524250	23.833342705635
777	(8, 9, -3)	1167.278957481237	568.917606718806	23.851993768212
778	(9, 8, -2)	1168.125644512641	569.330271709003	23.860642734616
779	(9, 3, -8)	1168.893998802822	569.704758271174	23.868488814149
780	(9, 4, -7)	1169.185864345162	569.847009996665	23.871468534564
781	(9, -6, 7)	1169.215097522721	569.861257892858	23.871766962101
782	(7, 9, -5)	1170.370146315442	570.424214665604	23.883555318788
783	(9, 2, -9)	1174.610570096976	572.490945787417	23.926783022116

Rank	s=(s1,s2,s3)	q_R (nm^-2)	kappa^2 (nm^-2)	kappa (nm^-1)
784	(6, -9, 8)	1174.740618010348	572.554329563129	23.928107521556
785	(9, 5, -6)	1175.486166723996	572.917700963892	23.935699299663
786	(9, -9, 1)	1175.567640346109	572.957410219170	23.936528783831
787	(7, 6, -9)	1180.402042459445	575.313639175931	23.985696553903
788	(5, 9, -8)	1187.309322350286	578.680163620879	24.055771939825
789	(4, 9, -9)	1187.351034045455	578.700493395211	24.056194491133
790	(9, 6, -5)	1187.794905939324	578.916831172853	24.060690579716
791	(9, -5, 9)	1193.367107542456	581.632654652657	24.117061484614
792	(9, 9, -1)	1199.291685618041	584.520222151392	24.176853024151
793	(9, -8, 4)	1204.862987555593	587.235607145104	24.232944665168
794	(9, 7, -4)	1206.112081991145	587.844400623549	24.245502688613
795	(6, 9, -7)	1208.565252120656	589.040045991787	24.270147218173
796	(9, -7, 6)	1209.039807211145	589.271338387325	24.274911707096
797	(7, -9, 7)	1210.486217738019	589.976301335563	24.289427768796
798	(8, -9, 5)	1211.297927246703	590.371918705406	24.297570222255
799	(9, -9, 2)	1213.820930047140	591.601599670500	24.322861666969
800	(9, -6, 8)	1226.351389013795	597.708793397165	24.448083634452
801	(9, 8, -3)	1230.437694879461	599.700409315979	24.488781295033
802	(8, 5, -9)	1234.519128878652	601.689650746160	24.529363031807
803	(7, -8, 9)	1238.155309972429	603.461881310421	24.565461145894
804	(8, 9, -4)	1238.393325494913	603.577887189347	24.567822190608
805	(9, 4, -8)	1240.288222783768	604.501437145974	24.586610932497
806	(9, 5, -7)	1245.756533782011	607.166625604976	24.640751319815
807	(8, -7, 9)	1249.438173066678	608.961011940145	24.677135407906
808	(7, 9, -6)	1251.118823356565	609.780140507935	24.693726744012
809	(9, -8, 5)	1254.872953736062	611.609858123528	24.730747221294
810	(9, 9, -2)	1260.771744604270	614.484857250144	24.788805079111
811	(9, -7, 7)	1265.344107321627	616.713371383408	24.833714409717
812	(6, 8, -9)	1269.744217348330	618.857931644387	24.876855340746
813	(8, -9, 6)	1272.690963216827	620.294139841553	24.905704965761
814	(9, 7, -5)	1274.718466287979	621.282320248183	24.925535505746
815	(5, 9, -9)	1281.864274896880	624.765100697356	24.995301572443
816	(7, -9, 8)	1284.094314878390	625.851995137658	25.017034099542
817	(7, 7, -9)	1285.840868273719	626.703244079746	25.034041704842
818	(9, 8, -4)	1298.212087795704	632.732826432389	25.154181092462
819	(6, 9, -8)	1298.948238189226	633.091617205909	25.161311913450
820	(9, -9, 4)	1306.714537097473	636.876817101465	25.236418468187
821	(9, -8, 6)	1310.345262465954	638.646388611386	25.271454026458
822	(8, 9, -5)	1314.970036058012	640.900447169323	25.316011675802
823	(9, 4, -9)	1316.852923771796	641.818143804716	25.334130018706
824	(9, 5, -8)	1321.489243389449	644.077829755494	25.378688495576
825	(9, -7, 8)	1327.110749981533	646.817683888925	25.432610638488
826	(8, 6, -9)	1330.154227673046	648.301038003432	25.461756380961
827	(9, 6, -7)	1332.133999843596	649.265954948006	25.480697693509
828	(7, 9, -7)	1337.329842947111	651.798345859701	25.530341671425
829	(8, -9, 7)	1339.546341736375	652.878640487134	25.551489985657
830	(9, 7, -6)	1348.787193134236	657.382519382253	25.639471901392
831	(9, -9, 5)	1361.354854446774	663.507845081098	25.758646025774
832	(7, -9, 9)	1363.164754568184	664.389968449187	25.775763198190
833	(8, -8, 9)	1364.637751432842	665.107889256013	25.789685714565
834	(9, -8, 7)	1371.279913745269	668.345198608679	25.852373171697
835	(9, 8, -5)	1371.448823261370	668.427523058234	25.853965325618

Rank	s=(s1,s2,s3)	q_R (nm ⁻²)	kappa ² (nm ⁻²)	kappa (nm ⁻¹)
836	(9, -7, 9)	1394.339735190863	679.584275903877	26.068837256462
837	(8, 9, -6)	1397.009089170535	680.885286658733	26.093778696439
838	(9, 9, -4)	1400.118890224998	682.400965975950	26.122805476747
839	(7, 8, -9)	1401.086490712728	682.872562718281	26.131830450971
840	(9, 5, -9)	1402.684295546311	683.651313415447	26.146726629072
841	(8, -9, 8)	1411.864062805347	688.125420642150	26.232144796836
842	(9, 6, -8)	1412.497060619867	688.433936099734	26.238024622668
843	(9, 7, -7)	1428.318262529917	696.144998025756	26.384559841425
844	(7, 9, -8)	1429.003205087081	696.478830720901	26.390885372054
845	(8, 7, -9)	1435.596123092176	699.692138995425	26.451694444693
846	(9, -8, 8)	1437.676907574008	700.706288115406	26.470857336237
847	(9, 8, -6)	1450.147901276460	706.784499193513	26.585418920783
848	(9, 9, -5)	1477.985976859498	720.352439603005	26.839382250771
849	(8, 9, -7)	1484.510484832481	723.532405657578	26.898557687311
850	(9, -9, 7)	1487.022516793647	724.756739568669	26.921306423884
851	(8, -9, 9)	1489.644126423741	726.034480306601	26.945027005119
852	(9, -8, 9)	1509.536243952170	735.729657131567	27.124336989714
853	(9, 7, -8)	1513.311674475020	737.569756178694	27.158235512984
854	(7, 9, -9)	1526.138909776474	743.821595091536	27.273092877258
855	(9, 8, -7)	1534.309321840973	747.803754838227	27.346000710126
856	(8, 8, -9)	1550.844815136041	755.862953722137	27.492961894313
857	(9, -9, 8)	1558.049861791219	759.374606076606	27.556752458819
858	(8, 9, -8)	1577.474223043851	768.841804165857	27.727996757174
859	(9, 7, -9)	1603.767428969547	781.656793841067	27.958125721176
860	(9, 8, -8)	1623.933084954910	791.485289992375	28.133348360840
861	(9, 9, -7)	1650.107177776766	804.242225385417	28.359164751195
862	(8, 9, -9)	1675.900303804643	816.813482183570	28.579948953481
863	(9, 8, -9)	1719.019190618270	837.829104655957	28.945277760905
864	(9, 9, -8)	1744.361292059535	850.180537540776	29.157855503119

11. Locked Mass Bridge

11.1 Face-area ratios

The three primitive face areas carry the locked projective bridge. In the terminal geometry they satisfy

$1000\ A_{bc}/A_{ab} = 1836.152673426000 = \eta_N.$
$100\ A_{bc}/A_{ac} = 137.035999177000 = c_N = \alpha_N^{-1}.$
$10\ A_{ac}/A_{ab} = 13.399053419929 = \eta_N\ \alpha_N.$

Table 11.1. Locked projective Mass-Bridge values.

Quantity	Locked value	Computed from B
η_N	1836.15267342600092	1836.15267342600092
α_N^{-1}	137.035999177000008	137.035999177000008
$\eta_N\ \alpha_N$	13.399053419929000	13.399053419929222
α_N	0.007297352564331425	inverse of c_N
μ_{ep}	0.999455679424762	$\eta_N/(1+\eta_N)$

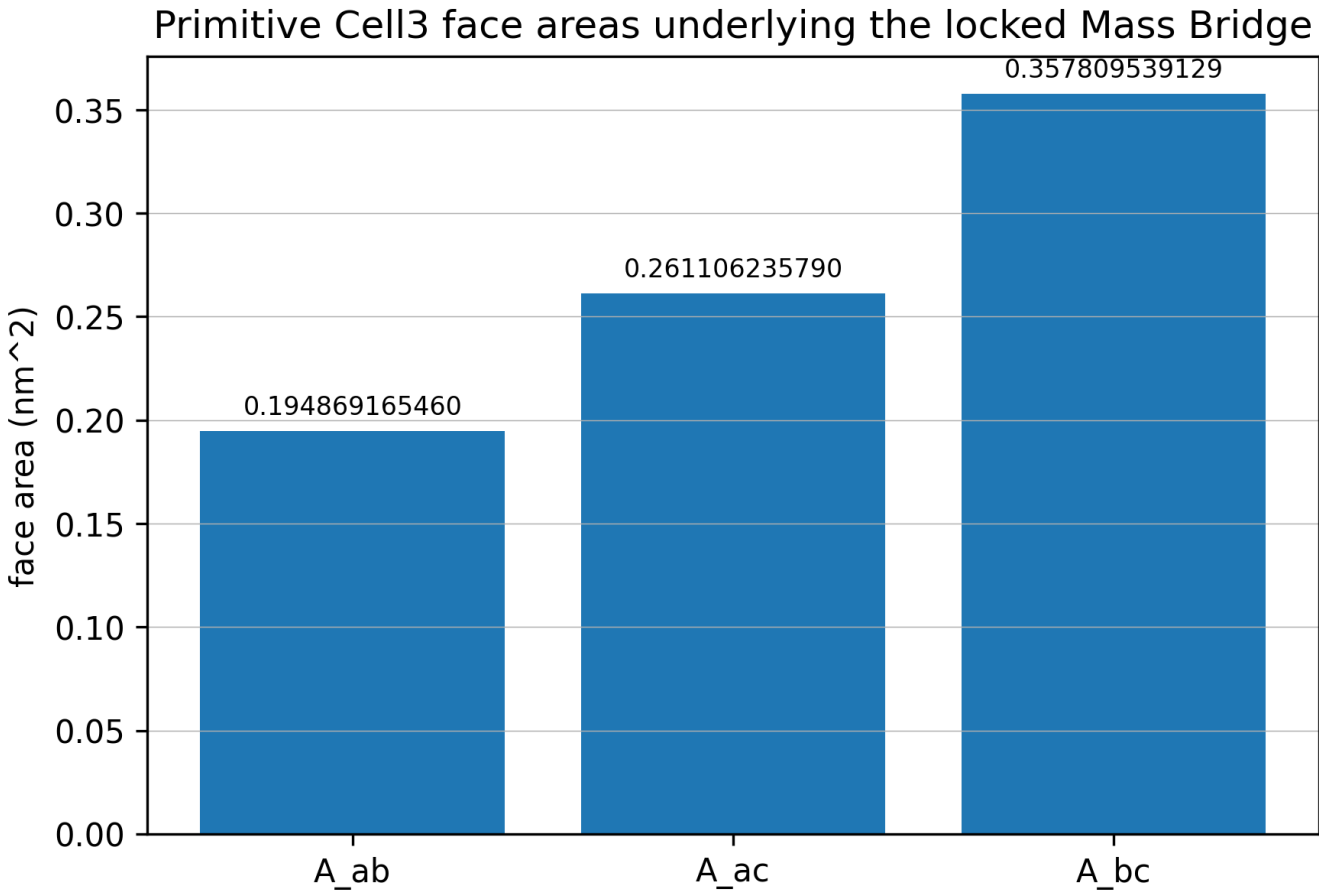


Figure 11.1. The three actual primitive face areas from B. Their projective ratios, after the fixed registry factors 10, 100, and 1000, produce the locked Mass-Bridge values.

11.2 Physical witness triangle

The corresponding physical witness relations are $\lambda_{\bar{e}}/\lambda_{\bar{p}}=\eta$, $\lambda_{\bar{e}}/r_e=\alpha^{-1}$, and $r_e/\lambda_{\bar{p}}=\eta\ \alpha$. Normalizing the reduced proton wavelength to one gives the three projective side ratios 1, $\eta\ \alpha$, and η .

Mass-Bridge physical witness triangle normalized to reduced proton wavelength

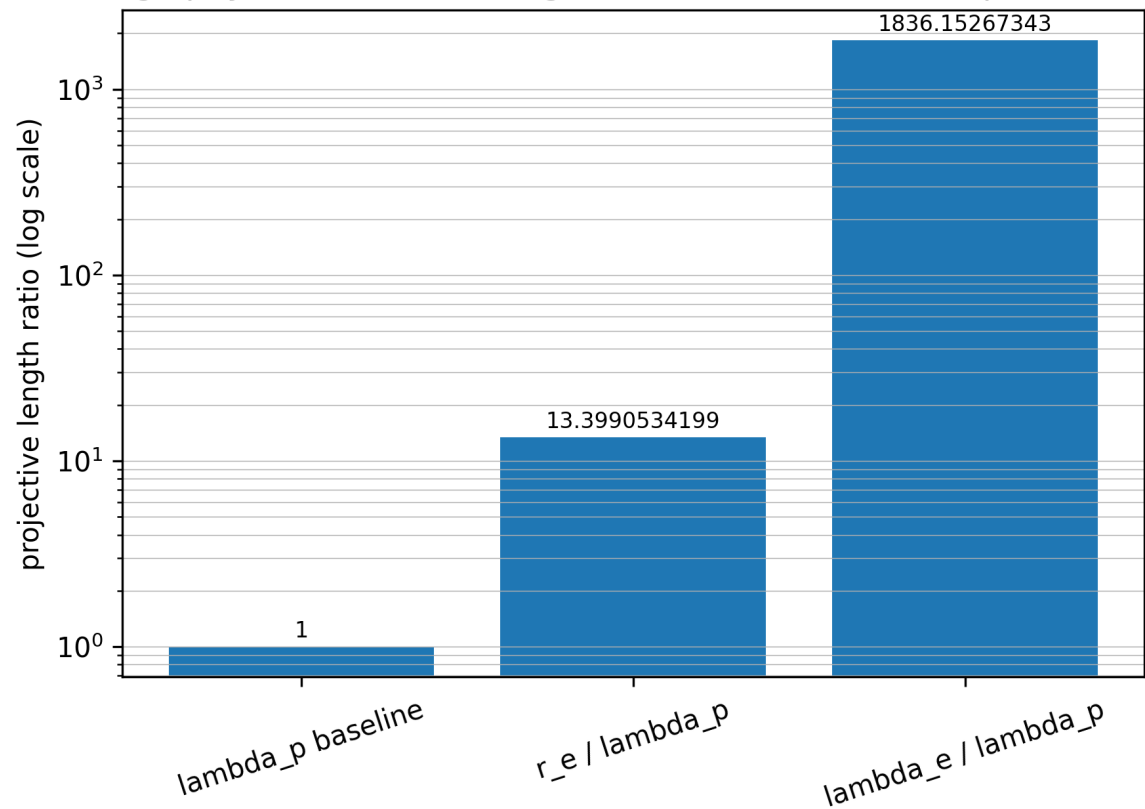


Figure 11.2. Physical witness ratios normalized to the reduced proton wavelength. A logarithmic ordinate is required because the projective range extends from 1 to 1836.152673426.

The bridge is projective. Its invariant content resides in the ratios and the registry factors 10, 100, and 1000. A common scale change of B changes all three areas by the same square factor and leaves every bridge ratio unchanged.

12. Positive Address Energy and the Phase Metric

12.1 Independent address-energy form W

Table 12.1. Positive-definite address-energy matrix W, in the stored nm² normalization.

W	column 1	column 2	column 3
row 1	0.08335330263345708	0.13750057660358808	0.10255921122195821
row 2	0.13750057660358811	0.69240004284854761	0.55080090445346952
row 3	0.10255921122195823	0.55080090445346952	0.58774765235339133

The eigenvalues are

0.050269586792171 nm²; 0.094205549207267 nm²; 1.219025861835957 nm²

All are positive, so every nonzero address-rate vector has positive quadratic phase cost. W is not inferred from M and is not a synonym for the direct or reciprocal metric.

12.2 Push-forward into physical velocity space

The centered half-step coordinate convention is $x=B q/2$, so $\dot{q}=2 B^{-1} v$. Substitution into $\dot{q}^T W \dot{q}$ gives

$$\Phi(\dot{q}) = \dot{q}^T W \dot{q} = v^T G_{\text{phase}} v; \quad G_{\text{phase}} = 4 B^{-T} W B^{-1}.$$

Table 12.2. Physical-space phase metric G_{phase}.

G _{phase}	column 1	column 2	column 3
row 1	2.9764106380574855	1.2431507795645353	-2.0803600355182303
row 2	1.2431507795645358	2.5922286635481870	-2.3178839937876110
row 3	-2.0803600355182308	-2.3178839937876106	6.7735709133237076

Its eigenvalues are

1.429328753509671; 2.112545026540889; 8.800336434878821

Therefore $v^T G_{\text{phase}} v$ is strictly positive for every nonzero physical velocity. The directional function $f_G(\hat{n})=\sqrt{\hat{n}^T G_{\text{phase}} \hat{n}}$ is an ellipsoidal indicatrix of the fixed field.

Unit phase-cost ellipsoid: $\mathbf{v}^T \mathbf{G}_{\text{phase}} \mathbf{v} = 1$

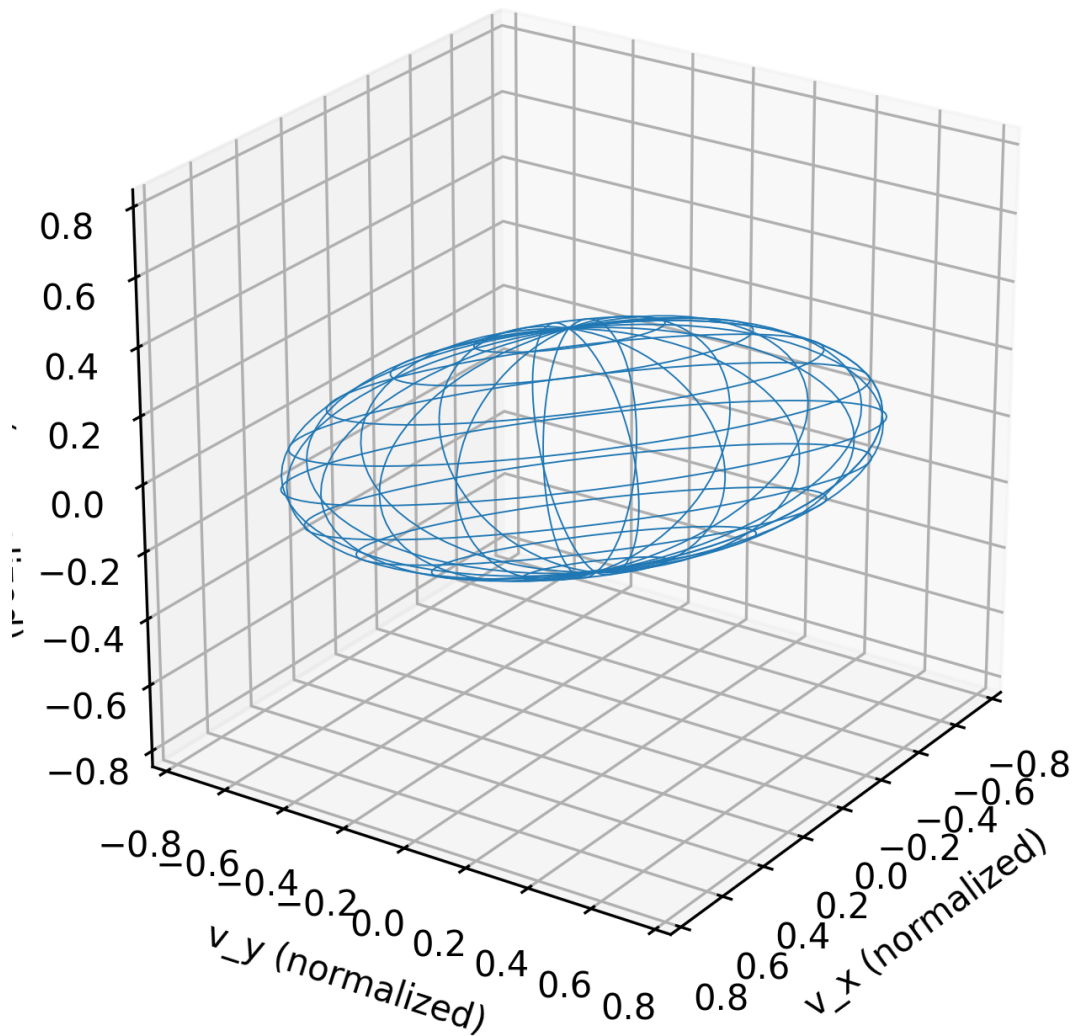


Figure 12.1. The unit phase-cost ellipsoid $\mathbf{v}^T \mathbf{G}_{\text{phase}} \mathbf{v} = 1$. Its principal semi-axes are the inverse square roots of the three positive eigenvalues of $\mathbf{G}_{\text{phase}}$. The orientation is computed from the eigenvectors in fixed ICRS coordinates.

13. Observer Rendering and Continuum Force Reduction

13.1 Mixed observer-motion term

For a local motion v_R superposed on a reference motion V_{ref} , the cross term of the quadratic phase energy is

$$\epsilon_{cross} = \kappa_F v_R^T G_{phase} V_{ref}.$$

The current calibrated matter-coupling coefficient is $\kappa_F=3.0740508997020345e-11$. The stored vector and apsidal channels are $3.04273994475465e-11$ and $3.10536185464942e-11$.

13.2 General mass- and shape-reduced force law

$$F_K = M_{active} \kappa_F H [\omega \times (G_{phase} V_{ref})].$$

M_{active} is the active mass, ω is the angular-velocity vector, and H is the dimensionless shape or occupancy tensor. The cross product fixes the force orthogonal to both the local rotation axis and the phase-weighted reference velocity. Reversing ω reverses the force. Setting ω to zero removes the rotational contribution.

13.3 Thin rim and circular oscillator

$$F_{rim} = (\kappa_F M \omega / 2) \hat{n} \times (G_{phase} V_{ref}).$$

$$F_{bar_K} = M_{active} \pi \kappa_F f C_{pol} H [\hat{n} \times (G_{phase} V_{ref})].$$

For equal-amplitude orthogonal channels at plus or minus 90 degrees phase, $C_{pol}=+1$ or -1 . Linear polarization gives $C_{pol}=0$. This polarity is a dynamical circulation sign and must not be confused with the static sheet polarity χ .

13.4 Laminar normal and CMB direction are different vectors

Table 13.1. Intrinsic and observer directions.

Vector	x	y	z	Role
N	0.876976866375	0.047223124232	-0.478206600101	intrinsic seven-sheet normal
p_obs	-0.970770855836	0.207348009111	-0.120874929482	external observer/traversal direction

The projective angular separation is 38.394897 degrees. The observer direction is neither a primitive face normal nor the laminar normal. The operational baseline is a declared SGA-2020/ICRS/J2000 observer packet; the older 369.82 km/s value remains a historical witness packet and does not alter B, W, the sheet complex, or the hierarchy.

14. Lucas-Supergroup Energy Bands

14.1 Interface-minus-exterior residual

For a complete block with N Cell3 volumes per axis, the bulk volume count, internal interfaces, and exposed primitive faces are

$$V(N)=N^3; \quad I(N)=3(N-1)N^2; \quad E_{\text{boundary}}(N)=6N^2.$$

The archived interface-minus-exterior residual is

$$\Delta(N)=I(N)-E_{\text{boundary}}(N)=3N^2(N-3).$$

At the Lucas Supergroup scale $N=9$, $V=729$, $I=1944$, $E_{\text{boundary}}=486$, and $\Delta=1458=2 \times 729$.

14.2 Similar assemblies of Lucas Supergroups

For an $m \times m \times m$ geometrically similar assembly of Lucas Supergroups, $N=9m$. Substitution yields

$$V_m = 729 m^3; \quad \Delta_m = 729 m^2(3m-1).$$

Let ϵ_3 be the energy quantum per Cell3 occupancy and $\epsilon_{LS}=729 \epsilon_3$. The consistent band law is

$$E_m = 729 m^2(3m-1) \epsilon_3 = m^2(3m-1) \epsilon_{LS}.$$

The dimensionless band index is $B_m=m^2(3m-1)$. Its first values are 2, 20, 72, 176, 350, 612, 980, and 1472. The third finite difference is constant: $\Delta^3 B_m=18$.

14.3 Modular and recursive structure

$$B_m \bmod 6 = 0 \text{ when } 3 \text{ divides } m; \quad B_m \bmod 6 = 2 \text{ otherwise.}$$

The recursive-only sequence $m=3^r$ contains $m^3=27^r$ Lucas Supergroups. This separates the complete geometric recursion from arbitrary positive integer assembly multipliers.

14.4 Electron-carrier energy rendering

With the electron-carrier correction, ϵ_3 is approximately 0.335 eV per Cell3 occupancy. Therefore $\epsilon_{LS}=729 \epsilon_3$ is approximately 244.215 eV, and the first Supergroup residual band is $E_1=2 \epsilon_{LS}$, approximately 488.430 eV.

A notation of the form $E_m=B_m \epsilon_3$ is consistent only when ϵ_3 has been redefined to mean the Supergroup quantum ϵ_{LS} . Applying that line to a per-Cell3 quantum omits the required factor 729. The complete formula above preserves the Cell3 normalization.

This band law is an exact geometric interface-minus-exterior accounting rule. It is not identical to total bulk energy and, by itself, is not a complete nuclear mass or decay law.

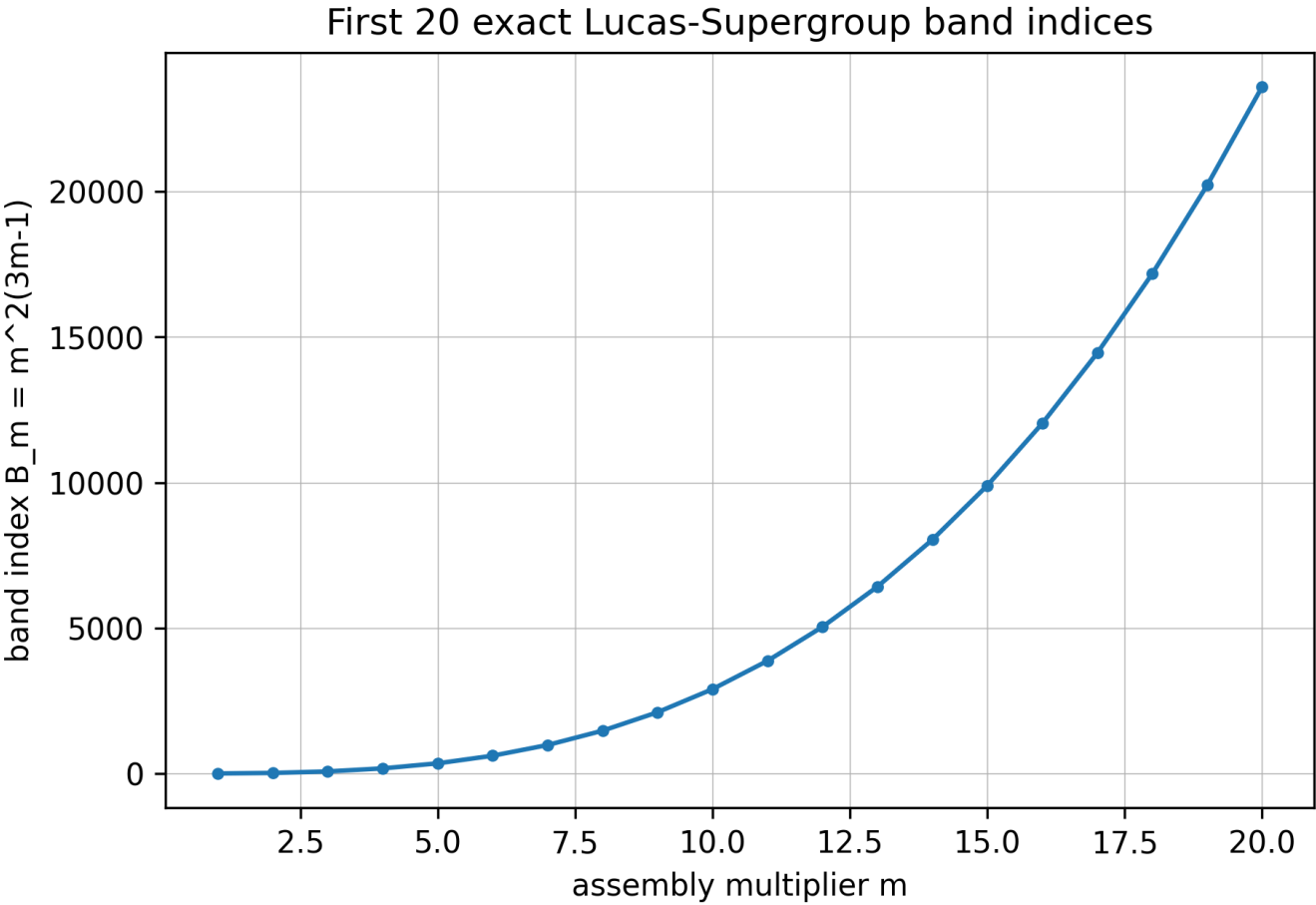


Figure 14.1. The first twenty exact band indices $B_m=m^2(3m-1)$. The cubic law produces a constant third finite difference rather than a uniformly spaced ladder.

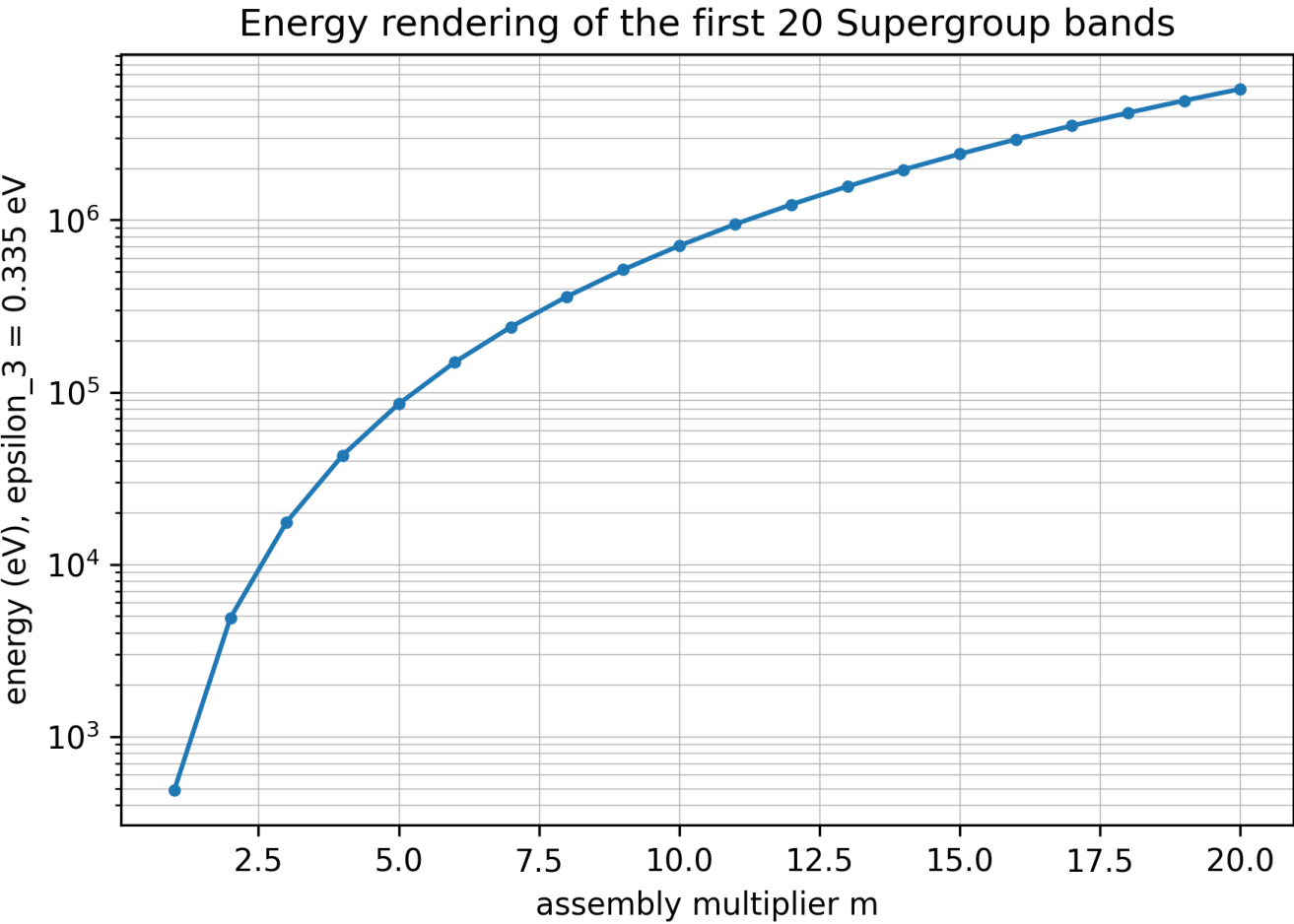


Figure 14.2. The same first twenty bands rendered in electronvolts using epsilon_3=0.335 eV and the required factor 729. A logarithmic ordinate displays the cubic range clearly.

14.5 Numerical band table

Table 14.1. First thirty Lucas-Supergroup energy bands.

m	m^3 Supergroups	B_m	729 B_m Cell3 quanta	Energy (eV)	B_m mod 6
1	1	2	1,458	488.430000	2
2	8	20	14,580	4884.300000	2
3	27	72	52,488	17583.480000	0
4	64	176	128,304	42981.840000	2
5	125	350	255,150	85475.250000	2
6	216	612	446,148	149459.580000	0
7	343	980	714,420	239330.700000	2
8	512	1,472	1,073,088	359484.480000	2
9	729	2,106	1,535,274	514316.790000	0
10	1,000	2,900	2,114,100	708223.500000	2
11	1,331	3,872	2,822,688	945600.480000	2
12	1,728	5,040	3,674,160	1230843.600000	0
13	2,197	6,422	4,681,638	1568348.730000	2
14	2,744	8,036	5,858,244	1962511.740000	2
15	3,375	9,900	7,217,100	2417728.500000	0
16	4,096	12,032	8,771,328	2938394.880000	2
17	4,913	14,450	10,534,050	3528906.750000	2

m	m^3 Supergroups	B_m	729 B_m Cell3 quanta	Energy (eV)	B_m mod 6
18	5,832	17,172	12,518,388	4193659.980000	0
19	6,859	20,216	14,737,464	4937050.440000	2
20	8,000	23,600	17,204,400	5763474.000000	2
21	9,261	27,342	19,932,318	6677326.530000	0
22	10,648	31,460	22,934,340	7683003.900000	2
23	12,167	35,972	26,223,588	8784901.980000	2
24	13,824	40,896	29,813,184	9987416.640000	0
25	15,625	46,250	33,716,250	11294943.750000	2
26	17,576	52,052	37,945,908	12711879.180000	2
27	19,683	58,320	42,515,280	14242618.800000	0
28	21,952	65,072	47,437,488	15891558.480000	2
29	24,389	72,326	52,725,654	17663094.090000	2
30	27,000	80,100	58,392,900	19561621.500000	0

15. Most Complete Internal Hamiltonian

15.1 Typed assembly

The full internal-energy architecture is the typed sum

$$H = H_{\text{rest}} + H_W + H_{\text{Emily}} + H_{\text{A2}} + H_{\text{sheet}} + H_{\text{Bloch}} + H_{\text{CMB}} + H_{\text{Coulomb}} + H_{\text{carry}}.$$

Table 15.1. Internal Hamiltonian terms.

Term	Formula	Function
H_W	$\kappa_W \sum_L (1/2) L^T W L n_L$	positive address-energy cost
H_Emily	$-J_E \sum_{\langle L, L_p \rangle} (a_L^\dagger a_{L_p} + a_{L_p}^\dagger a_L)$	nearest-neighbor hopping on the 27-state incidence graph
H_A2	$g_K U_K (\psi_1, \psi_2)$	same-sheet A2 interaction
H_sheet	$\Delta_\lambda \lambda^2 + \Delta_\chi (1-\chi)/2$	grade and polarity splitting
H_Bloch	$\sum_{N,\rho} \hbar \omega_{N,\rho} (b^\dagger b + 1/2)$	collective reciprocal modes
H_CMB	$\gamma \dot{W} \dot{E}$	observer-motion coupling
H_Coulomb	$\alpha \hbar c \sum_{i<j} P_i P_j / r_{ij}$	charge-sector interaction
H_carry	recursive carry and motif-transition energy	cross-scale transition structure

15.2 Geometry and constitutive closure

The geometry fixes B, M, R, W, the node registry, incidence, sheet labels, polarity, cubical boundaries, Hodge sectors, and reciprocal levels. The parameters κ_W , J_E , g_K , Δ_λ , Δ_χ , and γ , together with occupation, pairing, spin-assignment, decay, and boundary rules, form the constitutive layer. The typed Hamiltonian is therefore structurally complete while preserving the distinction between exact geometry and physical state assignment.

This is a complete typed assembly template, but it is not a parameter-free physical Hamiltonian. The exact Supergroup spectrum establishes the geometric state space on which shell, pairing, isomer, and decay models must operate, but it does not by itself establish a one-to-one observed nuclear hierarchy.

The exact Supergroup spectrum supplies the geometric state space on which shell, pairing, isomer, and decay models operate. In particular, the 1,216 center circulations cannot be counted as protected harmonic states because the full Hodge complex makes them coexact.

16. Canonical Corrections and Type-Safety Rules

The following corrections are part of the canonical definition. Each resolves a recurrent identification of two mathematically different objects.

Table 16.1. Canonical corrections.

Conflation	Canonical resolution
27 Lucas nodes interpreted as 27 Cell3 volumes	One Cell3 has 27 half-step states; a Lucas Group has 27 complete Cell3 volumes.
6,859 states interpreted as corner nodes	A Supergroup has 1,000 unrefined corner nodes and 6,859 half-step states.
A and B used interchangeably for direct basis	Terminal convention is $B=[a\ b\ c]$; inverse or inverse transpose is reciprocal.
M, R, W, G_phase, and Q treated as aliases	They are distinct typed tensors with different units and roles.
CMB vector identified with a Cell3 normal	CMB is external traversal direction; Cell3 and laminar normals derive from B.
Seven Emily zero modes treated as Laplacian/Hodge modes	Adjacency nullity is 7; connected Laplacian nullity is 1; global Hodge scalar nullity is 1.
1,216 center cycles treated as nontrivial topology	They are coexact after plaquettes are included; $\beta_1=0$.
55 sheet modes treated as physical zero modes	Orientation lifts 54; one constant scalar mode remains.
Registry standing wave identified with PDE cavity mode	Registry cosine, Bloch, and hard-wall Helmholtz modes are distinct.
Native geometry identified with nanometre witness	Native geometry is primary; nanometre rendering is a scale witness.
Mass-Bridge values refitted to later constants	Projective ratios are locked; physical constants are downstream witnesses.
$E_m=B_m\epsilon_3$ used with a per-Cell3 quantum	Use $E_m=729\ B_m\epsilon_3 = B_m\epsilon_{LS}$.
Proton carrier treated as primitive	Current Earth-CMB branch is electron-carrier normalized.
Galaxy positions used as occupancy domain	The domain is projective galaxy-axis orientation on the unit sphere.
Early force reduction retained	Edition 1784915217 is superseded by corrected edition 1784985105.
Original 13-element result retained as controlling	It is superseded by strict linear-inertia composition.

16.1 Core type-safety rules

Rule 1. A direct address n and a reciprocal address s are both integer triples, but they act through B and B^{-T} respectively.

Rule 2. A registry node is a position in one refined Cell3; a complete Cell3 is a volume element in the hierarchy.

Rule 3. A sheet grade λ is an integer address homomorphism; a physical plane offset is $h\lambda$ in nanometres.

Rule 4. Static polarity χ , circular-polarization sign C_{pol} , basis handedness, and orientation signs of cubical cells are separate sign systems.

Rule 5. Adjacency nullity, graph-Laplacian nullity, center-graph cycle rank, and Hodge Betti numbers are operator-specific quantities.

Rule 6. A scale witness changes dimensional sizes but leaves all projective and combinatorial invariants unchanged.

17. Minimal Complete Axioms

The complete model reduces to eight axioms. Each is required: removing any one erases a distinct part of the observed construction.

Axiom 1 - Fixed frame and basis

There is one fixed ICRS/J2000 origin and one ordered invertible left-handed basis B . Observer motion does not rotate or redefine this basis. This axiom gives a stable meaning to all direction comparisons.

Axiom 2 - Lossless integer addressing

The global direct field is $B Z^3$. Direct and reciprocal quadratic forms are $M=B^T B$ and $R=M^{-1}$. This axiom preserves exact discrete addresses while allowing continuous physical coordinates.

Axiom 3 - Centered half-step Cell3 registry

One Cell3 is sampled by $\{-1,0,1\}^3$ through $p=B L/2$. This axiom creates the 27-node $8+12+6+1$ stratification and the eight congruent half-step child volumes.

Axiom 4 - Laminar grade and polarity

The maps $\lambda=i+j+k$ and $\chi=(-1)^\lambda$ define the seven A2-parallel sheets and their alternating sign structure. This axiom creates the 1,3,6,7,6,3,1 populations.

Axiom 5 - Local incidence

Nearest-neighbor motion changes one address coordinate by one. This axiom produces $P3 \times P3 \times P3$, the Emily spectrum, and the local direct-reciprocal structure factors.

Axiom 6 - Complete-cell recursion by three

Only complete Cell3 volumes aggregate. The hierarchy $n=3^g$ produces Cell3, Lucas Group, Lucas Supergroup, and higher bulk levels without confusing nodes with volumes.

Axiom 7 - Independent positive address-energy form

W is positive definite and independent of M . Together with B it produces $G_{\text{phase}}=4 B^T W B^{-1}$ and a positive directional phase cost.

Axiom 8 - Locked physical bridge and observer rendering

The face-cofactor ratios are fixed by the Mass Bridge. Physical force or observer-indexed energy additionally requires a declared observer packet, shape tensor, and coupling κ_F .

17.1 Canonical formula set

$\Lambda = B Z^3$; $M=B^T B$; $R=M^{-1}$.

$p(L)=B L/2$; $\lambda=1^T L$; $\chi=(-1)^\lambda$.

$A_{\text{Cell3}} = J \text{ tensor } I \text{ tensor } I + I \text{ tensor } J \text{ tensor } I + I \text{ tensor } I \text{ tensor } J$.

$n=3^g$; $F_n(t)=((n+1)+nt)^3$; $U_n=\{-n, \dots, n\}^3$.

Lucas Group: $n=3$; Lucas Supergroup: $n=9$; $G_2=27$ $G_1=729$ Cell3.

$(C_0, C_1, C_2, C_3)_{G_2}=(1000, 2700, 2430, 729)$.

$(\beta_0, \beta_1, \beta_2, \beta_3)=(1, 0, 0, 0)$.

$\kappa_s^2=(2 \pi/9)^2 s^T R s$; primitive projective shell count=864.

G_phase=4 B^-T W B^-1.
F_K=M_active kappa_F H [omega cross (G_phase V_ref)].
E_m=729 m^2(3m-1) epsilon_3 = m^2(3m-1) epsilon_LS.

17.2 Final canonical model

The smallest internally complete formulation preserves the actual Cell3 geometry, Lucas nodes, seven-sheet polarity complex, direct-reciprocal duality, Mass Bridge, factor-three hierarchy, complete Lucas-Supergroup topology, deterministic reciprocal spectrum, positive phase-energy law, and observer-rendered dynamics without importing superseded definitions or conflating local graph structure with global closure.

18. Worked Derivations

18.1 Locating a Lucas node and its sheet

Take $L=(1,0,-1)$. Its grade is $\lambda=1+0-1=0$ and its polarity is $\chi=+1$. Its physical position is

$$p = B(1, 0, -1)^T/2 = (0.168815279841, 0.167445347241, 0.326123452736) \text{ nm}.$$

The plane check gives $N \cdot p = 1.203e-17$ nm, equal to $h \lambda = 0$ within numerical precision. Because exactly two address components are nonzero, the point is an edge center.

18.2 Deriving a face spacing

The altitude between opposite bc faces is the cell volume divided by the bc face area.

$$d_a = |\det B|/|b \times c| = 0.117914891911745/0.357809539128562 = 0.329546529695447 \text{ nm}.$$

18.3 Reconstructing the Supergroup f-vector

For $n=9$, the formula $((n+1)+nt)^3$ gives

$$(10+9t)^3 = 10^3 + 3(9)(10^2)t + 3(9^2)(10)t^2 + 9^3 t^3.$$

Thus $C_0=1000$, $C_1=2700$, $C_2=2430$, and $C_3=729$. The alternating Euler sum is $1000-2700+2430-729=1$, matching contractibility.

18.4 Lowest reciprocal quadratic-form direction

The minimum primitive shell address is $s=(1, 4, 9)$. Direct substitution into $q_R=s^T R s$ gives $130.593523477839 \text{ nm}^{-2}$. Multiplying by $(2 \pi/9)^2$ gives $\kappa^2=63.649699460279 \text{ nm}^{-2}$ and $|\kappa|=7.978076175387 \text{ nm}^{-1}$.

18.5 A recursive energy band

For $m=3$, $B_3=3^2(9-1)=72$. The assembly contains $m^3=27$ Lucas Supergroups. With $\epsilon_3=0.335 \text{ eV}$, the energy is

$$E_3 = 729 \times 72 \times 0.335 \text{ eV} = 17583.480000 \text{ eV}.$$

18.6 Mass-Bridge ratio check

Using the computed face areas, $1000 A_{bc}/A_{ab}=1836.152673426000$, $100 A_{bc}/A_{ac}=137.035999177000$, and $10 A_{ac}/A_{ab}=13.399053419929$. These values reproduce the locked projective bridge at the printed precision.

19. Exact Verification Identities

The following identities provide a compact independent reconstruction and consistency test.

Table 19.1. Numerical and combinatorial verification identities.

Identity	Value or consequence
$M \cdot B^T \cdot B$	zero matrix
$R \cdot M$	identity matrix
sign det B	negative; ordered basis is left-handed
$8+12+6+1$	27 Lucas nodes
$1+3+6+7+6+3+1$	27 sheet states
$N \cdot p(L) \cdot h \cdot \lambda(L)$	zero for all 27 nodes
$(n+1)^3 - 3n(n+1)^2 + 3n^2(n+1) - n^3$	1 for every complete cubical block
$1000-2700+2430-729$	1
$1944-729+1$	1216 center cycles
rank B2 + rank B1	$1701+999=2700=C1$
rank B3 + rank B2	$729+1701=2430=C2$
19^3	6859 half-step Supergroup states
19^3-17^3	1946 outer-shell states
$1946-1944$	2
$(6859-1)/2$	3429 noncentral antipodal pairs
$973-109$	864 new primitive projective directions
$12 \cdot J_2(9)$	864
number of distinct q values	864 at 12 decimals
minimum eigenvalue W	positive
minimum eigenvalue G_phase	positive
$\Delta^3 B_m$	18
$B_m \bmod 6$	0 for $3 m$, otherwise 2

20. Formula Compendium and Symbol Index

Table 20.1. Formula compendium.

Symbol or expression	Definition
$B=[a\ b\ c]$	ordered direct Cell3 basis
$M=B^T B$	direct Gram metric
$R=M^{-1}$	reciprocal metric
$V_1= \det B $	Cell3 volume
$A_{ab}= a\ \text{cross}\ b $	ab face area
$d_c=V_1/A_{ab}$	spacing normal to ab face
$L_0=\{-1,0,1\}^3$	single-Cell3 Lucas registry
$p(L)=B\ L/2$	node position
$\lambda=1^T L$	sheet grade
$\chi=(-1)^\lambda$	sheet polarity
$N=B^T\ 1/ B^T\ 1 $	sheet normal
$h=1/(2 B^T\ 1)$	adjacent sheet spacing
$A_2=\ker \lambda$	same-sheet root lattice
$A_{\text{Cell3}}=J\ \text{tensor}\ I\ \text{tensor}\ I + \dots$	nearest-neighbor adjacency
$n=3^g$	complete-cell hierarchy
$F_n(t)=((n+1)+nt)^3$	cubical f-polynomial
$P_n(\lambda)$	half-step sheet population
$I_n=3(n-1)n^2$	internal interfaces
$E_n=6n^2$	exposed faces
$\beta_{\text{center}}=2n^3-3n^2+1$	center-graph cycle rank
$k_s=(2\pi/9)B^T\ s$	Supergroup reciprocal wavevector
$q_R=s^T R\ s$	reciprocal quadratic form
$G_{\text{phase}}=4B^T\ W\ B^{-1}$	physical phase metric
$F_K=M_{\text{active}}\ \kappa_F\ H[\omega\ \text{cross}\ G_{\text{phase}}\ V_{\text{ref}}]$	general continuum reduction
$B_m=m^2(3m-1)$	band index
$E_m=729B_m\ \epsilon_3$	Cell3-normalized band energy

Appendix A. Canonical Numerical Matrices

This appendix repeats the four controlling matrices in a compact machine-checkable decimal form.

Table A.1. B (nm).

B	column 1	column 2	column 3
row 1	0.3275012420558906	0.035330775843828398	-0.01012931762672882
row 2	-0.054897007884006442	0.20731985044019177	-0.38978770236625987
row 3	-0.037487502894729907	-0.54740108538850807	-0.68973440836726763

Table A.2. M (nm^2).

M	column 1	column 2	column 3
row 1	0.11167605789605009	0.020710333280605803	0.043937235096219773
row 2	0.020710333280605803	0.34387773239278702	0.2963927589615229
row 3	0.043937235096219773	0.2963927589615229	0.62777061007729584

Table A.3. R (nm^-2).

R	column 1	column 2	column 3
row 1	9.208025529457549	0.0015348024281926977	-0.64518803569993899
row 2	0.0015348024281927502	4.9033983123679867	-2.315175584416389
row 3	-0.64518803569993899	-2.315175584416389	2.7311712747116945

Table A.4. W (nm^2).

W	column 1	column 2	column 3
row 1	0.083353302633457083	0.13750057660358808	0.10255921122195821
row 2	0.13750057660358811	0.69240004284854761	0.55080090445346952
row 3	0.10255921122195823	0.55080090445346952	0.58774765235339133

Table A.5. G_phase (dimensionless).

G_phase	column 1	column 2	column 3
row 1	2.9764106380574855	1.2431507795645353	-2.0803600355182303
row 2	1.2431507795645358	2.592228663548187	-2.317883993787611
row 3	-2.0803600355182308	-2.3178839937876106	6.7735709133237076

Appendix B. Complete Node-to-Sheet Cross-Reference

Node	Address	Class	lambda	chi	Sheet offset (nm)	Antipode
L(-1,-1,-1)	(-1, -1, -1)	vertex	-3	-1	-0.453818064112	(1, 1, 1)
L(-1,-1,0)	(-1, -1, 0)	edge center	-2	1	-0.302545376075	(1, 1, 0)
L(-1,0,-1)	(-1, 0, -1)	edge center	-2	1	-0.302545376075	(1, 0, 1)
L(0,-1,-1)	(0, -1, -1)	edge center	-2	1	-0.302545376075	(0, 1, 1)
L(-1,-1,1)	(-1, -1, 1)	vertex	-1	-1	-0.151272688037	(1, 1, -1)
L(-1,0,0)	(-1, 0, 0)	face center	-1	-1	-0.151272688037	(1, 0, 0)
L(-1,1,-1)	(-1, 1, -1)	vertex	-1	-1	-0.151272688037	(1, -1, 1)
L(0,-1,0)	(0, -1, 0)	face center	-1	-1	-0.151272688037	(0, 1, 0)
L(0,0,-1)	(0, 0, -1)	face center	-1	-1	-0.151272688037	(0, 0, 1)
L(1,-1,-1)	(1, -1, -1)	vertex	-1	-1	-0.151272688037	(-1, 1, 1)
L(-1,0,1)	(-1, 0, 1)	edge center	0	1	-0.000000000000	(1, 0, -1)
L(-1,1,0)	(-1, 1, 0)	edge center	0	1	-0.000000000000	(1, -1, 0)
L(0,-1,1)	(0, -1, 1)	edge center	0	1	-0.000000000000	(0, 1, -1)
L(0,0,0)	(0, 0, 0)	volume center	0	1	0.000000000000	(0, 0, 0)
L(0,1,-1)	(0, 1, -1)	edge center	0	1	0.000000000000	(0, -1, 1)
L(1,-1,0)	(1, -1, 0)	edge center	0	1	0.000000000000	(-1, 1, 0)
L(1,0,-1)	(1, 0, -1)	edge center	0	1	0.000000000000	(-1, 0, 1)
L(-1,1,1)	(-1, 1, 1)	vertex	1	-1	0.151272688037	(1, -1, -1)
L(0,0,1)	(0, 0, 1)	face center	1	-1	0.151272688037	(0, 0, -1)
L(0,1,0)	(0, 1, 0)	face center	1	-1	0.151272688037	(0, -1, 0)
L(1,-1,1)	(1, -1, 1)	vertex	1	-1	0.151272688037	(-1, 1, -1)
L(1,0,0)	(1, 0, 0)	face center	1	-1	0.151272688037	(-1, 0, 0)
L(1,1,-1)	(1, 1, -1)	vertex	1	-1	0.151272688037	(-1, -1, 1)
L(0,1,1)	(0, 1, 1)	edge center	2	1	0.302545376075	(0, -1, -1)
L(1,0,1)	(1, 0, 1)	edge center	2	1	0.302545376075	(-1, 0, -1)
L(1,1,0)	(1, 1, 0)	edge center	2	1	0.302545376075	(-1, -1, 0)
L(1,1,1)	(1, 1, 1)	vertex	3	-1	0.453818064112	(-1, -1, -1)

Appendix C. Extended Hierarchy Values

Table C.1. Complete hierarchy census through g=5.

g	n	Cell3 volumes	C0	C1	C2	C3	Half-step states	Sheet s	Central sheet	Interface s	Exposed faces	Plaquett es	Center cubes	Center cycles
0	1	1	8	12	6	1	27	7	7	0	6	0	0	0
1	3	27	64	144	108	27	343	19	37	54	54	36	8	28
2	9	729	1,000	2,700	2,430	729	6,859	55	271	1,944	486	1,728	512	1,216
3	27	19,683	21,952	63,504	61,236	19,683	166,375	163	2,269	56,862	4,374	54,756	17,576	37,180
4	81	531,441	551,368	1,633,932	1,614,006	531,441	4,330,747	487	19,927	1,574,640	39,366	1,555,200	512,000	1,043,200
5	243	14,348,907	14,526,784	43,401,744	43,223,868	14,348,907	115,501,303	1,459	177,877	42,869,574	354,294	42,693,156	14,172,488	28,520,668

Appendix D. Machine-Readable Companion Structure

The companion file `K-Field_from_Cell3_through_the_Lucas_Supergroup_1785241222.json` contains the complete numerical definition, not merely a summary. It includes every Cell3 node, all 55 Supergroup sheet populations, all 864 primitive projective reciprocal directions with physical wavevectors, the first 100 energy bands, the hierarchy through $g=5$, the Mass-Bridge values, the phase metric, the Hamiltonian terms, minimal axioms, correction rules, and archival metadata.

Table D.1. JSON schema overview.

Top-level key	Contents
metadata	document identity, epoch, PI, classification, units, font constraints
archival_graph_integrity	graph counts and authoritative dependency set
canonical_object	minimal tuple and observer packet
basis	B and all derived metric geometry
metrics	M, R, W, G_phase, eigenvalues
cell3_registry	all 27 node records
seven_sheet_complex	sheet normal, spacing, seven sheets, roots, 13 channels
local_incidence	adjacency, Emily spectrum, Jack quotient
hierarchy	closed formulas and levels
lucas_supergroup	full $n=9$ census and 55 sheets
cubical_topology	boundary dimensions, ranks, Betti numbers
reciprocal_spectrum	all 864 directions and q/k values
mass_bridge	locked ratios and witnesses
observer_rendering	coupling constants and force laws
energy_bands	first 100 bands
internal_hamiltonian	typed Hamiltonian assembly
minimal_axioms	eight axioms
canonical_corrections	type-safety corrections

Appendix E. Archival Reference Inventory

The controlling records are listed here exactly as archival identifiers. Application records in galaxy geometry, standing waves, propagation, materials, rotors, ballistics, satellites, and solar-system harmonics attach as validations or rendered consequences; they do not add an independent geometric axiom.

Order	Authoritative machine-readable record
1	K_Field_native_unit_sphere_derivation_stack_1784476462.json
2	K-Field_entire_compact_form_1784476462_machine_readable.json
3	K-Field_Mass-Bridge_is_now_a_locked_projective_ratio_system_1785176592(1).json
4	K-Field_mass_bridge_drives_volume_filling_hierarchy_1785204375.json
5	K-Field_internal_energy_hierarchy_1785190900.json
6	K-Field_Lucas_Supergroup_Analysis_1785208014.json
7	K-Field_Lucas_Supergroup_Complete_Nuclear_Closure_Domain_1785235808.json
8	K-Field_Lucas_Supergroup_metric_Hodge_program_1785236325.json
9	K-Field_Lucas_Supergroup_exact_deterministic_reciprocal_field_spectrum_1785237526.json
10	K-Field_Lucas_Supergroup_extracted_definition_1785238606.json

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